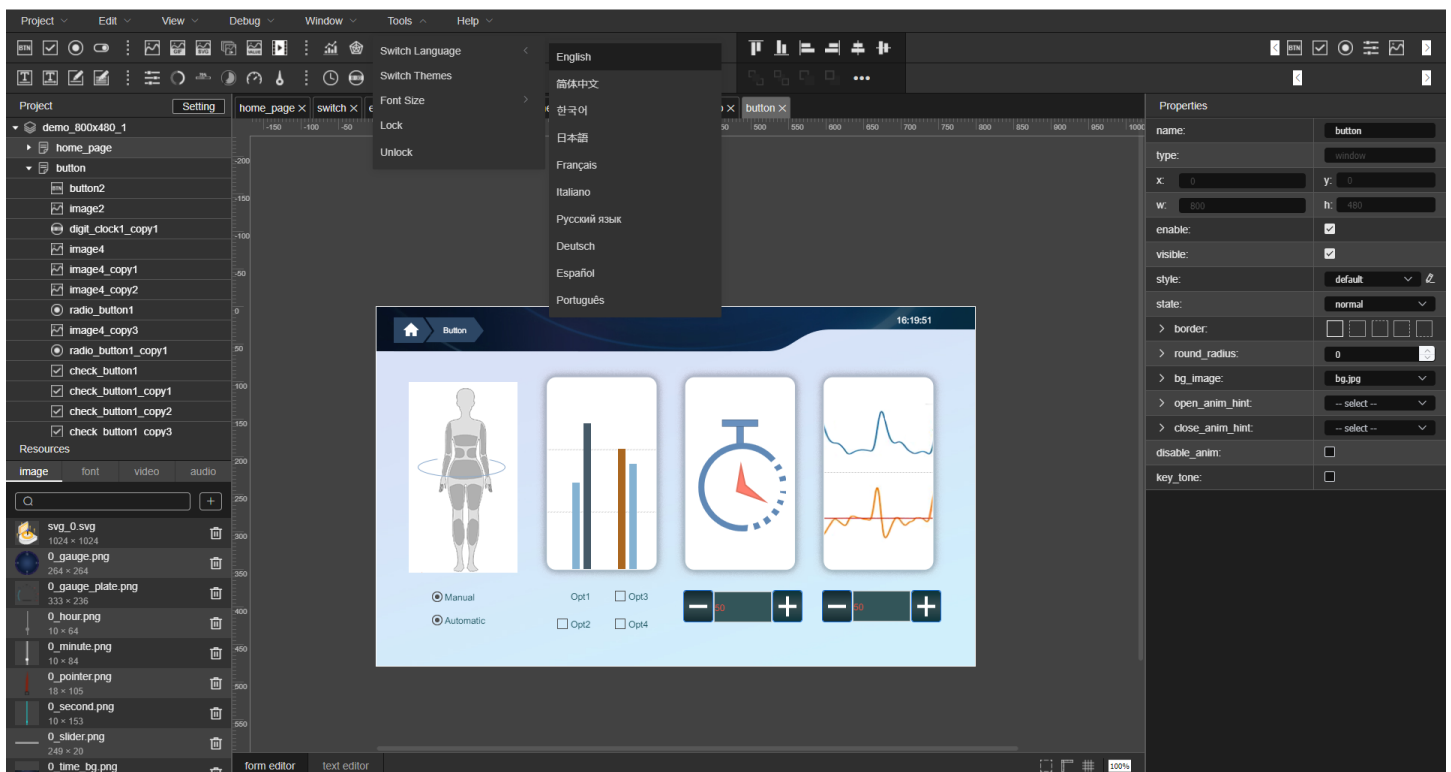


STONE Designer User Manual



Contents

Modification Record.....	
1. General Introduction of the Software	1
1.1 Software registration and login	1
1.1.1 Software registration and login	1
1.1.2 Software login	3
1.1.3 Forgot password	4
1.1.4 Setting	5
1.2 Software details introduction.....	6
1.3 Title bar area	7
1.3.1 Project.....	7
1.3.2 Edit.....	9
1.3.3 View	10
1.3.4 Debug.....	10
1.3.5 Window	11
1.3.6 Tools	11
1.3.7Help.....	12
1.4 Widgets	13
1.5 Debug Tool.....	13
1.6 Widget alignment area	13
1.7 Common widgets area	14
1.8 Customized common widgets area.....	14
1.9 Project area.....	15
1.9.1 Project name	15
1.9.2 Window name	15
1.9.3 Widget name	15
1.9.4 Icon	15
1.9.5 Setting	16
1.10 Resources area.....	18
1.10.1 Resources category	18
1.10.2 Fuzzy search.....	19
1.10.3 Add resources.....	19
1.10.4 Resource information.....	19
1.10.5 Delete resources.....	19
1.11 Properties area.....	20
1.12 Window view area	21
1.12.1 Opened window	21
1.12.2 Ruler.....	21

1.12.3 Gridlines.....	22
1.12.4 Window view.....	23
1.12.5 Tool bar.....	23
2.Project Download and Boot Logo Update.....	26
2.1 Project creation.....	26
2.2 Project download.....	28
2.2.1 Update project by Type A cable.....	29
2.2.2 Update project via U-DISK.....	31
2.3 Boot Logo Update.....	34
3.Window pages, Keyboards and Animations.....	37
3.1 Window pages and custom Keyboards.....	37
3.1.1 Window page.....	37
3.1.2 Create a new page.....	37
3.1.3 Different Window page types and application scenarios.....	38
3.1.4 keyboard.....	40
3.2 Animation and Page Animation.....	45
3.2.1 Page animation.....	45
3.2.2 Page animation type.....	45
3.2.3 Page animation properties.....	45
3.2.4 Widget Animation.....	46
3.2.5 Widget animation properties.....	47
4.Introduction of Properties.....	51
4.1 Common basic property settings.....	51
4.1.1 name.....	51
4.1.2 type.....	51
4.1.3 x/y.....	51
4.1.4 w/h.....	52
4.1.5 text.....	52
4.1.6 enable.....	52
4.1.7 visible.....	52
4.2 General style settings.....	53
4.2.1 border_color.....	53
4.2.2 border.....	53
4.2.3 border_width.....	53
4.2.4 round_radius.....	54
4.2.5 text_color.....	54
4.2.6 selected_text_color.....	55
4.2.7 tips_text_color.....	55
4.2.8 font_name.....	55
4.2.9 font_size.....	56

4.2.10 highlight_font_name	56
4.2.11 highlight_font_size	56
4.2.12 highlight_text_color	56
4.2.13 text_align	56
4.2.14 bg_color	57
4.2.15 bg_image	57
4.2.16 bg_image_draw_type	58
4.2.17 fg_color:	59
4.2.18 fg_image	59
4.2.19 fg_image_draw_type	59
4.2.20 mask_color:	60
4.2.21 icon	60
4.2.22 icon_at	61
4.2.23 margin	61
4.2.24 key_tone	62
4.2.25 selected_bg_color	62
4.2.26 selected_fg_color:	62
4.3 Customizing Style	63
5. Other properties for common widgets	66
5.1  window	66
5.2  button	66
5.2.1 user_define	66
5.2.2 target_name	67
5.2.3 min/max	67
5.2.4 step	67
5.2.5 format	68
5.2.6 increase_decrease	68
5.2.7 enable_long_press	68
5.2.8 long_press_time	68
5.2.9 repeat	69
5.2.10 enable_press_down/press_down_value	69
5.2.11 enable_press_up/press_up_value	69
5.2.12 enable_click/click_value	69
5.2.13 enable_long_press/long_press_value	70
5.3  check_button	70
5.3.1 value:	70
5.3.2 text	70
5.4  radio_button	70
5.4.1 value:	71

5.4.2 text	71
5.5  switch	71
5.5.1 value:	71
5.6  image	72
5.6.1 image	72
5.6.2 draw_type:	72
5.6.3 scale_x/y	73
5.6.4 anchor_x/y:	73
5.6.5 rotation	73
5.6.6 text	74
5.6.7 clickable	74
5.6.8 selectable	74
5.6.9 enable_press_down/press_down_value	74
5.6.10 enable_press_up/press_up_value	74
5.6.11 enable_click/click_value	75
5.7  gif	75
5.7.1 image	75
5.7.2 scale_x/y:	76
5.7.3 anchor_x/y	76
5.7.4 rotation	76
5.7.5 clickable	76
5.7.6 selectable	77
5.8  svg	77
5.8.1 scale_x/y	77
5.8.2 anchor_x/y	77
5.8.3 rotation	78
5.8.4 text:	78
5.8.5 clickable	78
5.8.6 selectable	78
5.9  image value	79
5.9.1 image	79
5.9.2 min/max:	79
5.9.3 value	79
5.9.4 format	79
5.10  image animation	80
5.10.1 auto_play	80
5.10.2 image	80
5.10.3 start_index	80

5.10.4		end_index	81
5.10.5		interval.....	81
5.10.6		format.....	81
5.10.7		delay:.....	81
5.10.8		loop	82
5.10.9		auto_play	82
5.11		video_image.....	82
5.11.1		loop	82
5.11.2		video_name	83
5.11.3		auto_play	83
5.11.4		frame_interval	83
5.11.5		audio_enable	83
5.11.6		audio_type	83
5.12		chart_view.....	84
5.12.1		top_series	84
5.12.2		x_axis	84
5.12.3		y_axis	87
5.12.4		bar_series.....	89
5.12.5		line_series	91
5.12.6		tooltip.....	94
5.13		radar chart.....	95
5.13.1		area_color	96
5.13.2		line_border_color	96
5.13.3		symbol_bg_color.....	96
5.13.4		symbol_border_color	96
5.13.5		line_border_width.....	97
5.13.6		symbol_round_radius	97
5.13.7		min/max:	97
5.13.8		origin_offset	97
5.13.9		point_num	98
5.13.10		data	98
5.14		pie_slice	98
5.14.1		max	98
5.14.2		value.....	98
5.14.3		start_angle	99
5.14.4		inner_radius	99

5.14.5	 uint	99
5.14.6	 counter_clock_wise	99
5.14.7	 show_text.....	99
5.14.8	 is_exploded.....	100
5.14.9	 pressed	100
5.14.10	 is_semicircle	100
5.14.11	 x_to/y_to	100
5.15	 qr.....	100
5.15.1	 value.....	101
5.16	 combo_box_ex	101
5.16.1	 auto_fix:.....	101
5.16.2	 input_type	101
5.16.3	 min/max:	102
5.16.4	 tips:.....	102
5.16.5	 text	102
5.16.6	 options:.....	102
5.16.7	 selected_index:.....	103
5.16.8	 item_height:	103
5.16.9	 action_text:.....	103
5.16.10	 readonly:	103
5.16.11	 password_visible:	104
5.16.12	 open_kb_when_focused/close_kb_when_blured	104
5.17	 spin_box	104
5.17.1	 auto_fix:.....	104
5.17.2	 input_type	104
5.17.3	 min/max:	105
5.17.4	 tips:.....	105
5.17.5	 text	105
5.17.6	 options:.....	106
5.17.7	 selected_index:.....	106
5.17.8	 item_height:	106
5.17.9	 action_text.....	106
5.17.10	 readonly	107
5.17.11	 password_visible	107
5.17.12	 open_kb_when_focused/close_kb_when_blured	107
5.18	 text selector	107
5.18.1	 selected_index.....	107
5.18.2	 visible_num	108

5.18.3	 y_speed_scale:	108
5.18.4	localize_options	108
5.18.5	loop_options	108
5.19	 slide_menu	109
5.19.1	min_scale	109
5.19.2	value	109
5.19.3	align_v	109
5.19.4	x/y_offset	109
5.20	 label	110
5.20.1	length	110
5.20.2	line_wrap	110
5.20.3	word_wrap	110
5.21	 hscroll label	110
5.21.1	loop:	110
5.21.2	yoyo:	111
5.21.3	duration	111
5.21.4	lull	111
5.21.5	text	111
5.22	 edit	112
5.22.1	auto_fix	112
5.22.2	input_type:	112
5.22.3	min/max:	112
5.22.4	tips:	112
5.22.5	text	113
5.22.6	action_text:	113
5.22.7	readonly	113
5.22.8	password_visible	113
5.22.9	close_kb_when_blured	114
5.23	 mledit	114
5.23.1	tips	114
5.23.2	keyboard	114
5.23.3	max_lines	114
5.23.4	wrap_word	115
5.23.5	scroll_line	115
5.23.6	cancelable	115
5.23.7	close_im_when_blured	115
5.23.8	open_im_when_focused	116
5.24	 slider	116

5.24.1		bar_size.....	116
5.24.2		dragger_size	116
5.24.3		slide_with_bar	117
5.24.4		min/max:	117
5.24.5		step	117
5.24.6		value.....	117
5.25		slider_circle	117
5.25.1		min/max:	118
5.25.2		step	118
5.25.3		start/end_angle	118
5.25.4		bg_line_width	118
5.25.5		fg_line_width	118
5.25.6		header_size	119
5.25.7		dragger_size	119
5.25.8		header_color	119
5.25.9		unit	119
5.25.10		line_cap.....	119
5.25.11		counter_clock_wise	120
5.25.12		show_text.....	120
5.26		progress circle	120
5.26.1		start angle	120
5.26.2		line width	120
5.26.3		unit	121
5.26.4		line_cap.....	121
5.26.5		counter_clock_wise	121
5.26.6		show_text.....	121
5.26.7		max	121
5.26.8		value.....	122
5.27		progress bar.....	122
5.27.1		vertical.....	122
5.27.2		show text.....	122
5.27.3		reverse	122
5.27.4		max	122
5.27.5		value.....	123
5.28		gauge.....	123
5.28.1		image:	123
5.28.2		draw_type:	123
5.29		gauge pointer.....	123

5.29.1		image	124
5.29.2		anchor_x/y:	124
5.29.3		angle:	124
5.30		time clock	124
5.30.1		image	124
5.30.2		dial_image	125
5.30.3		hour/minute/second_image:	125
5.30.4		hour/minute/second_anchor_x/y:	125
5.31		digit clock	126
5.31.1		format:	126
5.32		shape	127
5.32.1		line_color	127
5.32.2		line_width	127
5.32.3		line_soild	127
5.32.4		line_dotted	128
5.32.5		fill_color	128
5.33		table_view	128
5.33.1		table_client	128
5.33.2		scroll_bar_d	129
5.34		tab_view	129
5.34.1		pages	129
5.34.2		View	130
5.34.3		tab_button_group	130
5.34.4		tab_button	131
5.35		scroll_view	132
5.35.1		x/yoffset	132
5.35.2		x/yslidable	132
5.35.3		x/yspeed_scale	132
5.35.4		snap_to_page	133
5.35.5		move_to_page	133
5.35.6		recursive	133
5.35.7		focusable	133
5.35.8		x/y_offset	133
5.36		slide_view_view	134
5.36.1		slide_indicator	134
5.36.2		slide_view	136

5.37	 list_view.....	137
5.37.1	list_view.....	137
5.37.2	scroll_view	138
5.37.3	scroll_bar_m	140
5.37.4	list_item	140
5.38	 list_view_h	141
5.38.1	scroll_view	141
5.38.2	list_item	143
6.	Some Composite Widgets Usage	144
6.1	 chart_view.....	144
6.2	 tab_view.....	147
6.3	 scroll_view	148
6.4	 slide_view_view.....	150
6.5	 list_view_h	151
6.6	 list_view	152
6.7	Add subsidiary widgets	153
6.7.1	right click to add	153
6.7.2	drag to add	154
6.7.3	drag and drop to add	155
7.	Network port configuration instructions	157
7.1	Network port configuration	157
7.1.1	Network port configuration conditions	157
7.1.2	Network port configuration use range.....	157
7.2	Network port configuration steps	157
7.3	json field specific meaning	158
7.4	Communication implementation operation description.....	159
7.4.1	HMI's GUI project update	159
7.4.2	Server connection establishment and communication.....	159
8.	MQTT configuration instructions	161
8.1	MQTT introduction and development significance	161
8.2	MQTT configuration on Stone Designer software	162
8.3	Download and start of EMQX	164
8.4	MQTTX software installation and use	165
8.4.1	Software download	165
8.4.2	Software installation.....	165
8.4.3	Software use and device communication	166
8.4.4	Use MQTTX software to communicate with HMI	168
9.	Debug function introduction.....	170

9.1 Debug page.....	170
9.2 Detailed description of each section:	171
9.2.1 COM port Setting	171
9.2.2 Baud Rate Setting.....	171
9.2.3 Connection Status.....	171
9.2.4 Display Ratio	171
9.2.5 Command Input Area.....	172
9.2.6 Data Return Area	173
9.2.7 Debug Window.....	173
9.3 Debug Function Demo Example	174
9.3.1 Device-side effect simulation	174
9.3.2 user_define function	177
9.3.3 Simulation of instruction sending and receiving	182

Modification Record

Version	Date	Last modification	Modified by	Modification summary
v1.0	2022-01-04	2022-01-20		Develop the framework of the first version of the Chinese manual and fill in the content
v1.1	2022-04-18	2022-04-28		Add animations, enhance widgets, and add descriptions of child widgets.
V1.2	2022-7-04	2022-7-14		1. Replace or modify some UI interfaces and corresponding descriptions. 2. Add descriptions and instructions for chart and view widgets.
V1.3	2023-3-23			1. Add the instructions related to the network port 2. Add the instructions for modifying the logo 3. Modify the title of chapter 2 with the project download and boot logo update, and fill in the content.
V1.4	2023-11-2			Added Debug function description and demonstration and added some cases. Add 4.3 custom style description.
V1.5		2023-12-20		1. Add login and registration instructions. 2. Corrected many errors and added some pictures based on actual needs. 3. Modify settings related to settings and improve multiple settings. 4. Add MQTT field description. Added description of dynamic and static IP for network ports.
V1.6	2024-02-08	2024-02-20		1. Modify the settings related to settings. 2. Add the field description of MQTT. Add the description of dynamic and static IP of the network port.
V1.7	2025-01-03	2025-1-14		1. Streamline the common basic property settings of some controls; Optimize the first, fifth, sixth and seventh chapters; 2. Add descriptions of mledit, pie_slice, radar_chart, table_view, slider_circle, shape, and video_image controls. 3. Add descriptions related to switching languages;

1. General Introduction of the Software

The software adopts flat design, reasonable layout, simple and clear interface, and supports one-click switching between light and dark themes.

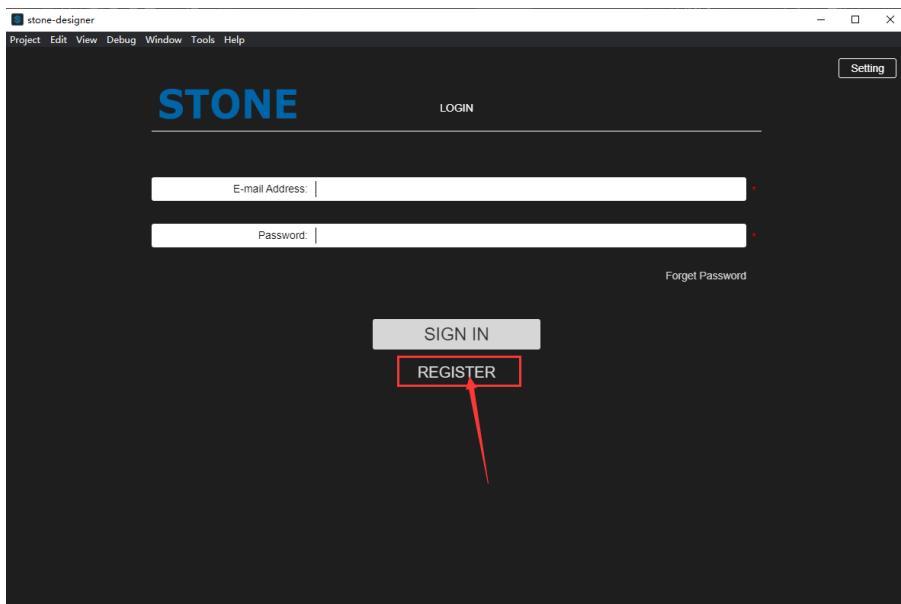
The software interface adapts to the screen size and adapts to a variety of resolution screens.

A variety of functions and different widgets can be used to develop the project in detail after the UI interface design.

1.1 Software registration and login

1.1.1 Software registration and login

Double-click to install the software and the login interface will pop up. When using it for the first time, the user needs to register the software:



Fill in the information as required

stone-designer

Project Edit View Debug Window Tools Help

Setting

STONE REGISTER

E-mail Address: 12322@abc.com

E-mail Verification Code: 123456 Send Code

Country / Region: +61 (Australia)

Phone Number: +61 1234567890

Enter Password:

Re-enter Password:

Company Name: company name1

Industry Category: technical production

SIGN UP

LOGIN

Wait for the system to pop up the Register Success prompt.

stone-designer

Project Edit View Debug Window Tools Help

Setting

STONE REGISTER

E-mail Address:

E-mail Verification Code: 10(s)

Country / Region:

Phone Number:

Enter Password:

Re-enter Password:

Company Name:

Industry Category:

Register Success.

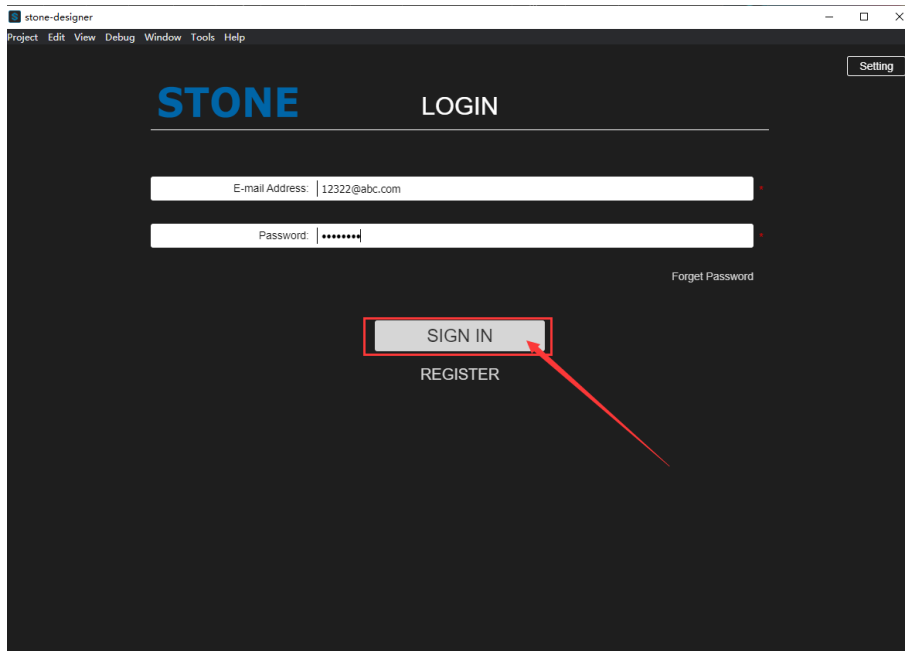
SIGN UP

LOGIN

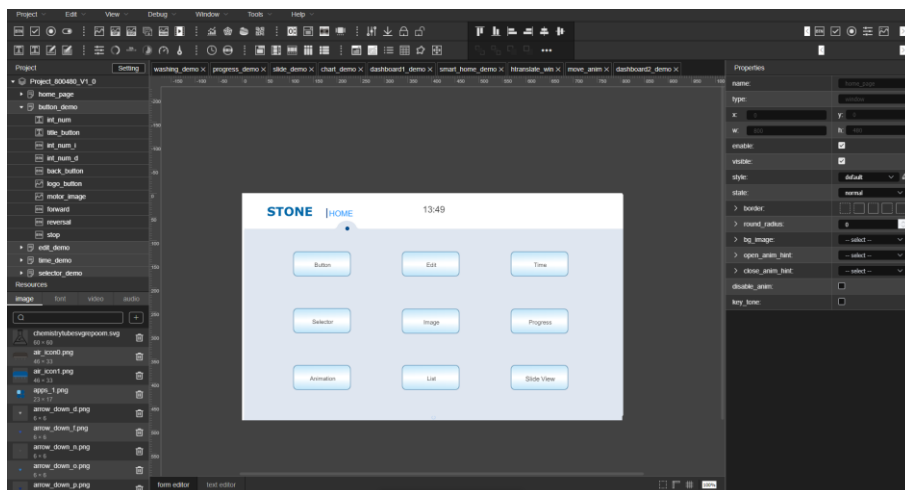
1.1.2 Software login

After registration is completed, click LOGIN and you will be returned to the login page, where you can log in with your registered email address.

Enter the email and password you just registered and click SIGN IN.

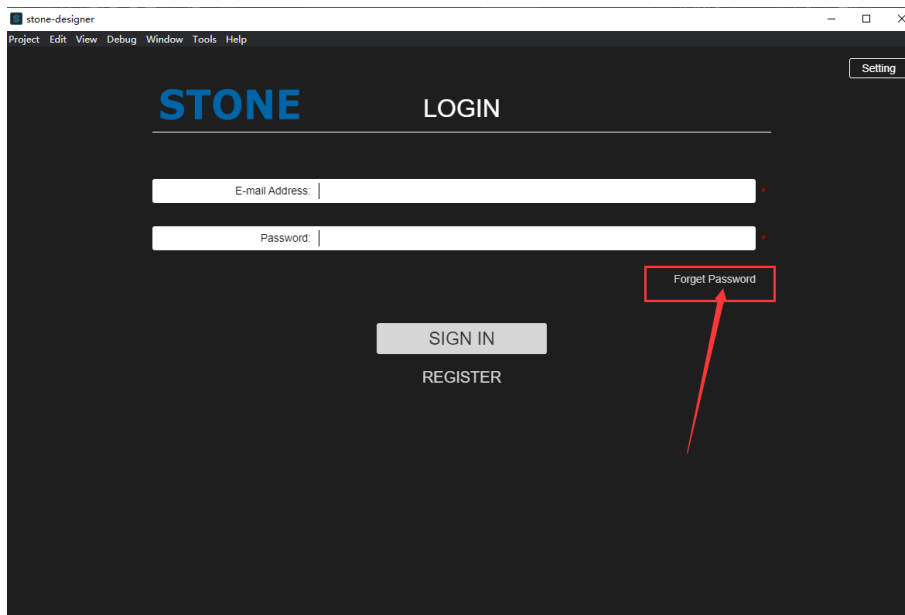


After successful login, the page will automatically switch to the GUI software page, where you can create or modify projects.

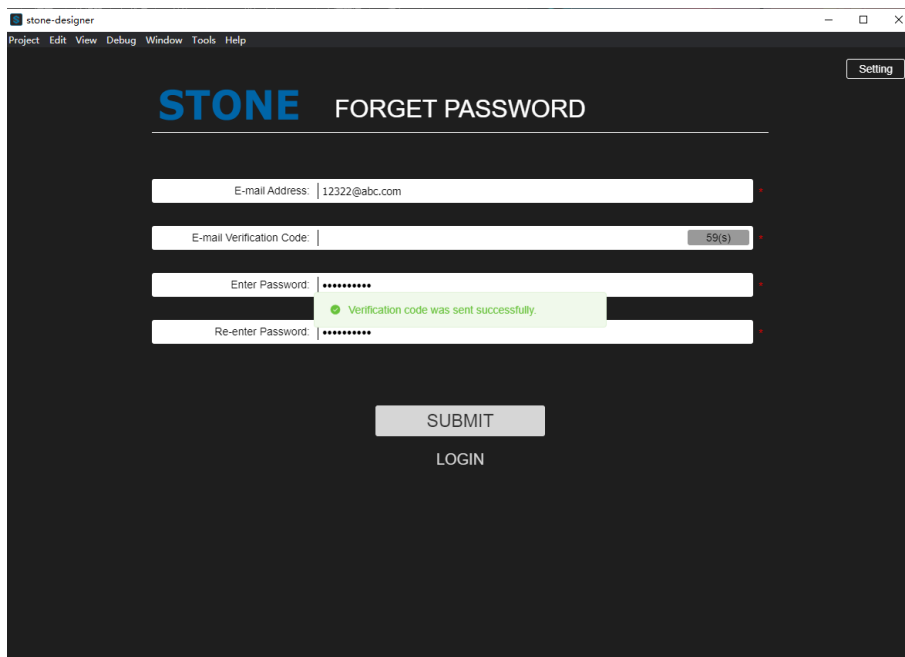


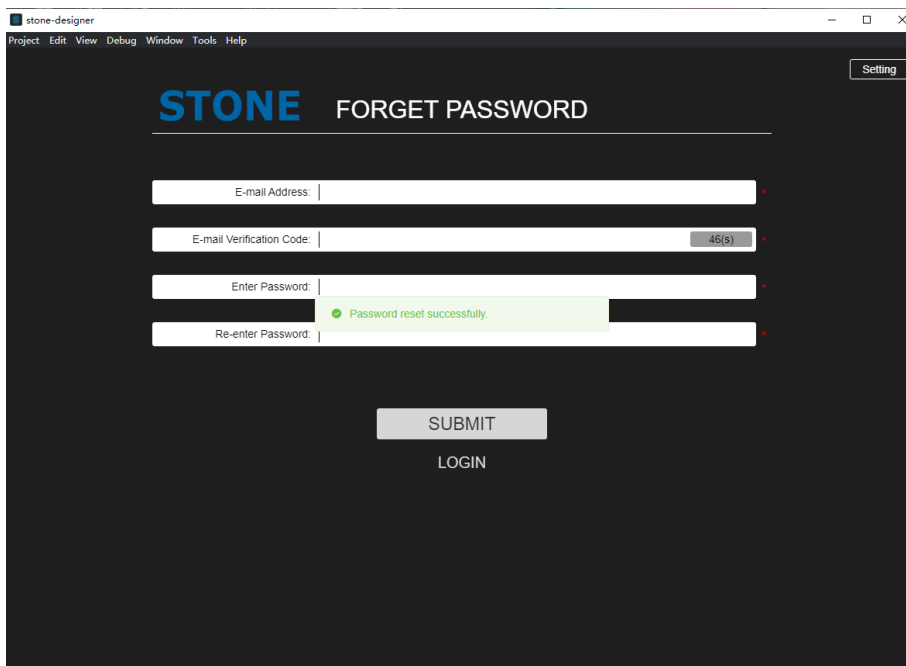
1.1.3 Forgot password

If you need to reset your password, you can reset it by clicking forget password on the homepage.



After entering the page, fill in the email information and new password.

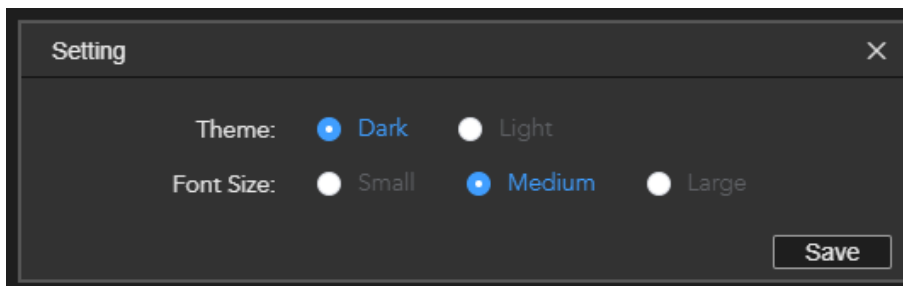




Then log in using your new password.

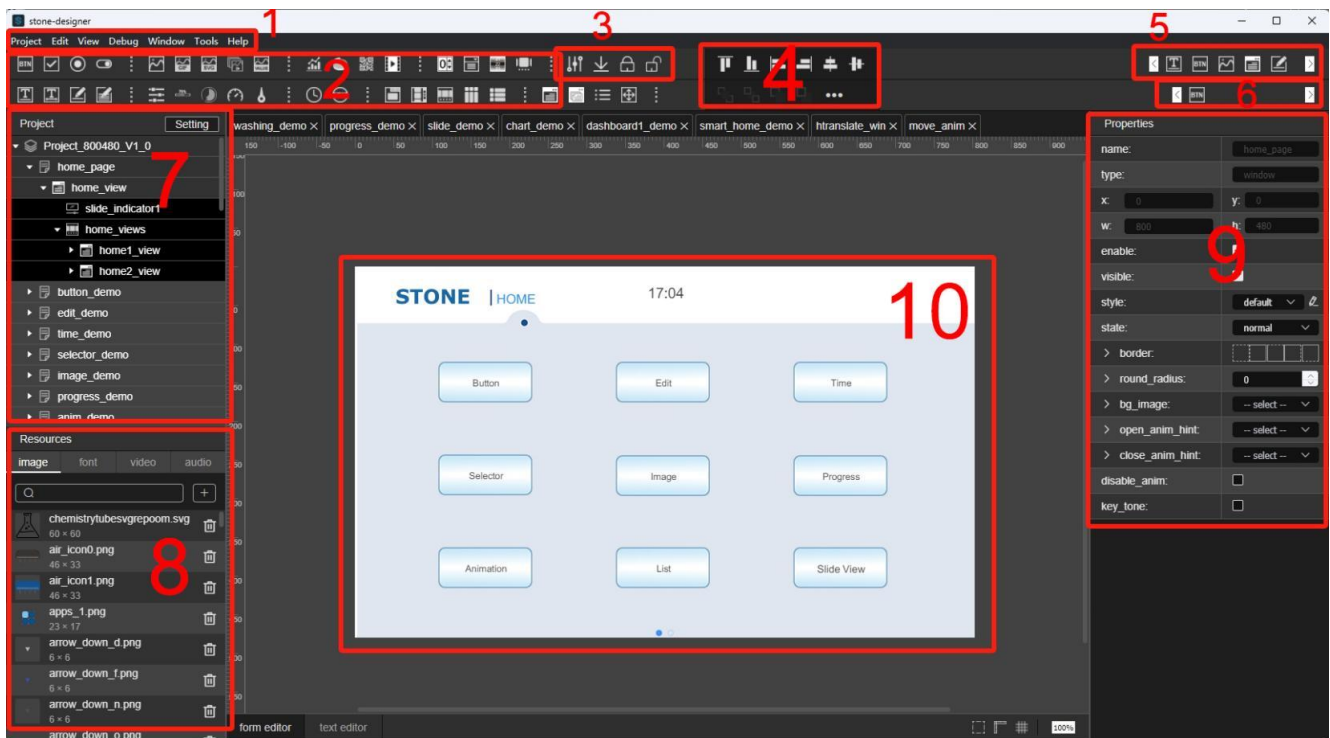
1.1.4 Setting

The Setting page supports switching themes and font sizes.



1.2 Software details introduction

The overall page design is shown in the figure below, and the interior is divided into ten different modules to be introduced one by one.

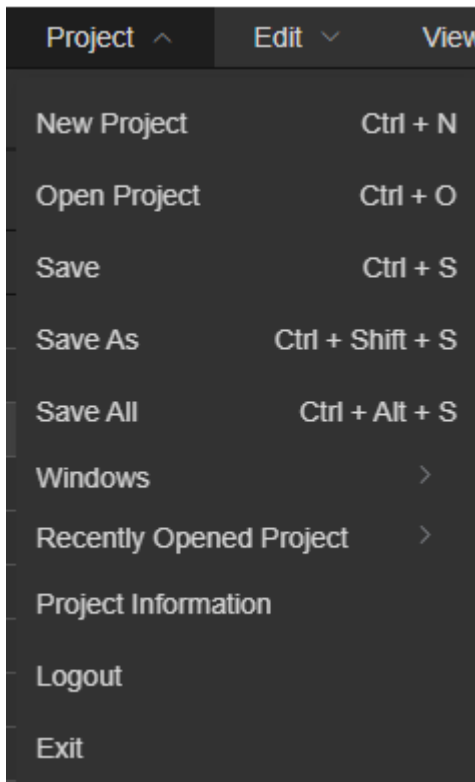


The above marked points are:

1: Title bar area 2: Widget area 3: Debugger area 4: Widget alignment area 5: Common widgets area 6: Customized common widgets area 7: Project area 8: Resources area 9: Properties area 10: Window view area

1.3 Title bar area

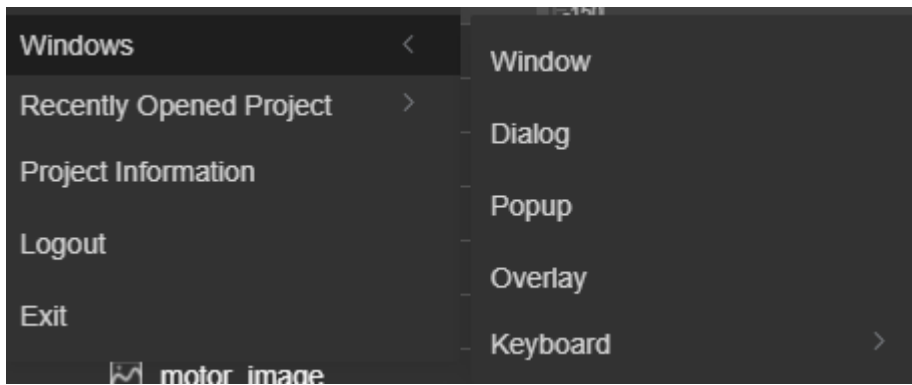
1.3.1 Project



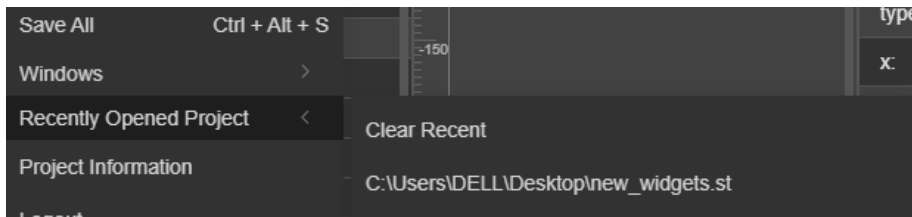
(1) New Project: create a new project, click to pop up the detailed setting information of the new project. Users can set baud rate, change the resolution, brightness of HMI and rotate the screen to 90°, 180° or 270° according to their own needs.



- (2) open project: select the corresponding UI (.st) file to open it.
- (3) save: save the current page.
- (4) save as: Save the file to the specified path.
- (5) save all: save the entire project.
- (6) windows: form function. After clicking, different forms can be selected to create, such as pop-up boxes, keyboards, etc.



(7) recently opened project: display recently opened projects, click to open.

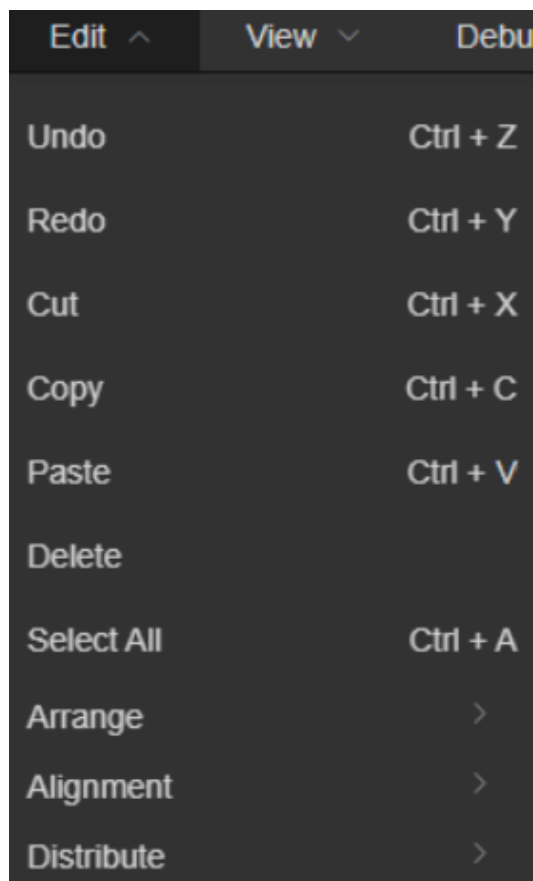


(8) project information: click to view information about the currently open project.

Information		×
Project Name:	Project_800480_V1_0	
Project Path:	C:\Users\DELL\Desktop\new_widgets.st	
Display Size:	800×480	
Baud rate:	115200	
Brightness:	100	
LCD rotation:	0	
enable widget buzzer:	Disable	
enable boot buzzer:	Disable	
enable touch screen:	Enable	
LAN:	Disabled	
MQTT server:	Disabled	
Screen saver enable:	Disabled	

(9) exit: Exit Software

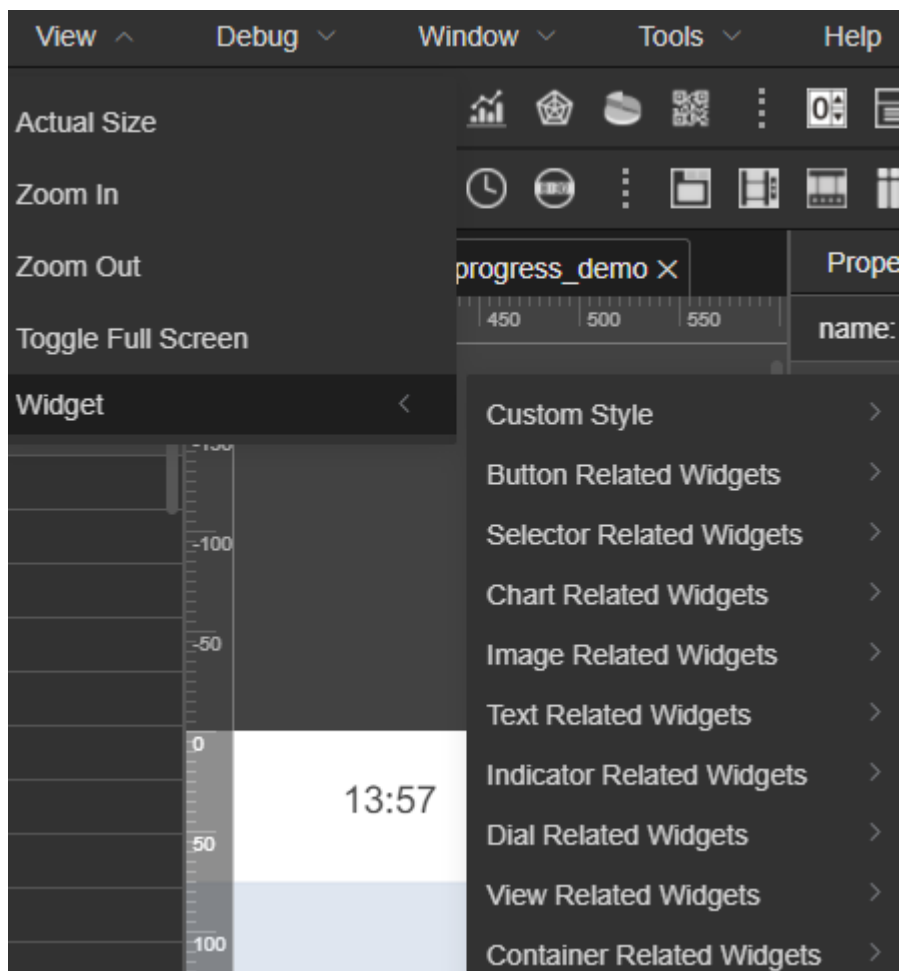
1.3.2 Edit



The corresponding shortcut keys can be used.

- (1) Undo: Click to undo the previous operation.
- (2) Redo: Click to resume the operation.
- (3) Cut: After selecting the widget, click cut to cut the widget, which needs to be used with paste.
- (4) Copy: After selecting the widgets, click copy to copy the widgets, which needs to be used with paste.
- (5) Paste: Paste widgets need to be used in conjunction with cut and copy.
- (6) Delete: Delete widgets.
- (7) Select all: Selects all widgets on the same page.
- (8) Arrange: Set the hierarchical relationship of widgets.
- (9) Alignment: set the alignment of multiple widgets. See chapter 1.4 for details..
- (10) Distribute: set the layout of multiple widgets.

1.3.3 View

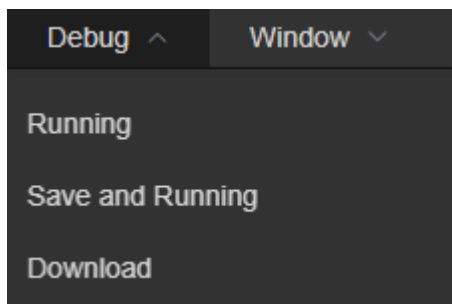


- (1) actual size: after clicking, the window will restore the original actual size.
- (2) zoom in/out: zoom in and out of the current window view size.
- (3) toggle full screen: display the item in full screen.
- (4) widget: All of the available widgets.

P.S. there is a ✓ after each widget, and the widget will be displayed on the upper computer only when it is checked.

If the button is not checked, its icon will not be displayed in the widget bar.

1.3.4 Debug

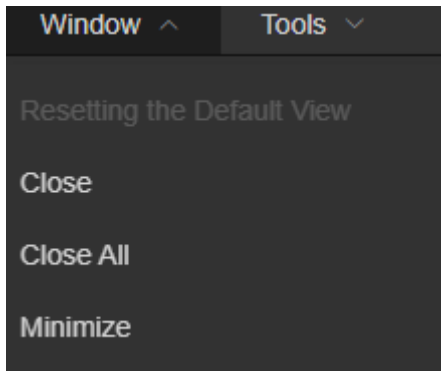


- (1) Running: simulate running the current project.

P.S. different from save and running. When it is not saved, the interface of the project when it was last saved is run.

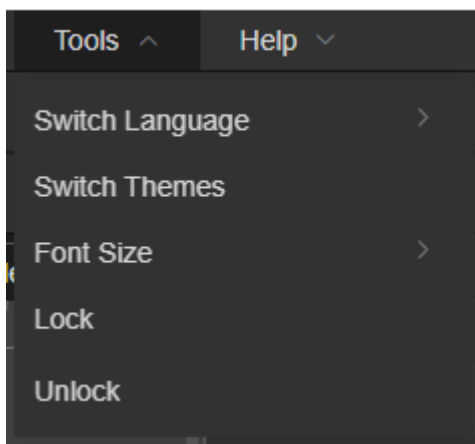
- (2) Save and Running: save and simulate running the current project.
- (3) Download: download the current project.

1.3.5 Window

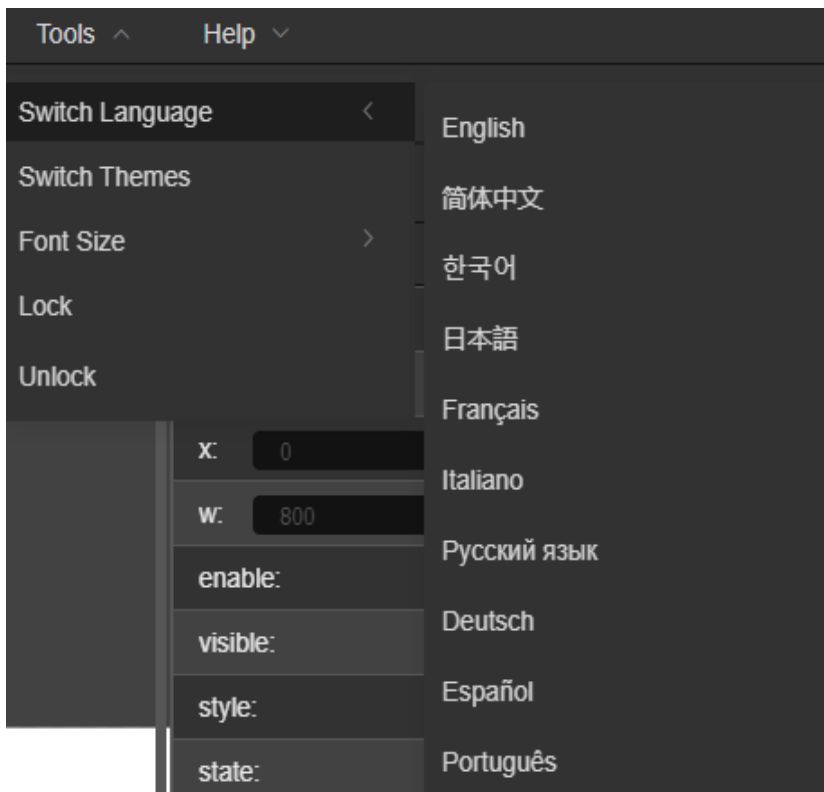


- (1) Resetting the Default View: reset default view.
- (2) Close: close the currently selected window page.
- (3) Close All: close all open windows page.
- (4) Minimize: minimize the software.

1.3.6 Tools



- (1) Switch Language:



(2) Switch Themes: Supports switching between dark theme and light theme.

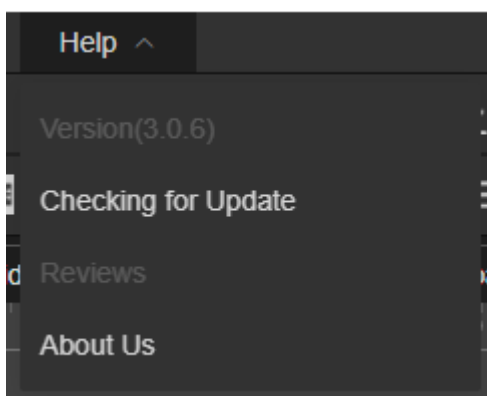
(3) Font Size : Modify the overall font size of the software.

(4) Lock: Lock the currently selected widget, support multiple selection.

(5) Unlock: Unlocks the currently selected widget, supports multiple selection.

PS. The firmware upgrade option will be opened later. When the connection between the PC and the HMI is normal, click to upgrade the firmware.

1.3.7 Help



(1) Version: Current version number.

(2) Checking for update: Check for updates.

(3) Reviews: Check what's updated.

(4) About us: Contact information, etc.

1.4 Widgets

Button Related Widgets: button, check_button, radio_button, switch;

Selector Related Widgets: spin_box, combo_box_ex, text_selector, slide_menu;

Chart Related Widgets: chart_view, radar_chart, pie_slice, qr;

Image Related Widgets: image, gif, svg, image_value, image_animation, video_image;

Text Related Widgets: label, hscroll_label, edit, mledit;

Indicator Related Widgets: slider, slider_circle, progress_bar, progress_circle, gauge, gauge_pointer;

Dial Related Widgets: digit_clock, time_clock;

View Related Widgets: tab_view, scroll_view, slide_view, list_view, list_view_h, table_view, shape;

Container Related Widgets: pages, view, list_item, dragger;

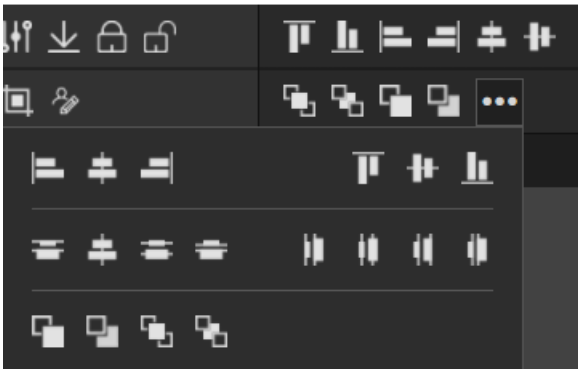
For detailed description of the properties, please refer to Chapter 6 and Chapter 7.

1.5 Debug Tool

For details about the Debug tool, please refer to [Chapter 11 Debug Function Introduction](#).

1.6 Widget alignment area

One-click alignment for multiple widgets.



When multiple widgets are selected, the hierarchical relationship related items will be grayed out and only the alignment can be set.



1.  Top alignment  Bottom alignment  Left alignment  Right alignment .
2.  Vertical top equally divided  Horizontal center alignment  Vertical center alignment
-  Vertical center equally divided  Vertical equally divided
3.  Horizontal left equally divided  Horizontal center equally divided  Horizontal right
-  Horizontal left equally divided  Horizontal equally divided
4.  Bring forward  Send backward  Bring to front  Send to back

1.7 Common widgets area

The top ten widgets are arranged according to the user's usage frequency, which is convenient for users to drag and drop quickly.

It will be refreshed every time the project is opened.



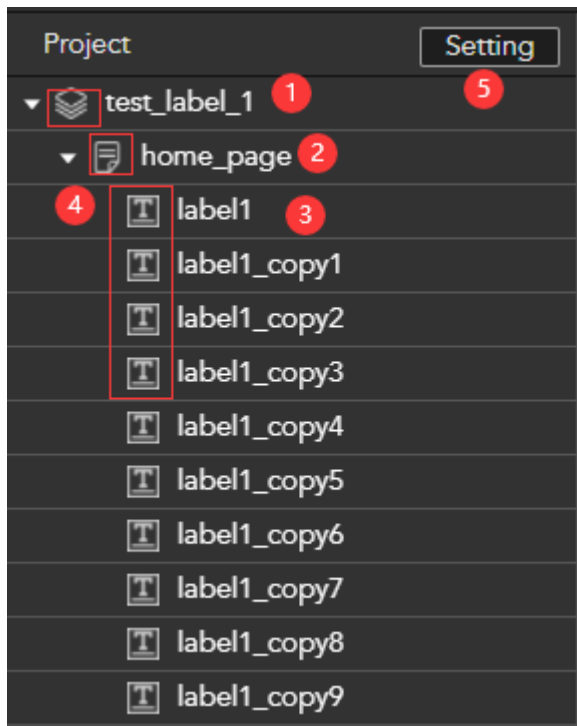
1.8 Customized common widgets area

Users can customize and drag the commonly used widgets they need here.

Double-click the widget to delete it.



1.9 Project area



1. Project name; 2.Window name; 3.Widget name; 4.Icon; 5.Setting.

1.9.1 Project name

User-defined name when creating a new project, which can be changed later.

P. S. Project name and file name are two different concepts.

1.9.2 Window name

The name of the window in the UI project, the name can be customized, but there must be a unique `home_page`.

For the convenience of sending and receiving instructions and other issues, avoid using invalid characters in the Json code to name widgets.

1.9.3 Widget name

The name of the widget in the project. The name can be customized.

For the convenience of sending and receiving instructions and other issues, avoid using invalid characters in the Json code to name widgets.

1.9.4 Icon

Different widgets have different corresponding icons and cannot be changed.

1.9.5 Setting

Setting can modify all attributes except path in new project. And you can set LAN, MQTT, and screen saver functions.

After modification, some properties will be automatically synchronized with HMI's *app._conf.json* file. You can find this file in HMI's flash memory.

Setting	
Project Name:	Project_800480_V1_0
Project Path:	C:\Users\DELL\Desktop\新demo合集
Display Size:	800x480
Baud Rate:	115200
Brightness:	100
LCD Rotation:	0
Enable widget buzzer:	☐
Enable boot buzzer:	☐
Enable touch screen:	☒
LAN	Disabled
MQTT Server	☐
Screen Server	
Screen saver enable:	☐
Save	

For some basic settings, please refer to the description in 1.2.1project.

For the rest of the settings, please refer to the following instructions:

Enable widget buzzer: Whether to allow the buzzer function (key_tone) of the UI to be turned on. The buzzer settings here have priority over the buzzer settings in widget properties. So if you set key_tone for the widget in a UI project but turn buzzer off in the Setting page, then it will not beep.

Enable boot buzzer: Whether there will be a buzzer after booting. This is proof of whether the boot is complete, and it is recommended to keep it on during testing.

Enable touch screen: Whether to enable touch mode. When unchecked, the HMI cannot perform touch operations.

LAN: For instructions on network ports, please refer to [10.1 Network Port Configuration Instructions](#).

LAN	Static IP
IP addr:	xxx.xxx.xxx.xxx
Net mask:	xxx.xxx.xxx.xxx
Broad cast:	xxx.xxx.xxx.xxx
Gate way:	xxx.xxx.xxx.xxx
DNS addr:	xxx.xxx.xxx.xxx
MAC addr:	xx:xx:xx:xx:xx:xx

MQTT: For MQTT configuration instructions, please refer to [10.5 MQTT Configuration Instructions](#).

MQTT Server	☒
Server addr:	xxx.xxx.xxx.xxx
Server port:	
Client id:	
User name:	
Password:	
Subscribe topic:	
Publish topic:	
Will topic:	
Will message:	

Screen Server: After checking, you can choose to turn on the screen saver of the HMI, and you can choose the screen saver page, screen brightness, etc.

Screen Server	
Screen saver enable:	☒
Screen saver targeting page:	edit_demo
Screen saver activation time:	0
Screen saver max brightness:	0
Screen saver min brightness:	0

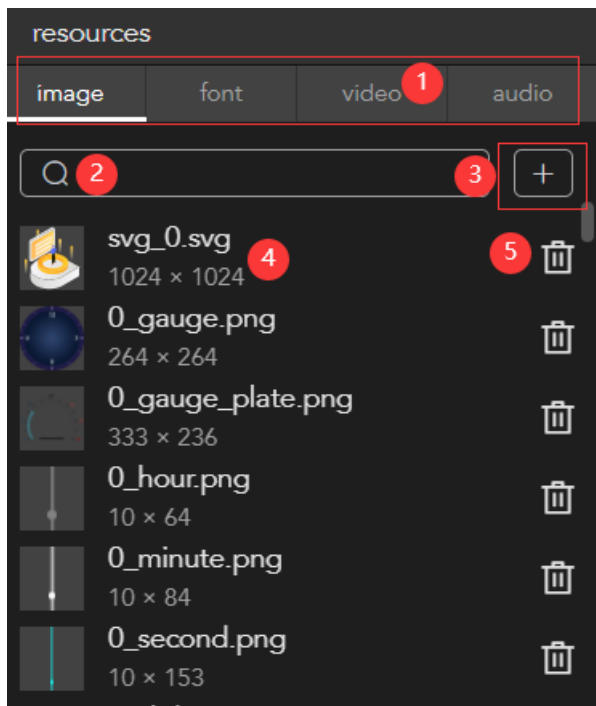
Screen saver targeting page: You can select one page from UI project as screen saver display page.

Screen saver activation time: The starting time of the screen saver page system. The unit is seconds.

Screen saver max brightness: After exiting the screensaver (clicking HMI screen or sending a command, etc.), the system will maintain the brightness. The unit is percentage.

Screen saver min brightness: The brightness when entering the screen saver, It should be greater than or equal to the value set by Screen saver max brightness. The unit is percentage.

1.10 Resources area



The resources bar is mainly used to store various resources. Resources must be put into resources before use it.

It is recommended to name resources using a mixture of letters and numbers.

The above correspond to: 1. Resources category; 2. Fuzzy Search; 3. Add resources; 4. Resource Information; 5. Delete resources.

P.S.

1. Resources with the same name and format will be replaced.
2. The longest resources file is limited to 31 bytes, excluding the suffix name.

1.10.1 Resources category

There are four different categories: image / font / video / audio

image supported file types: (*.png;*.jpg;*.jpeg;*.gif;*.svg;*.bmp)

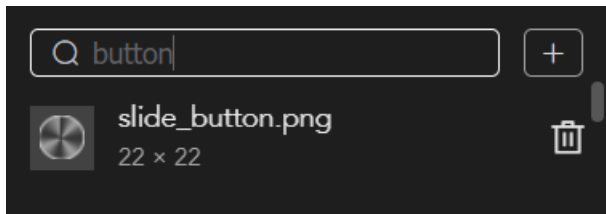
font supported file types: (*.ttf)

video supported file types: (*.mp4)

audio supported file types: (*.wav;*.mp3)

1.10.2 Fuzzy search

Resources support fuzzy search, perform fuzzy search by entering characters in the search box.

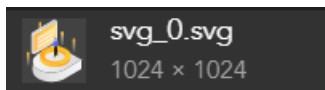


1.10.3 Add resources

Under different categories, resources in corresponding formats can be added.

Supported file types are listed in [1.10.1 Resources category](#).

1.10.4 Resource information



Resources added to resources will display thumbnail and actual size (image only), resource name.

1.10.5 Delete resources

Click delete to delete this resource in the resources library.

There will be no confirmation prompt, click delete to delete directly. Please check carefully the resources that need to be deleted.

1.11 Properties area

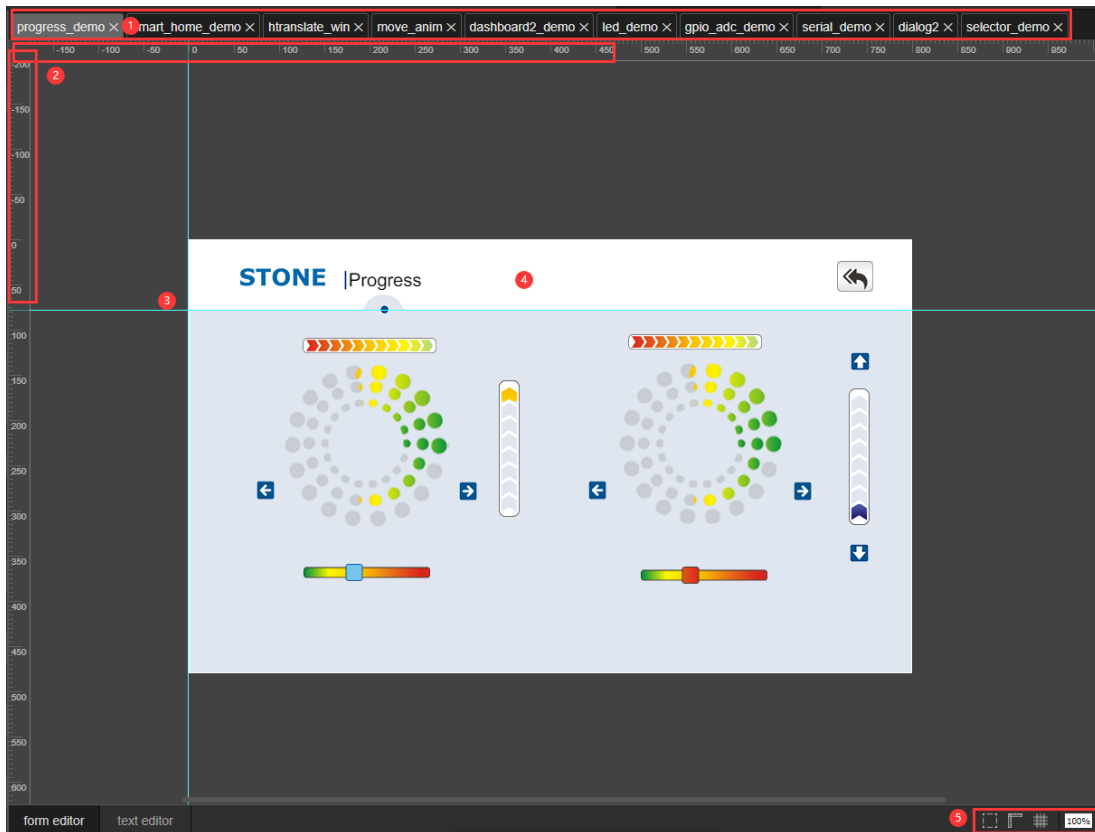
properties

name:	home_page
type:	window
x:	0
y:	0
w:	800
h:	480
enable:	☒
visible:	☒
style:	
state:	normal
border:	
border_width:	1
border_color:	
round_radius:	0
top_left:	0
top_right:	0
bottom_left:	0
bottom_right:	0
bg_image:	bg
bg_color:	
bg_image_draw_type:	请选择
key_tone:	☐

Properties area correspond to various properties of the widget, see [4. Introduction of Properties](#)

1.12 Window view area

The window view is shown below.

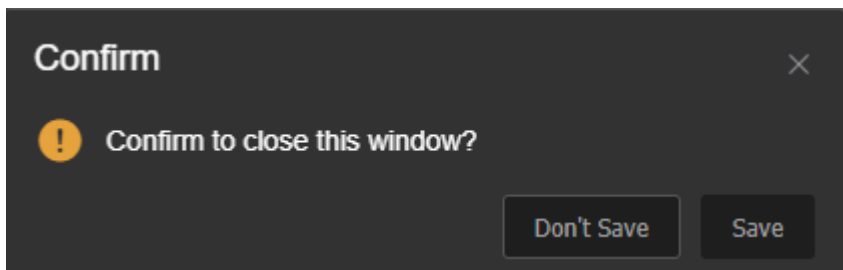
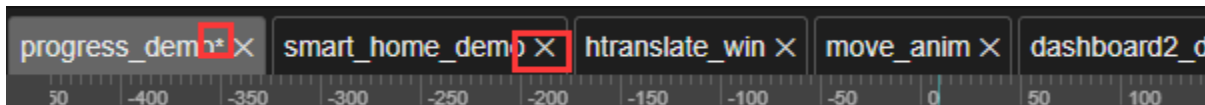


1. Opened Windows; 2. Ruler; 3. Grid lines; 4. Window page; 5. Tool bar.

1.12.1 Opened window

The windows recently opened in this UI project, an * in the upper right corner indicates pages with unsaved items.

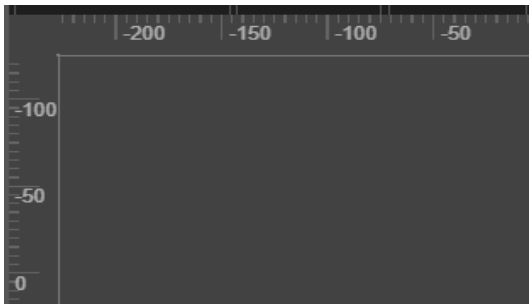
User can click the x to close that window.



1.12.2 Ruler

The ruler of the overall project allows users to view the current coordinates more clearly.

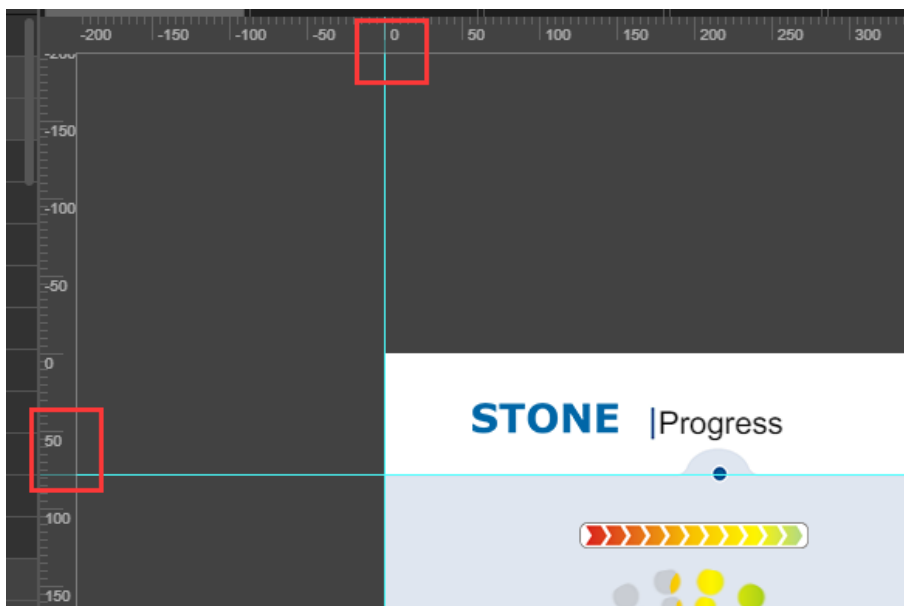
The ruler icon in the lower right corner can be clicked to switch to not display the ruler.



1.12.3 Gridlines

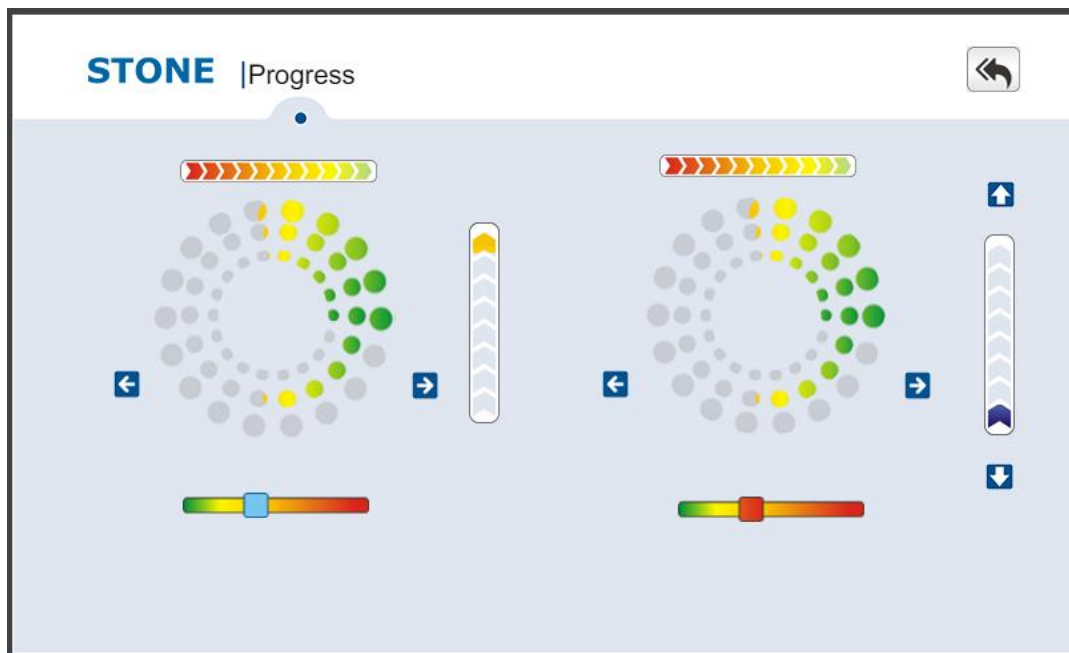
It can help users to locate coordinate points.

Click the grid lines icon in the lower right corner to switch to not display the grid lines.

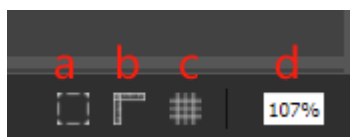


1.12.4 Window view

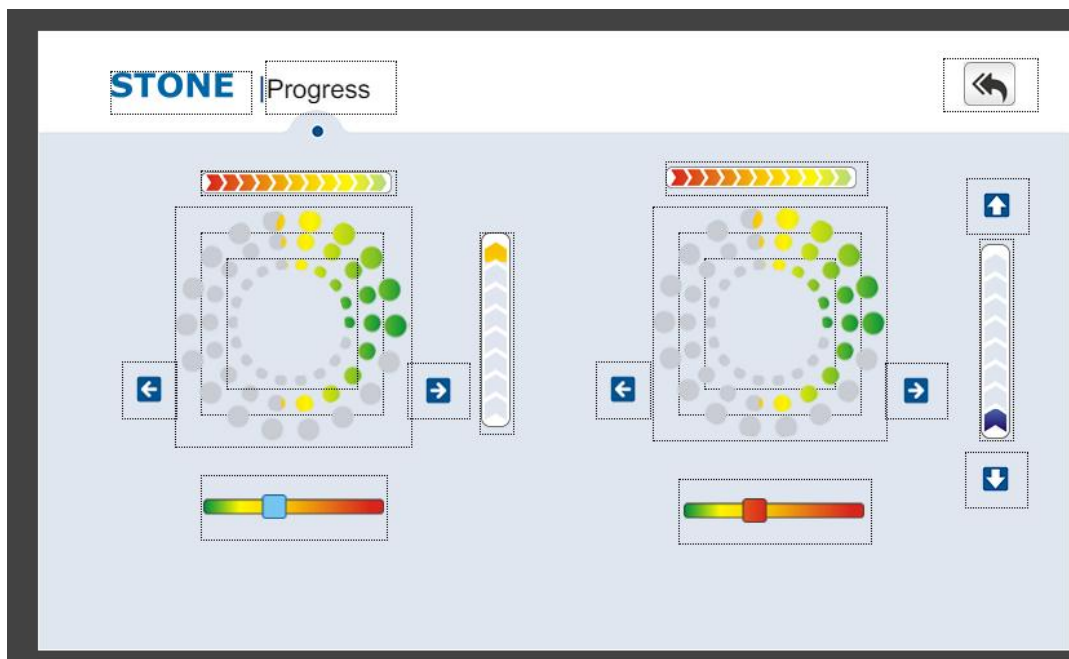
The window page is shown here. Users can design UI projects in this area.



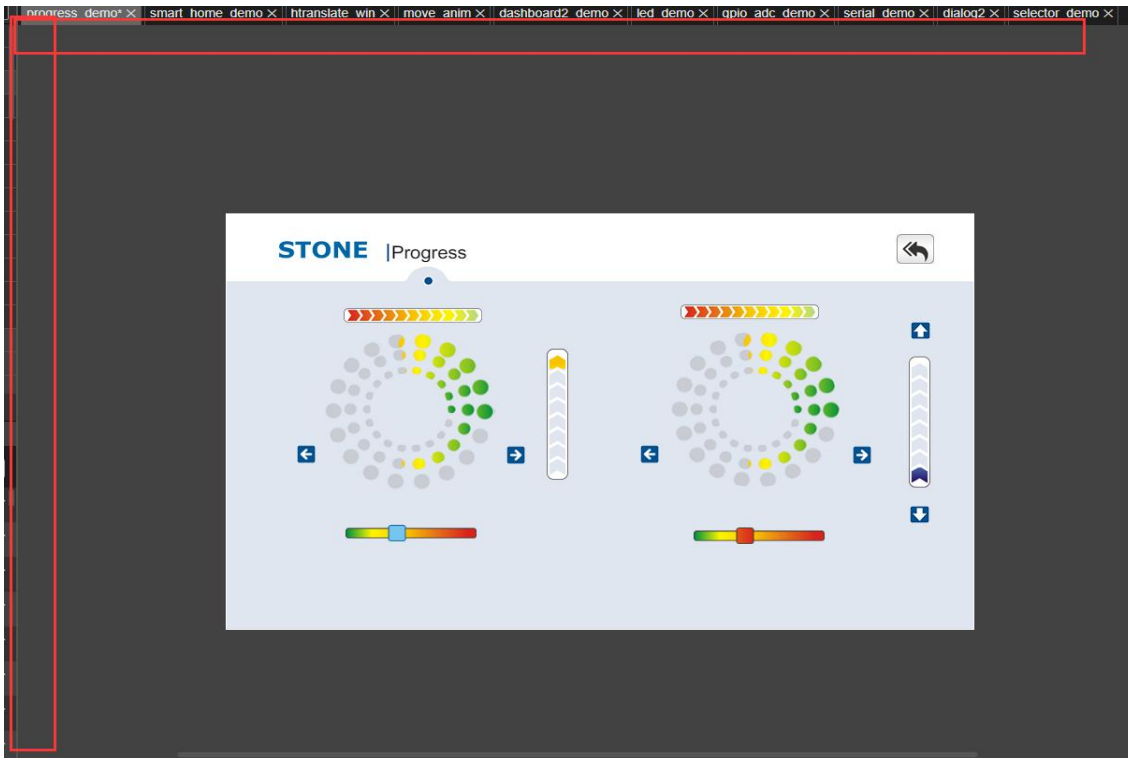
1.12.5 Tool bar



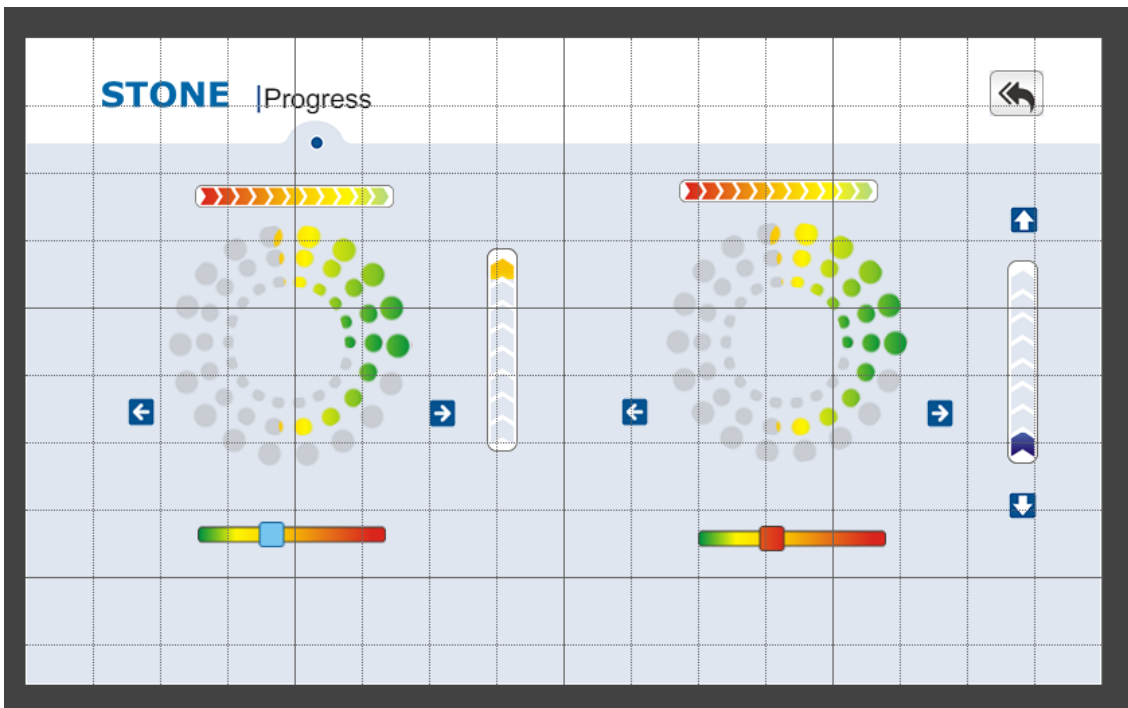
a: Show border lines for all widgets. After clicking, all the widgets can be circled, which is convenient for users to find.



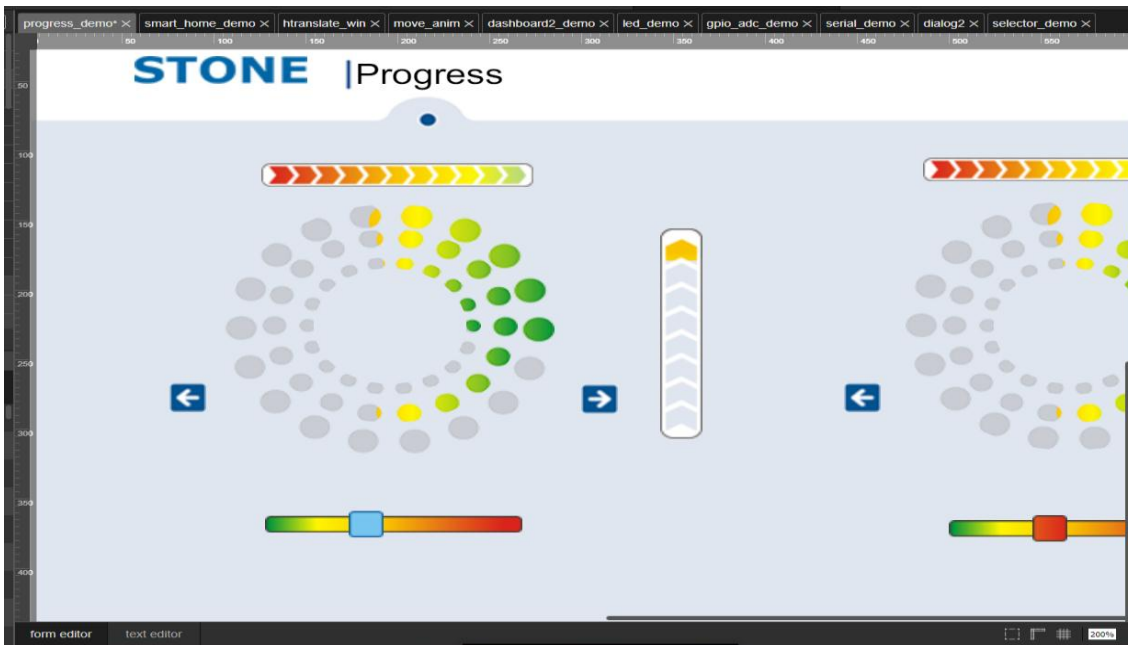
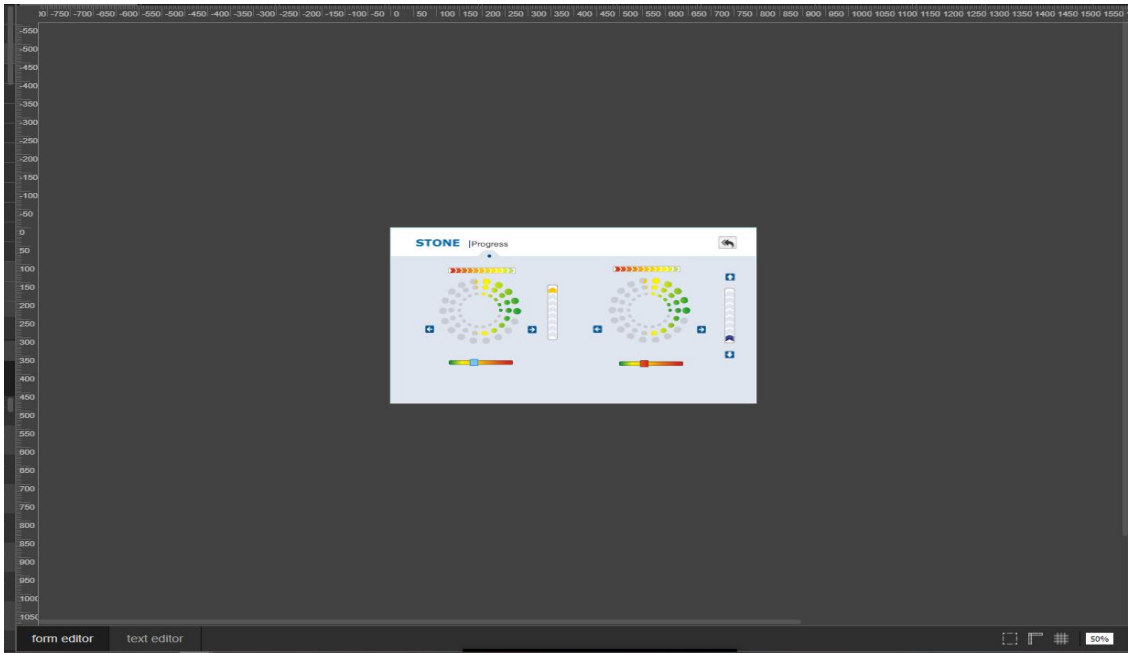
b: Ruler. After clicking, you can choose whether to display the ruler in the window bar.



c: Gridlines: After clicking, you can choose whether to display the gridlines in the window bar.



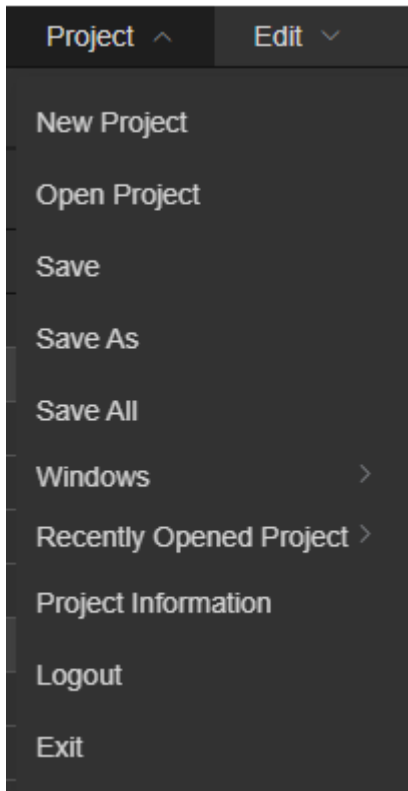
d: Window page Zoom ratio. Display the thumbnail size of the current view. 100% is the original size.



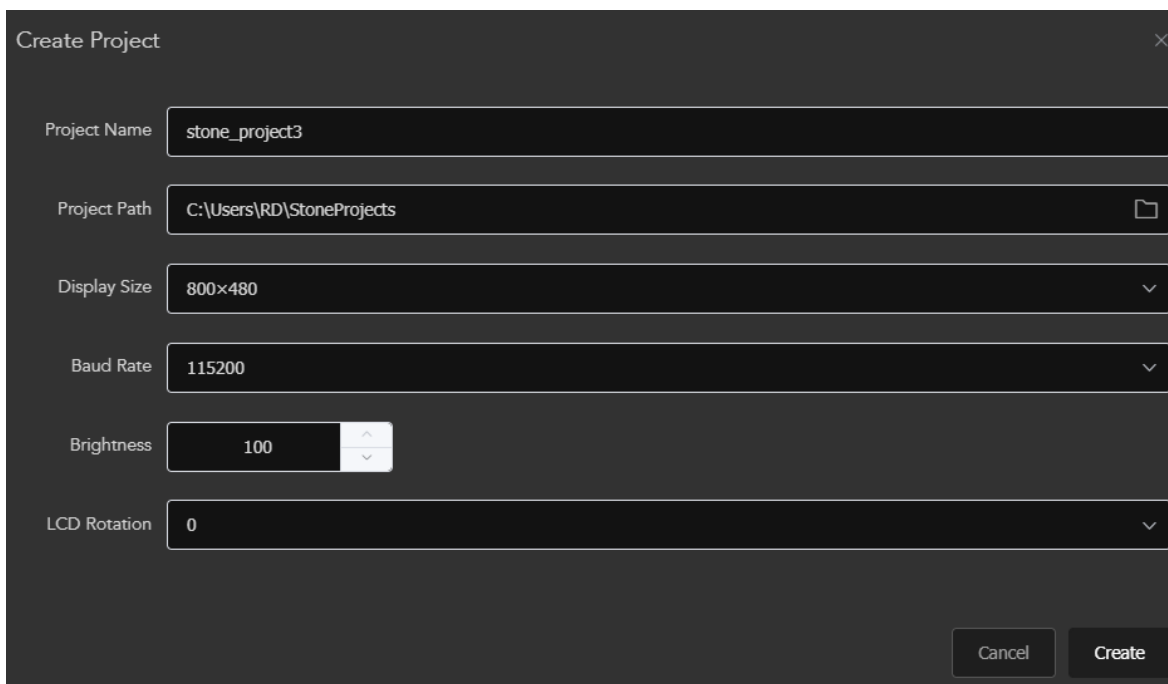
2.Project Download and Boot Logo Update

2.1 Project creation

(1) Open the GUI software, right-click the new project in the project or the shortcut key ctrl+n.

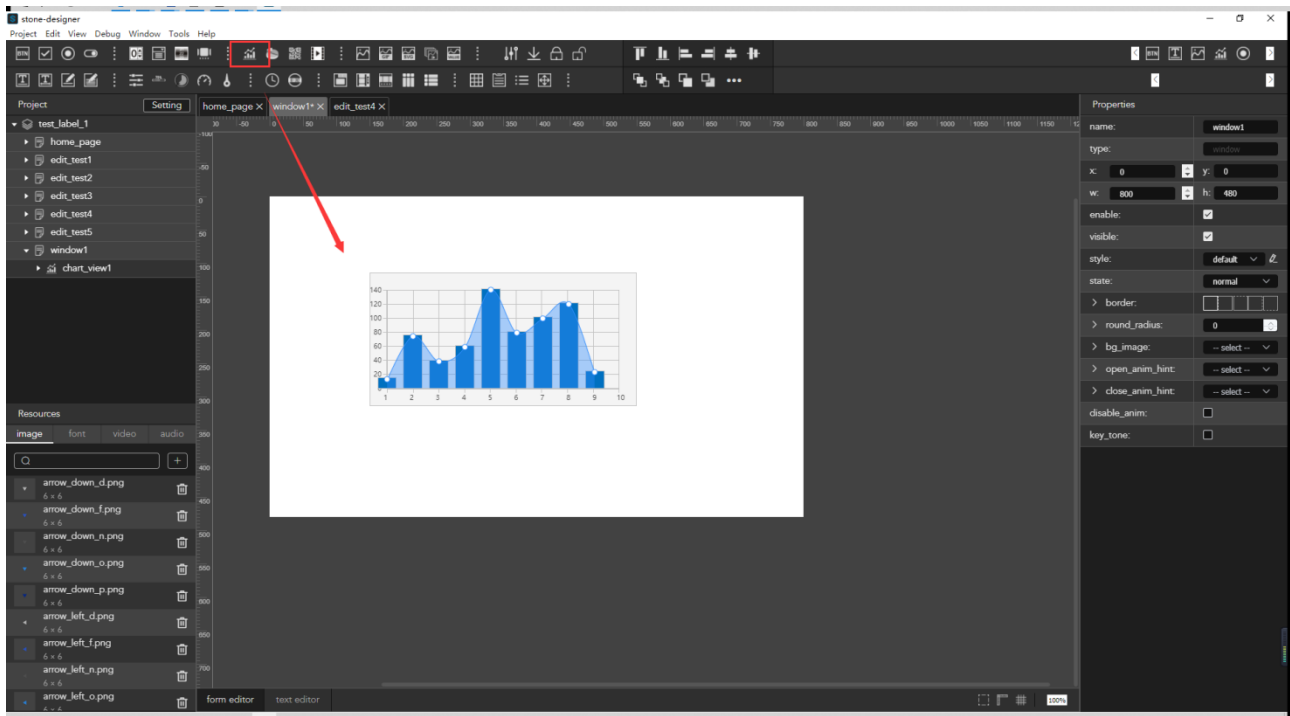


Modify the project information as needed:

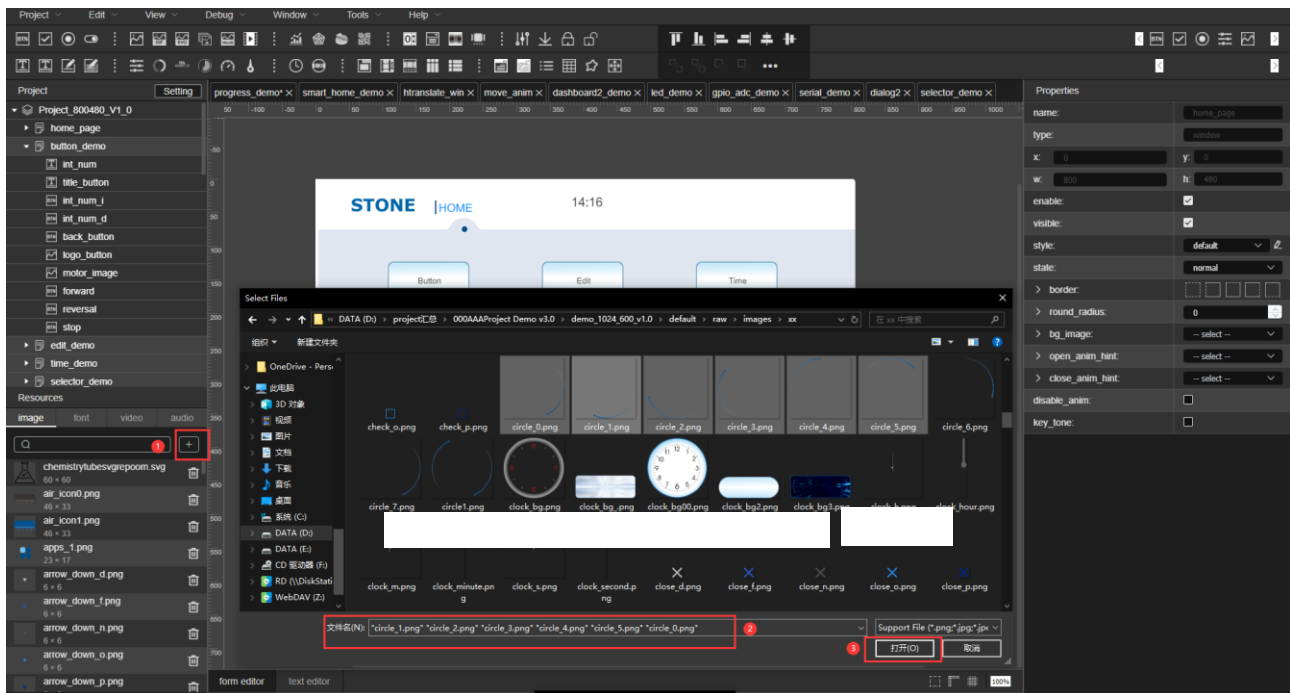
A screenshot of a 'Create Project' dialog box. It contains the following fields: 'Project Name' with the text 'stone_project3'; 'Project Path' with the text 'C:\Users\RD\StoneProjects' and a folder icon; 'Display Size' with a dropdown menu showing '800x480'; 'Baud Rate' with a dropdown menu showing '115200'; 'Brightness' with a slider set to '100'; and 'LCD Rotation' with a dropdown menu showing '0'. At the bottom right are 'Cancel' and 'Create' buttons.

After setting, click create to create the project.

(2) According to the user's needs, drag the different widgets to the window pages to design the interface.



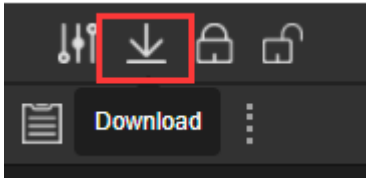
At the same time, add resources according to your own needs. Taking image as an example:



(3) After editing, save the project. The system will generate a **.st file** under the set Project Path. This file is the project file.

2.2 Project download

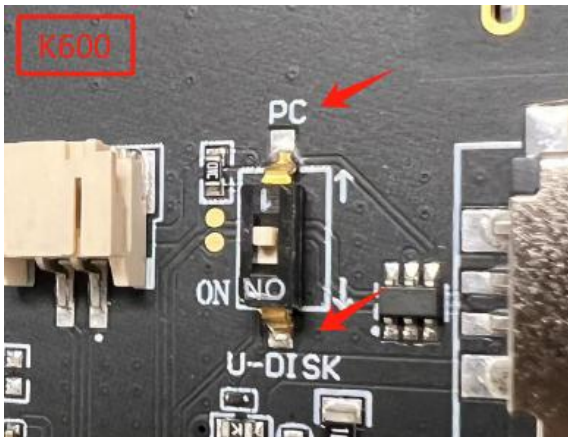
When the project is completed, there is two ways for users to download designed UI projects to HMI:



(1) **Type A double USB cable:** Directly connect the HMI with PC through a Type A double USB cable, then click download button on GUI design software and download UI project directly to HMI 's flash memory's root directory. Please refer [2.2.1 Update project by Type A cable](#)

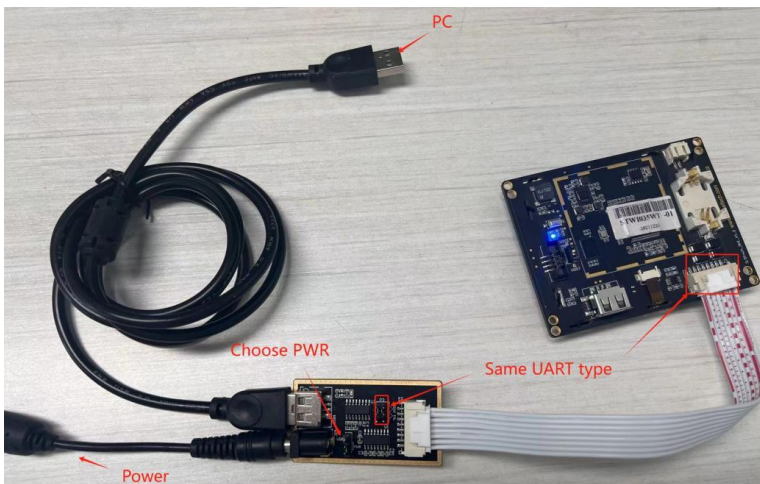
(2) **USB flash drive:** Download UI files to a specific path on U disk and plug into HMI. Please refer [2.2.2 Update project via U-DISK](#)

When selecting a specific update method, please ensure that the K600 on the HMI PCB is also set to the corresponding mode. The old version's "DEVICE" corresponds to the new version's "PC", and the old version's "HOST" corresponds to the new version's "U-disk":



In additional:

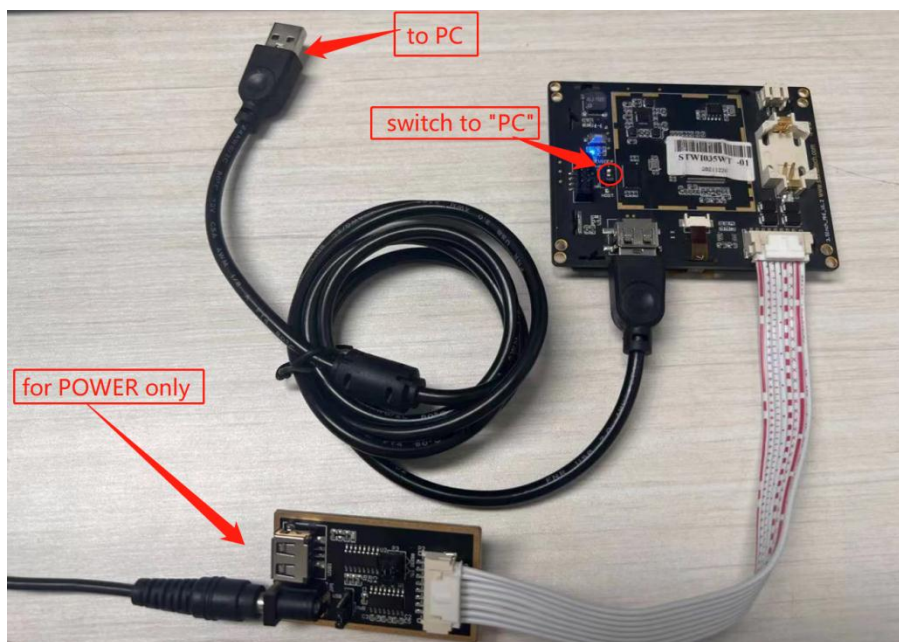
(1) The Type A cable can be used to do the command test with serial port tool on the PC.



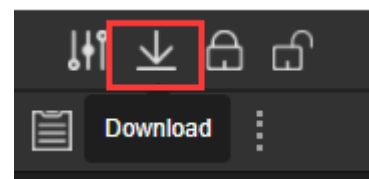
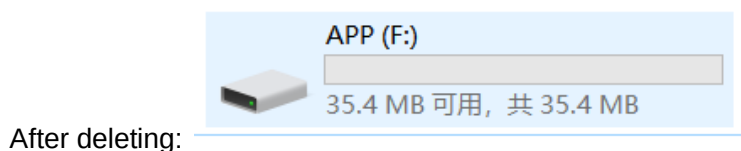
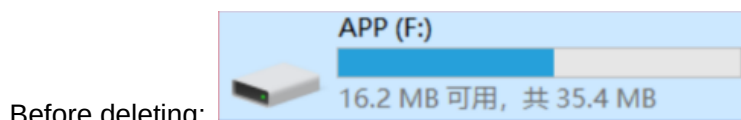
(2) The USB flash drive can be used to update the boot logo and firmware.

2.2.1 Update project by Type A cable

(1) After editing, save the project. Now you can switch HMI's K600 as “**PC**” mode, then connect the HMI with PC by a type A double USB cable



***The HMI flash memory need to be deleted or formatted before downloading.**

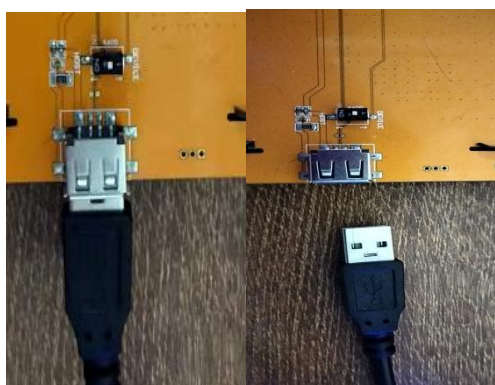


Click download on GUI software and select the flash memory of the U disk:

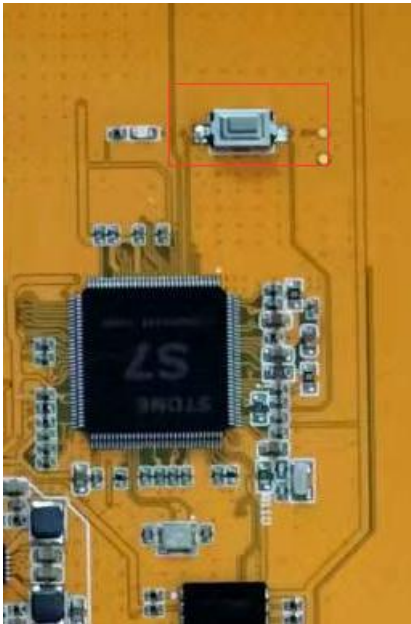
(2) There will be a prompt message after the download is successful. And the UI will be stored in the flash memory of HMI in the following path.



(3) After Download Successfully, safely remove HMI device from PC and unplug the double USB cable.



(4) Click the reset button of the HMI to restart.

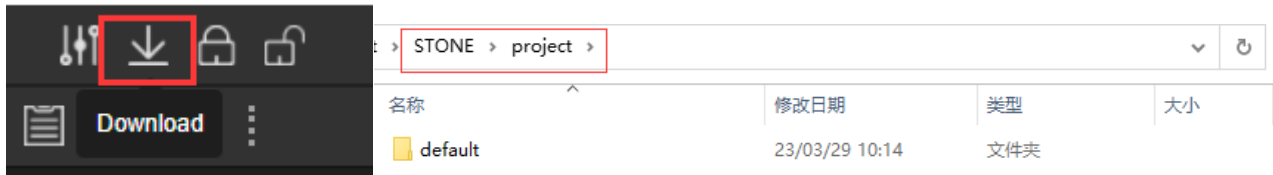


The HMI is restarted, and the project is successfully burned to the HMI.

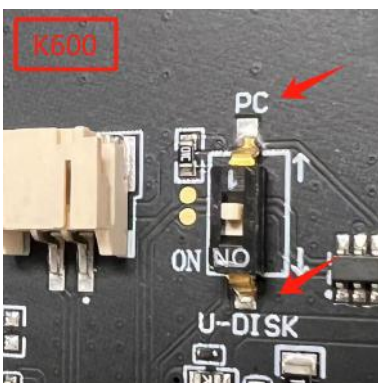
P. S. The user can use U disk to update the UI project but for changing boot logo is recommended to use the U-disk update method, the method reference through the [boot logo update](#).

2.2.2 Update project via U-DISK

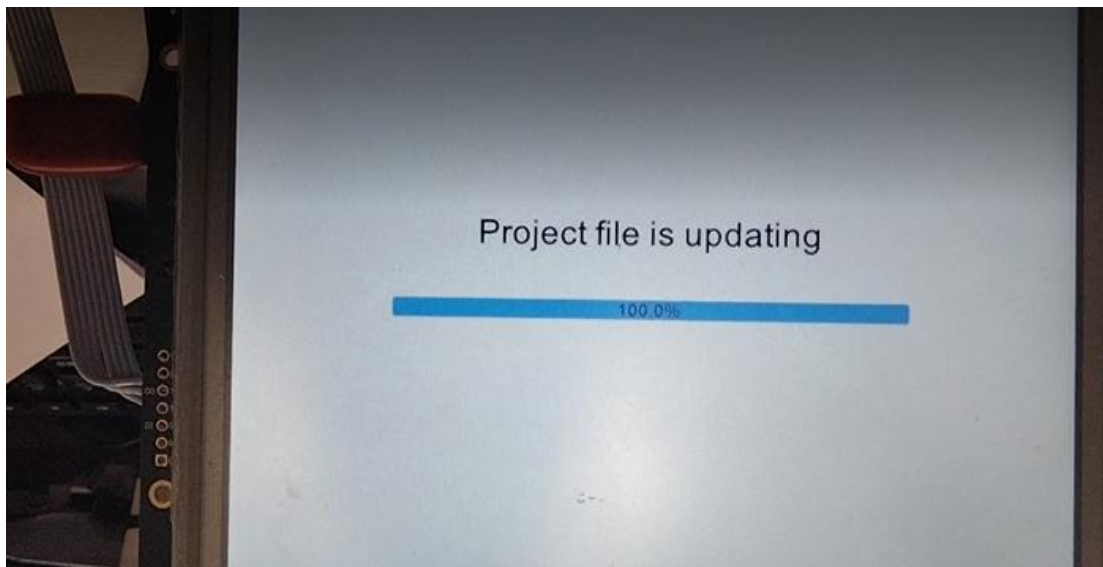
1. Create \STONE\project on a **formatted** U disk. Then click the Download on GUI software and download the completed UI project under \STONE\project.



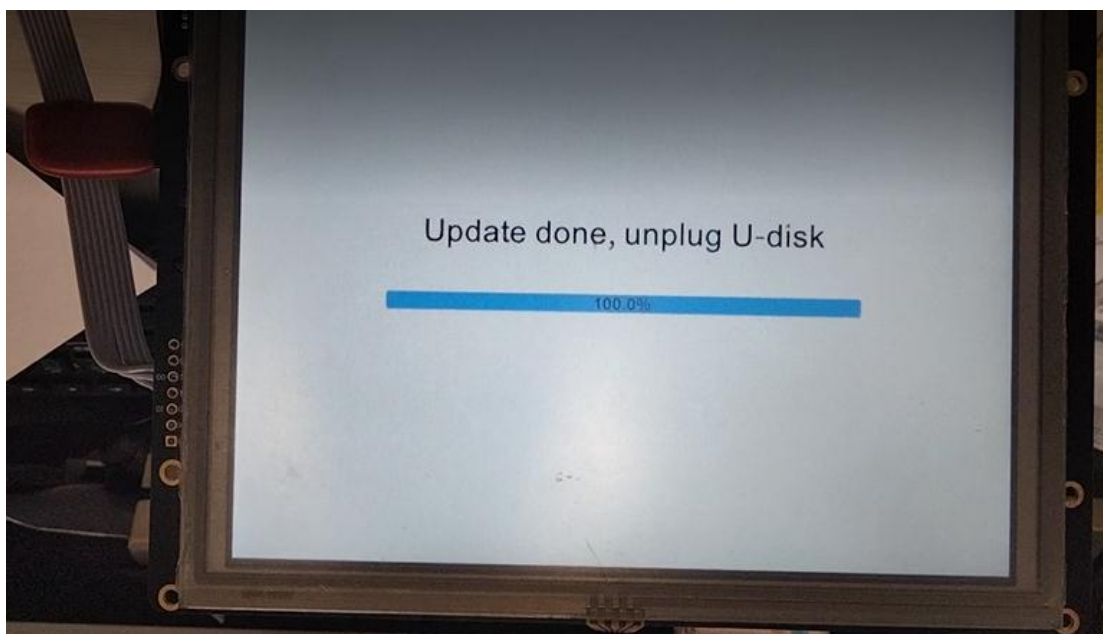
2. In the power-on state, turn the K600 switch on the HMI's PCB board to the **U-disk** mode and insert the prepared U-disk. The screen will prompt the update progress, there will be corresponding prompts on the screen during the update project, follow the prompts to operate, see the figure below.



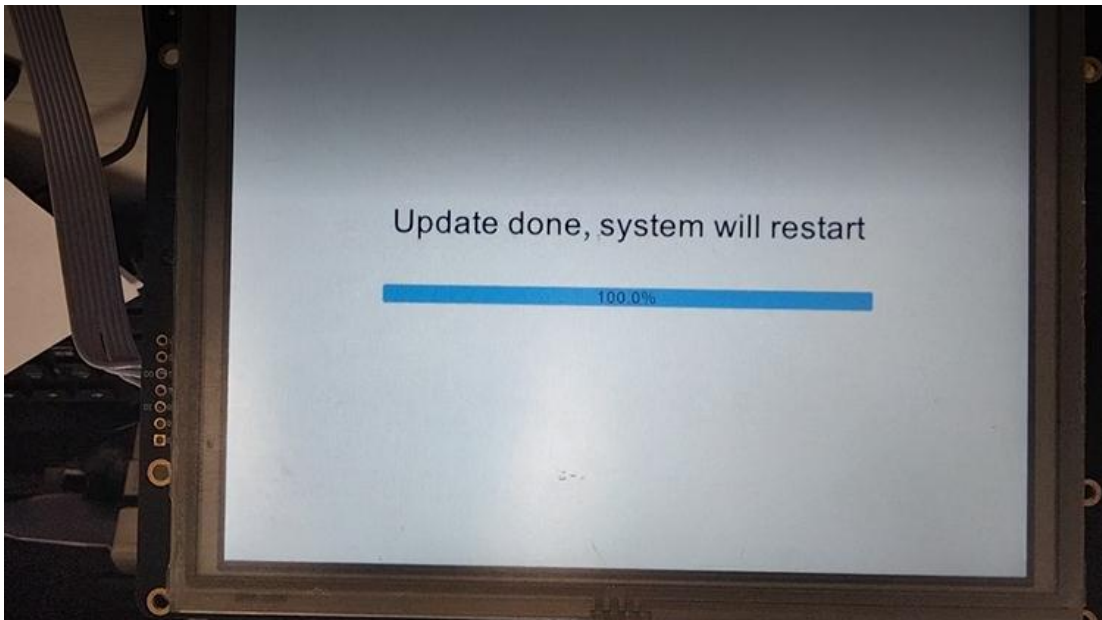
Project Update Screen Prompt.



Screen prompt after update is complete.



Screen prompt after unplugging the U disk after the update.



3. After unplugging the U disk, the HMI will automatically reboot, and the project update will be successful after the reboot is completed.

Special Notes:

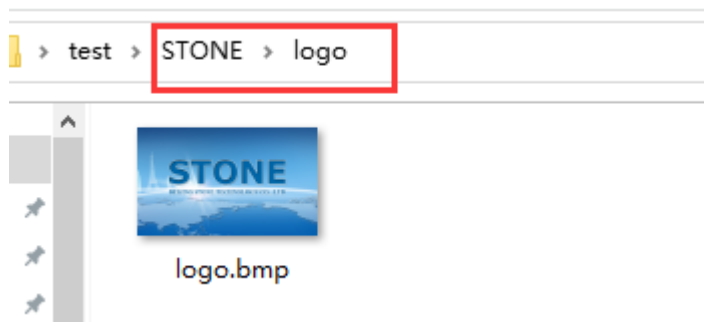
U-disk needs to be in **FAT32 format** with no hidden partitions.

Ensure the path is correct, the path is named correctly (note the case) and the file integrity can be updated successfully.

Some older version before January 2022 cannot be updated UI project directly through U disk, the user need to update the product's firmware. Please seek help from staff in this situation.

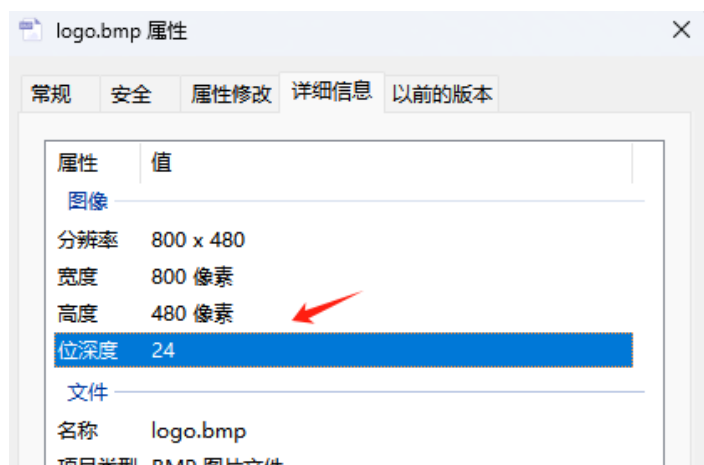
2.3 Boot Logo Update

1. Create the file path \STONE\logo (note the case), rename the logo file that needs to be updated to logo.bmp, and copy it to the following folder that was created.

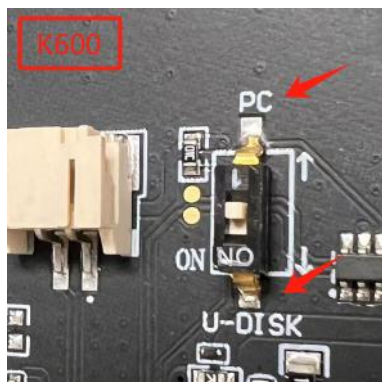


Special Note: Resolution is not mandatory and can be determined by the actual size of the project and HMI.

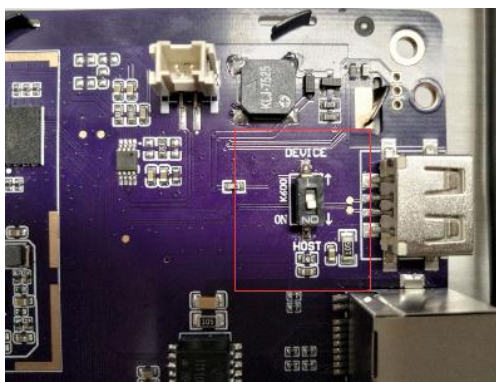
The name is unique, logo.bmp. **bit depth is fixed to 8 or 24.**



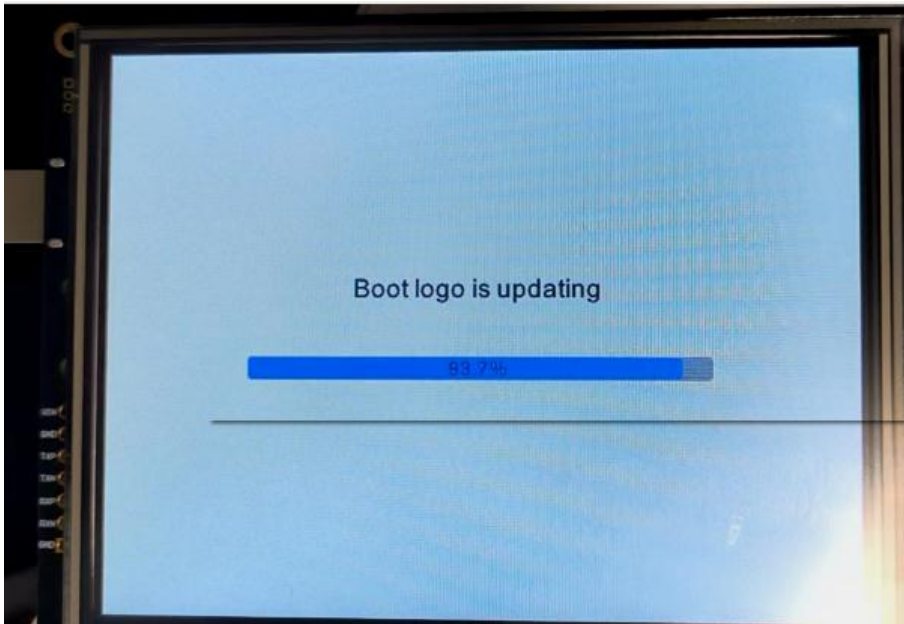
2. In the power-on state, turn the K600 to U-disk (or HOST for old version) and insert the prepared U-disk. The screen will prompt the update progress, there will be corresponding prompts on the screen during the update project, follow the prompts to operate, see the figure below.



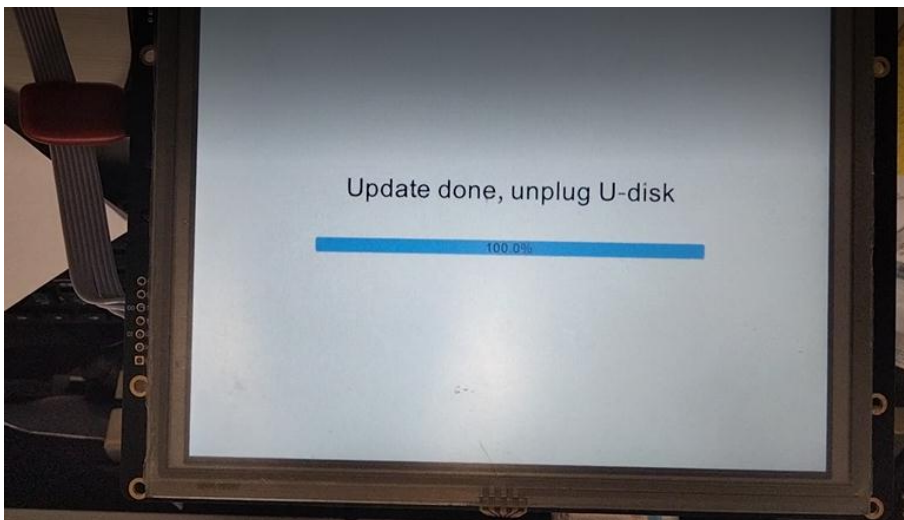
or



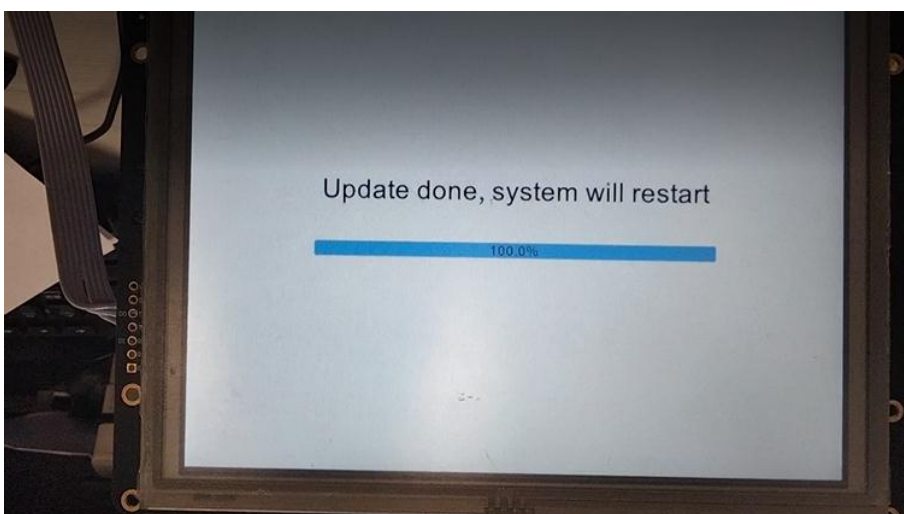
Screen Prompt for Logo update.



After the update is completed, the screen prompts are as follows:



Screen prompt after unplugging the USB flash drive:



3. After unplugging the U disk, the HMI will automatically reboot and the new logo will be used during the reboot.

Special Notes:

U-disk needs to be in FAT32 format with no hidden partitions.

The name is unique, logo.bmp. **bit depth is fixed to 8 or 24.**

Ensure the path is correct, the path is named correctly (note the case) and the file integrity can be updated successfully.

Some older version before January 2022 cannot be updated UI project directly through U disk, the user need to update the product's firmware. Please seek help from staff in this situation.

3.Window pages, Keyboards and Animations

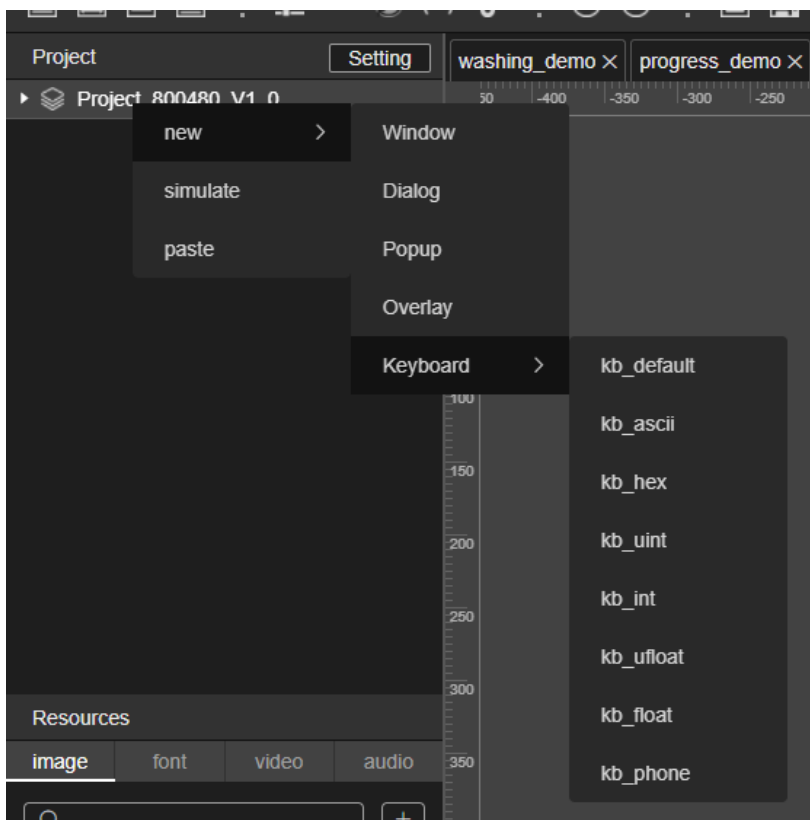
3.1 Window pages and custom Keyboards

3.1.1 Window page

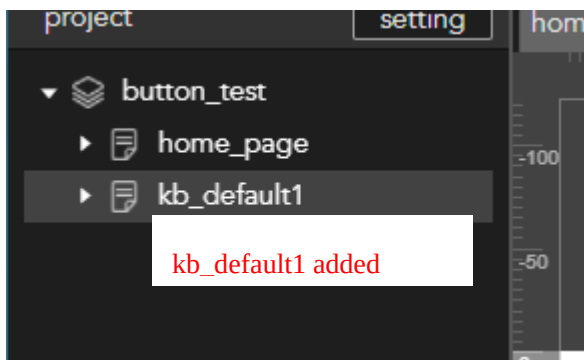
The software provides different page types for users to choose according to their own needs. Including Window, Dialog, Popup, Overlay and different Keyboards.

3.1.2 Create a new page

Select the project name in the project on the left, and right-click to select a different type to add.



The added type will exist in the project bar on the left.

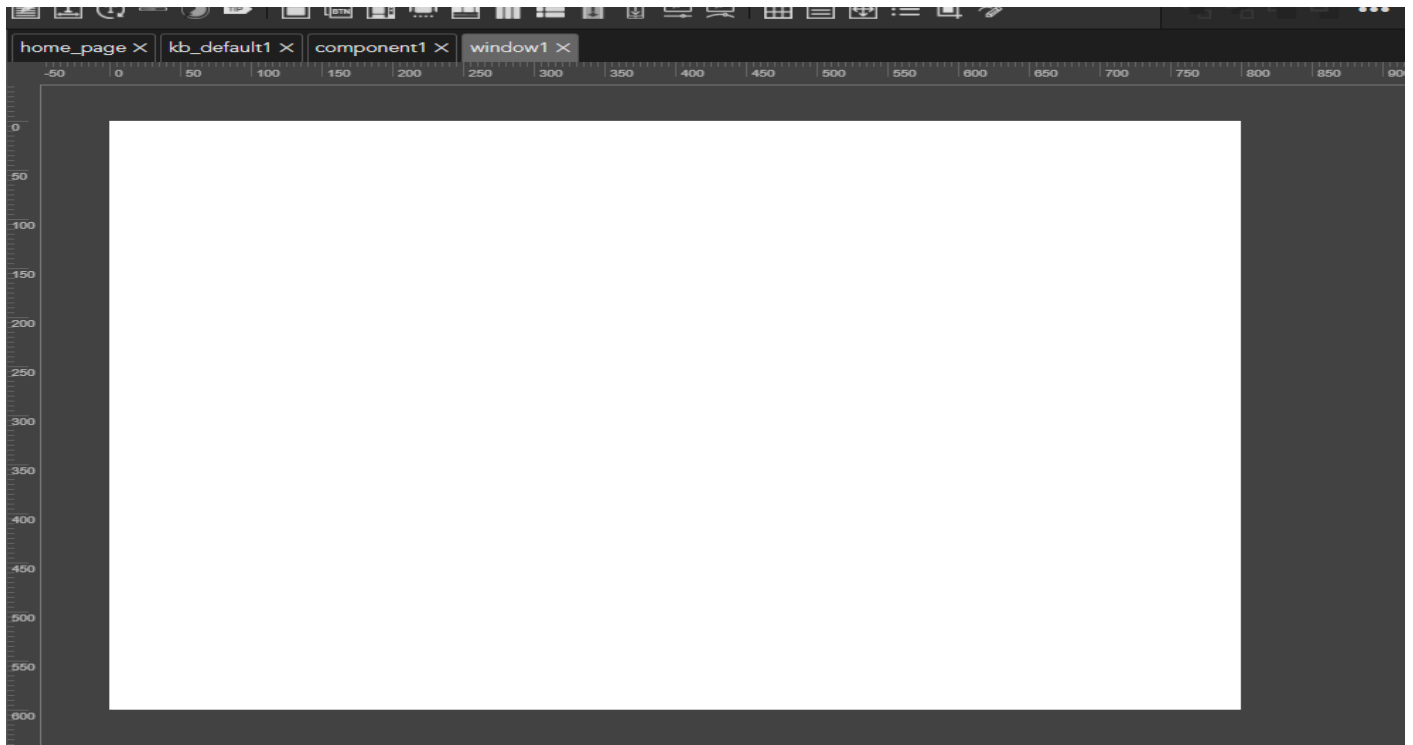


3.1.3 Different Window page types and application scenarios

3.1.3.1 Window

As one of the basic widgets, you can drag and drop widgets on the window to create projects.

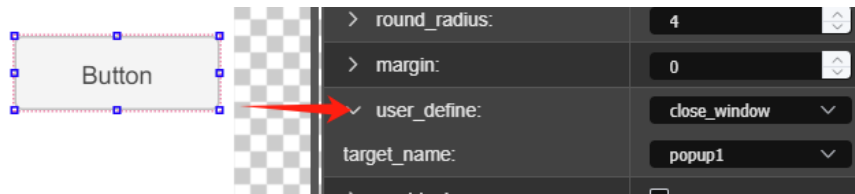
All widgets except the window can be dragged and dropped on the window.



3.1.3.2 Popup

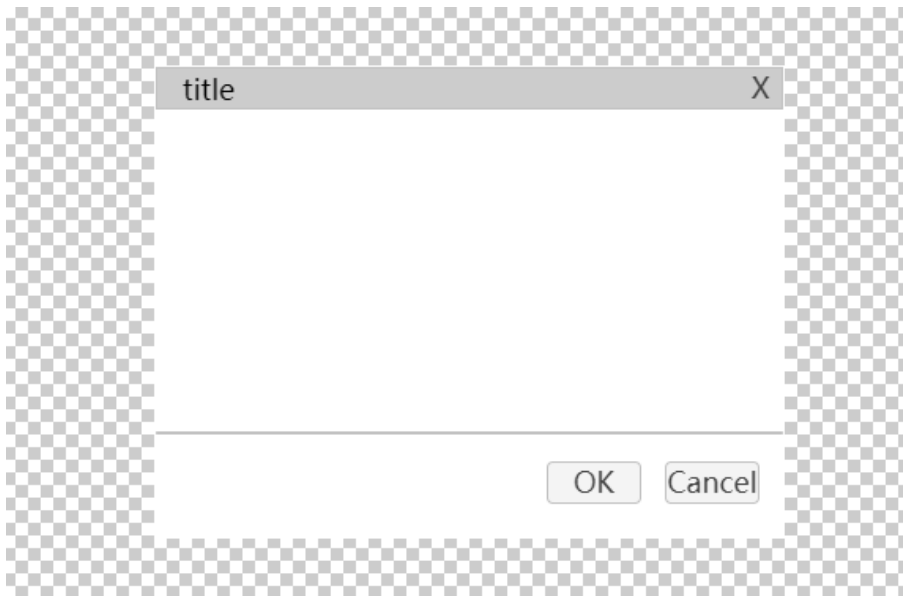
When performing a certain operation, you can choose to use popup with pop-up prompts. The opening and closing of popup needs to cooperate with the `user_define` function of the button.





3.1.3.3 Dialog

The dialog is a dialog box. Compared with the normal pop-up box, it completes the title and the prompt below. It needs to be opened and closed with the user_define function of the button. When the button is clicked, the pop-up prompt appears. Such as warning, displaying current value, etc.



3.1.3.4 Overlay

Overlay is coverage. When performing an operation, you can choose to cover a certain area to form data hiding and other effects.

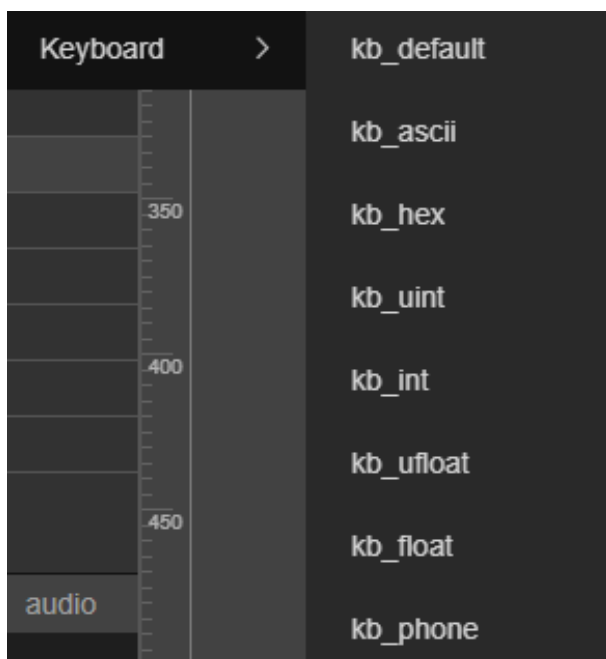


3.1.4 keyboard

3.1.4.1 introduction of keyboard

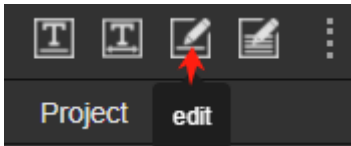
As the keyboard, considering that different users need different types of keyboards in different projects to meet the needs of different widgets. Therefore, users can design different keyboards according to their own needs. Set the `bg_color`, `bg_image`, etc. of each button of the keyboard according to your own needs. Afterwards, the user can link keyboard with different widgets through the settings of the widgets.

The currently supported keyboards are shown below.



3.1.4.2 Link Keyboard with other widgets

1. Choose a widget that supports text input, such as `edit` / `spin_box` / `combo_box_ex`.



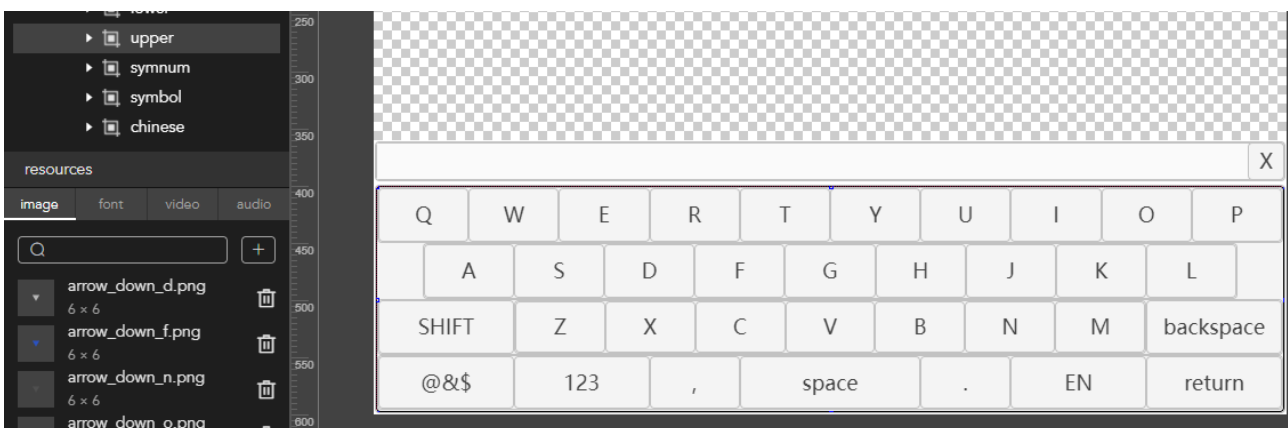
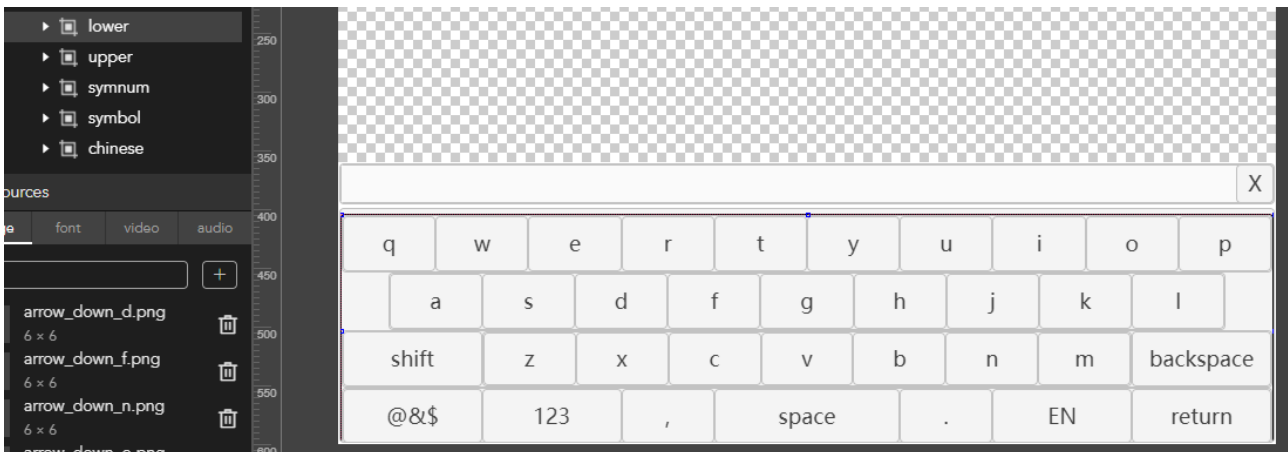
2. Find the keyboard option in the properties on the right. This keyboard (except the default keyboard) is only available after you add it in the project.

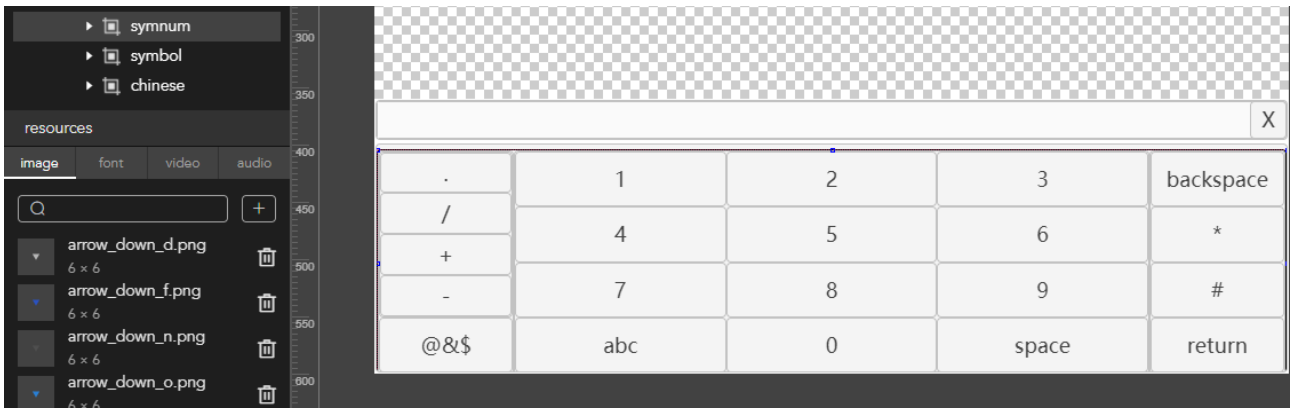


3. After setting, download the project to the HMI. After clicking the widget on the HMI screen, the set keyboard can be applied.

3.1.4.3 kb_default

kb_default is the default keyboard, all content can be set, kb_default includes uppercase keyboard, lowercase keyboard, numeric keyboard, symbol keyboard and Chinese keyboard.

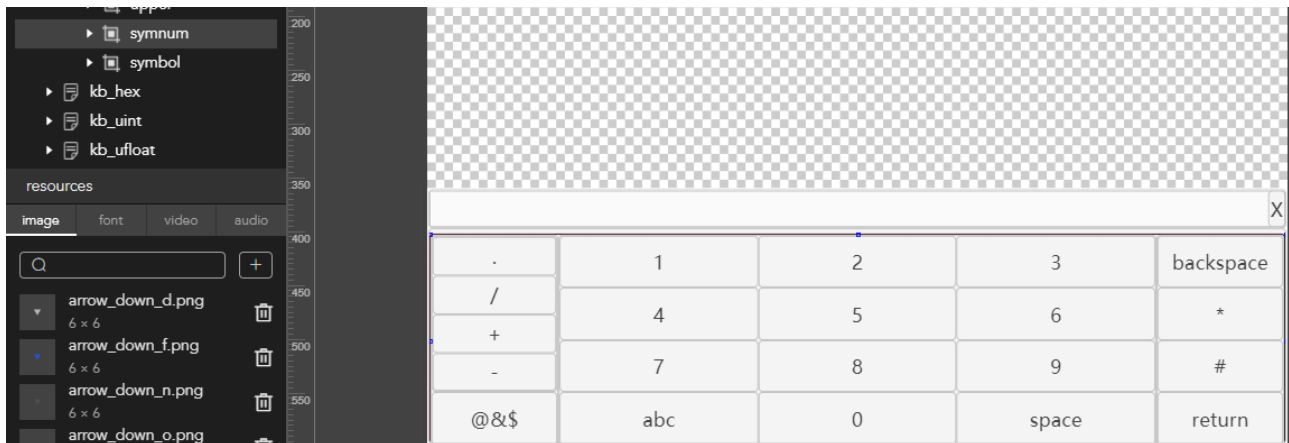




3.1.4.4 kb_ascii

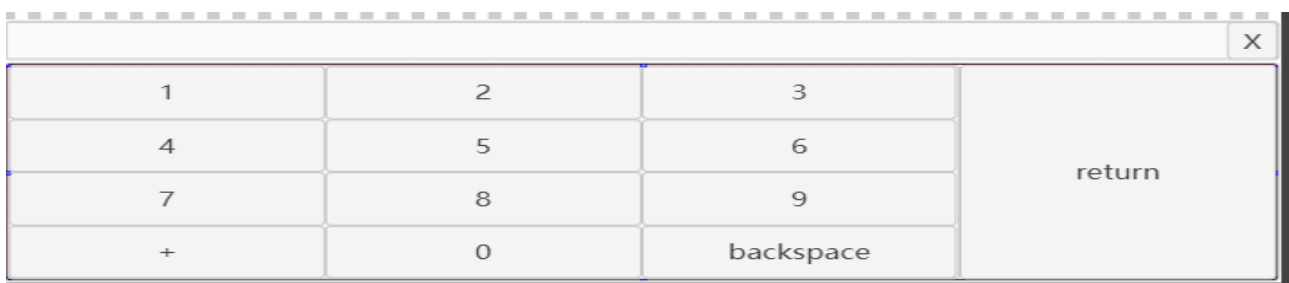
kb_ascii is a keyboard that can only input ASCII, and you can design pages according to your own needs. All legal ASCII input is supported.





3.1.4.5 kb_uint

kb_uint only allows the input of uint, and design the page according to your own needs.



3.1.4.6 kb_int

kb_int only allows the input of int, and design the page according to your own needs.

				X
1	2	3	return	
4	5	6		
7	8	9		
-	0	backspace		

3.1.4.7 kb_ufloat

kb_ufloat only allows input of ufloat, and design the page according to your own needs.

				X
1	2	3	backspace	
4	5	6		
7	8	9	return	
+	0	.		

3.1.4.8 kb_float

kb_float only allows the input of float, and design the page according to your own needs.

				X
1	2	3	backspace	
4	5	6		
7	8	9	return	
-	0	.		

3.1.4.9 kb_phone

kb_phone only allows the input of numeric numbers, and design the page according to your own needs.

			backsp	X
1	2	3	return	
4	5	6		
7	8	9		
*	0	#		

3.2 Animation and Page Animation

3.2.1 Page animation

Page animation is applied to window opening and window closing, and the properties involved are `open_anim_hint` and `close_anim_hint`. Set it according to your needs.

3.2.2 Page animation type

Page animations can only be applied to different pages. Page animation types are divided into the following types:



`htranslate`: left and right translate

`vtranslate` : up and down translate

`slide_up` : cover to the top

`slide_down` : cover to the bottom

`slide_left` : cover to the left

`slide_right` : cover to the right

3.2.3 Page animation properties

3.2.3.1 `anim_duration`

Animation duration.

When is 0, the default is 500ms.



- Default value: 0
- Data type: uint
- Whether it is editable: yes

3.2.3.2 anim_trend

Animation trends.

anim_trend:	cubic_out ▼
-------------	-------------

- Default value: cubic_out
- Data type: text
- Whether it is editable: yes

3.2.3.3 opacity_start_value

Transparency at the start of animation playback.

opacity_start_value:	0	↕
----------------------	---	---

- Default value: 0
- Data type: uint
- Whether it is editable: yes

3.2.3.4 opacity_end_value

Transparency at the end of animation playback.

opacity_end_value:	0	↕
--------------------	---	---

- Default value: 0
- Data type: uint
- Whether it is editable: yes

3.2.3.5 disable_anim

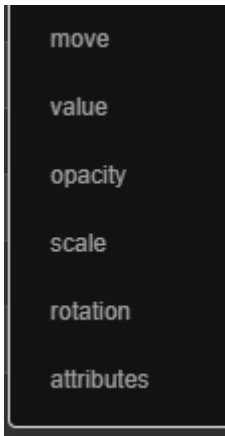
Whether to automatically destroy after the animation finishes playing.

disable_anim:	☐
---------------	--------------------------

- Default value: false
- Data type: bool
- Whether it is editable: yes

3.2.4 Widget Animation

Widget animation can only be applied to widgets other than Pages. The corresponding property is the animation_type in each widget, and the animation types of the widgets are divided into the following types:



move: control the widget to move within the specified coordinate range during the animation cycle.

value: control the change of the widget with the value property during the animation cycle.

opacity: control the transparency of the widget from the beginning to the end of the animation cycle.

scale: control the proportional scaling of the widget during the animation cycle.

rotation: control to perform a rotation of the widget around itself during the animation cycle.

Notice: scale and rotation are specific to image/gif/svg/image_value_image_animation.

3.2.5 Widget animation properties

3.2.5.1 name

The name saved after setting the animation properties for the widget.

name:	
-------	----------------------

- Default value:
- Data type: text
- Whether it is editable: yes

3.2.5.2 delay_start_time

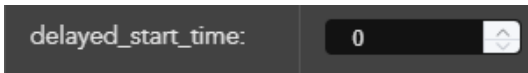
Delay start time.

delayed_start_time:	0
---------------------	--------------------------------

- Default value: 0
- Data type: uint
- Whether it is editable: yes

3.2.5.3 single_anim_duration

The duration of a single animation.

A UI control for the 'single_anim_duration' property. It consists of a label 'delayed_start_time:' on the left and a numeric input field on the right. The input field contains the value '0' and has a small up/down arrow icon on its right side.

- Default value: 0
- Data type: uint
- Whether it is editable: yes

3.2.5.4 multi_speed

The playback speed of the animation.

A UI control for the 'multi_speed' property. It consists of a label 'multi_speed:' on the left and a numeric input field on the right. The input field contains the value '1' and has a small up/down arrow icon on its right side.

- Default value: 1
- Data type: uint
- Whether it is editable: yes

3.2.5.5 single_anim_trend

Animation trends.

A UI control for the 'single_anim_trend' property. It consists of a label 'single_anim_trend:' on the left and a dropdown menu on the right. The dropdown menu shows the value 'cubic_out' and a downward arrow icon.

- Default value: cubic_out
- Data type: text
- Whether it is editable: yes

3.2.5.6 round_trip_time

Number of round trips.

Need to check, then the value will appear, 0 means permanent play.

A UI control for the 'round_trip_time' property. It consists of a label 'round_trip_time:' on the left and a checkbox on the right. The checkbox is currently unchecked.

- Default value: false
- Data type: bool/uint
- Whether it is editable: yes

3.2.5.7 repeat_time

Repeat times.

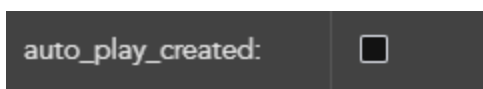
Need to check the front, the value will appear, 0 means permanent play.

The image shows a UI element for the 'repeat_time' property. It consists of a label 'repeat_time:' followed by a small square checkbox and a dark rectangular text input field.

- Default value: false
- Data type: bool/uint
- Whether it is editable: yes

3.2.5.8 auto_play_created

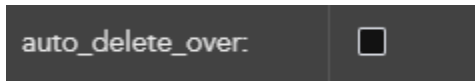
The animation plays automatically after it is created.

The image shows a UI element for the 'auto_play_created' property. It consists of a label 'auto_play_created:' followed by a small square checkbox.

- Default value: false
- Data type: bool
- Whether it is editable: yes

3.2.5.9 auto_delete_over

Destroyed automatically after the animation ends.

The image shows a UI element for the 'auto_delete_over' property. It consists of a label 'auto_delete_over:' followed by a small square checkbox.

- Default value: false
- Data type: bool
- Whether it is editable: yes

3.2.5.10 x_start/end

The starting x-coordinate and ending x-coordinate of the animation playback.

The image shows a UI element for the 'x_start' and 'x_end' properties. It consists of two rows. The first row has a label 'x_start:' followed by a numeric input field containing the value '0' and a small up/down arrow icon. The second row has a label 'x_end:' followed by a similar numeric input field containing the value '0' and an up/down arrow icon.

- Default value: 0/0
- Data type: int
- Whether it is editable: yes

3.2.5.11 y_start/end

The start y coordinate and end y coordinate of the animation playback.

y_start:	0	↑	↓
y_end:	0	↑	↓

- Default value: 0/0
- Data type: int
- Whether it is editable: yes

3.2.5.12 start/end_value:

The start and end values of animation playback.

start_value:	0
end_value:	0

- Default value: 0/0
- Data type: int
- Whether it is editable: yes

3.2.5.13 opacity_start/end_value

The start and end values of transparency.

opacity_start_value:	0	↑	↓
opacity_end_value:	0	↑	↓

- Default value: 0/0
- Data type: int
- Whether it is editable: yes

3.2.5.14 rotation_start/end_value

The start and end values of the rotation.

rotation_start_value:	0	↑	↓
rotation_end_value:	0	↑	↓

- Default value: 0/0
- Data type: int
- Whether it is editable: yes

4.Introduction of Properties

4.1 Common basic property settings

4.1.1 name

Widgets name. For a project, **each widget must have a unique name.**

For the convenience of sending and receiving instructions and other issues, avoid using invalid characters in the Json code to name widgets.

name:	blink1
-------	--------

- Default value: based on the widget type, assigned by the system
- Type of data: text
- Whether it is editable: Yes

4.1.2 type

Widget type (specified by the system, cannot be changed)

type:	button
-------	--------

- Default value: based on the widget type, assigned by the system
- Type of data: text
- Whether it is editable: No

4.1.3 x/y

Widget x/y coordinates (upper left corner).

Can be set to a negative value.

x: 646	y: 133
--------	--------

- Default value: based on the widget type, assigned by the system
- Type of data: int
- Whether it is editable: Yes

4.1.4 w/h

The width/height of the widget.

Unit: pixel.

w:	100	
h:	36	

- Default value: based on the widget type, assigned by the system
- Type of data: uint
- Whether it is editable: Yes

4.1.5 text

Initial text on the widget.

text:	BLINK1
-------	--------

- Default value: based on the widget type, assigned by the system
- Type of data: text
- Whether it is editable: Yes

4.1.6 enable

Whether the widget is enabled.

P.S. If this option is not selected, it will be displayed as "disabled" and cannot be operated.

enable:	☒
---------	-------------------------------------

- Default value: true
- Data type: bool
- Whether it is editable: Yes

4.1.7 visible

Determines whether the widget is visible.

Different from enable. If this option is not checked, the widgets will not be displayed but will still be operable.

visible:	☒
----------	-------------------------------------

- Default value: true
- Data type: bool
- Whether it is editable: Yes

4.2 General style settings

4.2.1 border_color

The border color of the widget. Most widgets allow user to define border color.

The desired translucent effect can be achieved by adjusting the alpha value.



- Default value: based on the widget type, assigned by the system
- Type of data: hex
- Whether it is editable: Yes

4.2.2 border

The border type of the widget.

5 types are available: full border, transparent on the left, transparent on the top, transparent on the right, and transparent on the bottom. The user can **double click on the options to remove this feature**.



- Default value: all (full border)
- Data type: text
- Whether it is editable: Yes

4.2.3 border_width

The border width of the widget.

Unit: pixel.



- Default value: 1 (different widgets with different default values)
- Data type: uint
- Whether it is editable: Yes

4.2.4 round_radius

The corner arc of the widget border.

Unit: pixel.



Click the arrow on the left to expand more detailed settings:

The upper left radian, the upper right radian, the lower left radian, and the lower right radian.



- Default value: 4 (different widgets with different default values)
- Data type: uint
- Whether it is editable: Yes

4.2.5 text_color

The text color of the widget. Most widgets allow user to define text color.

The desired translucent effect can be achieved by adjusting the alpha value.



- Default value: based on the widget type, assigned by the system
- Type of data: hex
- Whether it is editable: Yes

4.2.6 selected_text_color

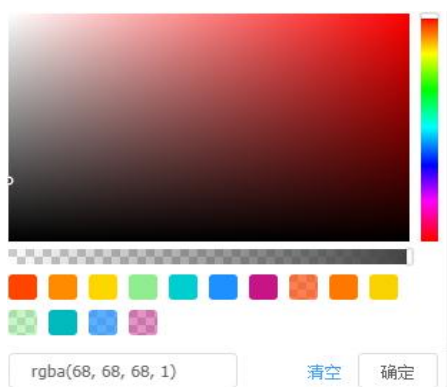
The color of the selected text.



- Default value: based on the widget type, assigned by the system
- Data type: hex
- Whether it is editable: yes

4.2.7 tips_text_color

The color of the hint text.

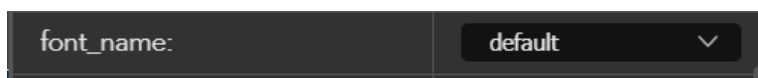


- Default value: based on the widget type, assigned by the system
- Data type: hex
- Whether it is editable: yes

4.2.8 font_name

The font of the widget text. The font should be added to the resource list, otherwise it cannot be selected in font_name.

It is recommended to avoid naming fonts with invalid characters in Json code.



- Default value: default (cn normal)
- Type of data: text
- Whether it is editable: Yes

4.2.9 font_size

The font size of the widget text

font_size:	18
------------	----

- Default value: 18
- Type of data: uint
- Whether it is editable: Yes

4.2.10 highlight_font_name

Font for highlighted areas.

Fonts must be uploaded in recourse before they can be used.

highlight_font_name:	default
----------------------	---------

- Default value: default (cn normal)
- Data type: text
- Whether it is editable: yes

4.2.11 highlight_font_size

The font size of the highlighted area.

highlight_font_size:	35
----------------------	----

- Default value: 18
- Data type: unit
- Whether it is editable: yes

4.2.12 highlight_text_color

Font color for highlighted areas.

highlight_text_color:	
-----------------------	--

- Default value: based on the widget type, assigned by the system.
- Data type: hex
- Whether it is editable: yes

4.2.13 text_align

Defines the alignment style for the widget text.

text_align_v: vertical alignment style

text_align_h:	center
---------------	--------

text_align_h: horizontal alignment style

text_align_v:	middle ▼
---------------	----------

- Default value: middle (text_align_v); center (text_align_h)
- Data type: text
- Whether it is editable: yes

4.2.14 bg_color

The background color of the widget. Most widgets allow user to define background color. It also includes the color of the unslidden part of the slider of the Indicator Related Widgets (slider, slider_circle, progress_bar, progress_circle), and the background color when the switch widget is not opened.

The desired translucent effect can be achieved by adjusting the alpha value.



- Default value: based on the widget type, assigned by the system
- Type of data: hex
- Whether it is editable: Yes

4.2.15 bg_image

Defines the widgets' background image by name. It also includes the background image of the unslidden part of the slider of the Indicator Related Widgets (slider, slider_circle, progress_bar, progress_circle), and the background image when the switch widget is not opened.

(1) if the image is not defined, the background will appear transparent.

(2) The picture should be uploaded to the resources list on the left before it can be referenced in the widget.

bg_image:	home_btn ▼
-----------	------------

- Default value: based on the widget type, assigned by the system
- Type of data: text
- Whether it is editable: Yes

4.2.16 bg_image_draw_type

Defines the background image drawing type for the widgets' background image.

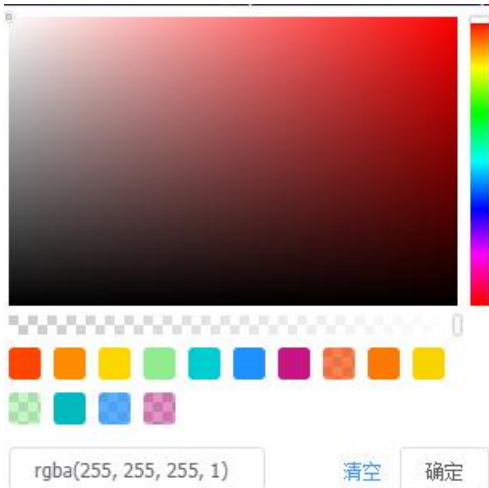


- (1) default: the picture will be displayed in its original size and aspect ratio and aligned in the upper left corner of the display area of this image widget
- (2) Icon: the picture will be tiled and displayed in the entire display area according to the original aspect ratio. When the aspect ratio of the actual display area is inconsistent with the aspect ratio of the original display area, the image will be enlarged or reduced according to the longer side.
- (3) center: the picture will be displayed in the center of the display area with the original size and aspect ratio, and the excess part will not be scaled.
- (4) scale: the picture will be displayed in the original aspect ratio. When the aspect ratio of the actual display area is inconsistent with the aspect ratio of the original display area, the picture will be enlarged or reduced according to the shorter side. It will be aligned at the top.
- (5) auto: similar to the default. The picture will be displayed at its original size and aspect ratio, aligned at the top left corner of the display area of this image widget.
- (6) scale_auto: the picture will be displayed in the original aspect ratio. When the aspect ratio of the actual display area is inconsistent with the aspect ratio of the original display area, the picture will be enlarged or reduced according to the shorter side.
- (7) scale_down: similar to the center. The picture will be displayed in the center of the display area at its original size and aspect ratio.
- (8) scale_w: width zoom display. The image is scaled by the width of the target rectangle, the width is scaled by this ratio, and the excess will not be displayed. The picture will be displayed in its original aspect ratio. When the aspect ratio of the actual display area is inconsistent with the aspect ratio of the original display area, the picture will be enlarged or reduced according to the shorter side. It will be aligned at the top.
- (9) scale h: height zoom display. The image is scaled by the height of the target rectangle, and the width is scaled by this ratio. The excess part will not be displayed
- (10) repeat: the picture is tiled in the space.

- Default value: default
- Type of data: text
- Whether it is editable: Yes

4.2.17 fg_color:

The foreground color of the widget when the switch is in the false state.



- Default value: based on the widget type, assigned by the system
- Data type: hex
- Whether it is editable: yes

4.2.18 fg_image

Define the widget's foreground image by name. It also includes the foreground of the unslid part of the slider of the Indicator Related Widgets (slider, slider_circle, progress_bar, progress_circle), and the foreground image when the switch widget is not opened.

(1) if the image is not defined, the background will appear transparent.

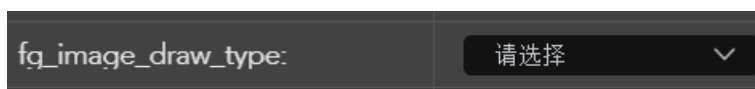
(2)The picture should be uploaded to the resources list on the left before it can be referenced in the widget.



- Default value: based on the widget type, assigned by the system
- Data type: text
- Whether is it editable: Yes

4.2.19 fg_image_draw_type

Define the drawing type of the widget's foreground image.



(1) default: the picture will be displayed in its original size and aspect ratio and aligned in the upper left corner of the display area of this image widget

(2) Icon: the picture will be tiled and displayed in the entire display area according to the original aspect ratio. When the aspect ratio of the actual display area is inconsistent with the aspect ratio of the original display area, the image will be enlarged or reduced according to the longer side.

(3) center: the picture will be displayed in the center of the display area with the original size and aspect ratio, and the excess part will not be scaled.

(4) scale: the picture will be displayed in the original aspect ratio. When the aspect ratio of the actual display area is inconsistent with the aspect ratio of the original display area, the picture will be enlarged or reduced according to the shorter side. It will be aligned at the top.

(5) auto: similar to the default. The picture will be displayed at its original size and aspect ratio, aligned at the top left corner of the display area of this image widget.

(6) scale_auto: the picture will be displayed in the original aspect ratio. When the aspect ratio of the actual display area is inconsistent with the aspect ratio of the original display area, the picture will be enlarged or reduced according to the shorter side.

(7) scale_down: similar to the center. The picture will be displayed in the center of the display area at its original size and aspect ratio.

(8) scale_w: width zoom display. The image is scaled by the width of the target rectangle, the width is scaled by this ratio, and the excess will not be displayed. The picture will be displayed in its original aspect ratio. When the aspect ratio of the actual display area is inconsistent with the aspect ratio of the original display area, the picture will be enlarged or reduced according to the shorter side. It will be aligned at the top.

(9) scale h: height zoom display. The image is scaled by the height of the target rectangle, and the width is scaled by this ratio. The excess part will not be displayed

(10) repeat: the picture is tiled in the space.

- Default value: based on the widget type, assigned by the system
- Type of data: text
- Whether it is editable: Yes

4.2.20 mask_color:

Define the mask color for the entire widget. The color will gradually change from the edge to the center.

mask_color:	
-------------	-------------------------------

- Default value: based on the widget type, assigned by the system
- Data type: hex
- Whether it is editable: Yes

4.2.21 icon

Use the image which already uploaded in the resource to replace the widget icon.

(1) The default icon will be displayed as transparent.

(2) The images should be uploaded to the resources list on the left before it can be used in the widget.

icon:	none
-------	-----------------------------------

- Default value: based on the widget type, assigned by the system
- Type of data: text
- Whether it is editable: Yes

4.2.22 icon_at

Define the position of the icon on the widget

auto: the middle left of the widget.

left: the middle left of the widget.

center: the center of the widget. right: the middle right of the widget.

top: the middle top of the widget.

bottom: the middle bottom of the widget.

icon_at:	auto ▼
----------	--------

- Default value: auto
- Type of data: text
- Whether it is editable: Yes

4.2.23 margin

Defines the position and distance of the text inside the widget, calculated from its original position (defined by the text alignment and the widget itself).

> margin:	0	↕
-----------	---	---

Click to expand and adjust the four directions individually.

Unit: pixel.

top: increase the distance of the current text to the top of the original position.

bottom: increase the distance of the current text to the bottom of the original position.

left: increase the distance of the current text to the left of the original position.

right: increase the distance of the current text to the right of the original position.

▼ margin:	0	↕
left:	0	↕
top:	0	↕
right:	0	↕
bottom:	0	↕

- Default value: 0
- Type of data: uint
- Whether it is editable: Yes

4.2.24 key_tone

Click whether to trigger the background sound.



- Default value: true (different widgets with different default values)
- Data type: bool
- Whether it is editable: yes

4.2.25 selected_bg_color

The background color of the widget when it is selected. For the Switch widget, it is the foreground color when it is turned on.



- Default value: based on the widget type, assigned by the system
- Data type: hex
- Whether it is editable: yes

4.2.26 selected_fg_color:

The foreground color of the widget when it is selected. For the Switch widget, it is the background color when it is turned on.

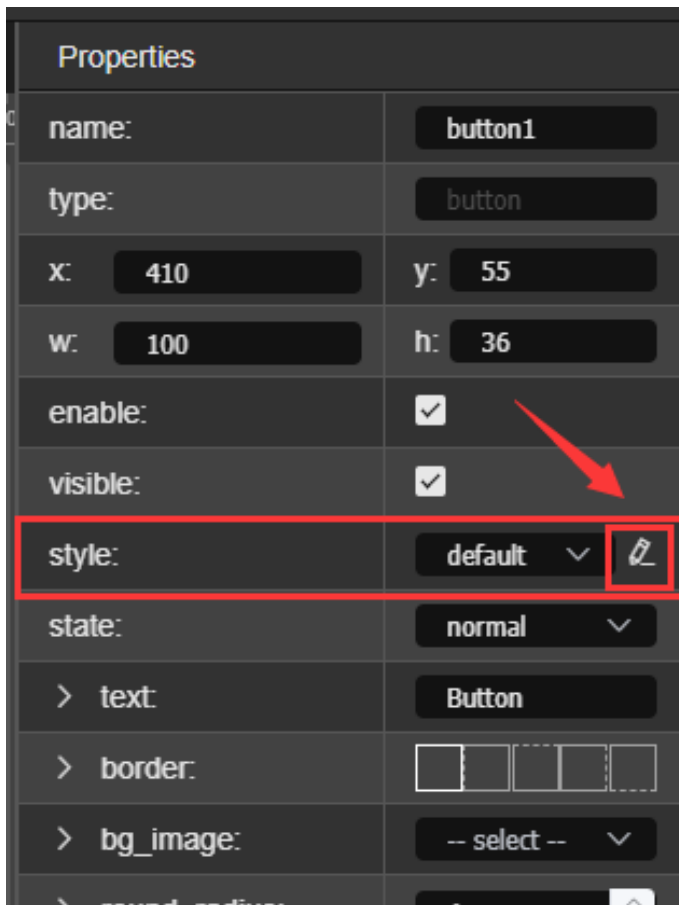


- Default value: 0xffffffff
- Data type: hex
- Whether it is editable: yes

4.3 Customizing Style

Custom style is applicable to every widget, the purpose of custom style is to facilitate the user to set a style according to their own needs, so that it can be applied to multiple widgets of the same type.

After selecting a widget, click the button on the right side of the style to enter the custom style setting page.



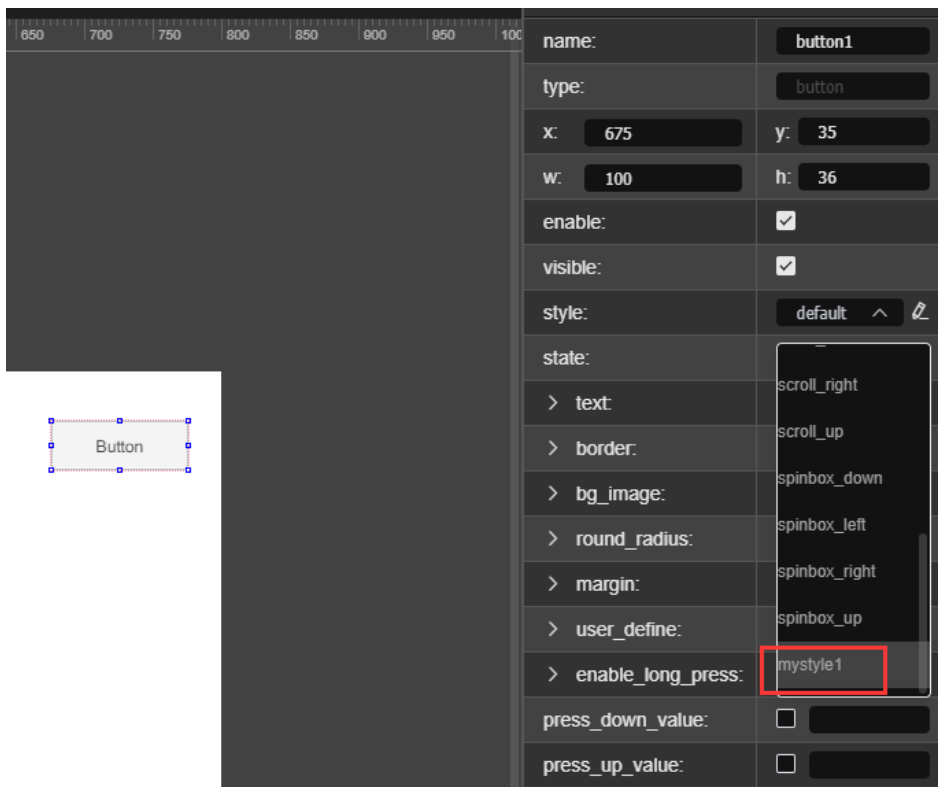
Properties	
name:	button1
type:	button
x:	410
y:	55
w:	100
h:	36
enable:	☒
visible:	☒
style:	default 
state:	normal
> text:	Button
> border:	
> bg_image:	-- select --
> round_radius:	

After saving, the customized style can be applied to all widgets of the same type, different widgets need to be set separately.

Set the name according to the requirement, it is recommended not to repeat the name with the existing style, set the relevant parameters and click save.



You can then set it in the properties of different widgets, using the button as an example:



5. Other properties for common widgets

5.1 window

As the most basic widget, the window widget needs to be added by right-clicking on the project name

5.2 button

The use of the button widget is very wide, the general use is as follows:

- (1) The button can be used to jump to the page (window), pop-up box (dialog), etc. after clicking.
- (2) The button button can be used to increase or decrease the value after it is bundled with the widget.
- (3) Clicking on the button generates a return value, which can be programmed with different external controller by the user to achieve more linkages.

5.2.1 user_define

The button's user_define has five different options: none

open_window : clicking on this widget will open a new window for the custom set target_name.

home_window : clicking this widget will return to the project's home_page.

back_window : clicking this widget will return to the previous window and close the current window.

close_window : click this widget to close the current window (home_page cannot be closed), the default target_name is the window where it is located, and it can be modified to other windows.

set_value : Set incremental and decremental adjustment for other widget. Set the bundle target_name for addition and subtraction.



- Default value: none
- Data type: text
- Whether it is editable: yes

5.2.2 target_name

The name of the object to set.

target_name:	none
--------------	------

user_define:	set_value
target_name:	edit1
min:	0
max:	100
step:	1
format:	%d
increase_decrease:	++

(1) If user_define is set to open_window, the target name will be chosen from the window names already created in the project. If set to home_window/back_window, you don't have to choose a target name.

(2) If set to set_value, you need to select the widgets that have been generated in the page. And drag the widget to the page before you can select the widget for set_value.

- Default value: none
- Data type: text
- Whether it is editable: yes

5.2.3 min/max

The minimum/maximum numerical limit of the corresponding widget.

Options are only displayed if user_define is set to set_value.

min:	0
max:	100

- Default value: 0/100
- Data type: float
- Whether it is editable: yes

5.2.4 step

How much value will be increased or decreased by one click.

Options are only displayed if user_define is set to "set_value".

step:	1
-------	---

- Default value: 0/100
- Data type: float
- Whether it is editable: yes

5.2.5 format

Define the data format type.

%d only displays the integer part. %f will be displayed as a floating-point number.

format:	%d
---------	----

- Default value: %02d
- Data type: text
- Whether it is editable: yes

5.2.6 increase_decrease

Define increment or decrement after click.

++: After the button is clicked, the value of the defined widget is incremented. The incremental value will be defined by step.

--: Each trigger decrements the value of the defined widget after the button is clicked. The decremental value is defined by step.

increase_decrease:	++
--------------------	----

- Default value: ++
- Data type: text
- Whether it is editable: yes

5.2.7 enable_long_press

Whether to enable long press events.

When repeat is greater than 0, the button function will be triggered when the time of pressing the button is exceeded.

If the widget is kept pressed, it will fire repeatedly after the set time interval.

enable_long_press:	☐
--------------------	--------------------------

- Default value: false
- Data type: bool
- Whether it is editable: yes

5.2.8 long_press_time

The time to trigger the long press.

The unit is millisecond.

long_press_time:	1,000
------------------	-------

- Default value: 1000
- Data type: uint
- Whether it is editable: yes

5.2.9 repeat

Repeat trigger time interval, in millisecond. If the value is 0, repeat does not trigger.

When repeat is greater than 0, the button function will be triggered when the time of pressing the button exceeds. And if the widget is kept pressed, it will fire repeatedly after the set time interval.

repeat:	0
---------	---

- Default value: 0
- Data type: uint
- Whether it is editable: yes

5.2.10 enable_press_down/press_down_value

Whether to allow the return value to be sent when the button is pressed. / Custom return value sent when the button is pressed.

If enable_press_down is not checked, the set custom return value is invalid.

press_down_value:	☐
-------------------	---

- Default value: false/
- Data type: bool/hex
- Whether it is editable: yes

5.2.11 enable_press_up/press_up_value

Whether to allow the return value to be sent when the button is released. / Custom return value sent when the button is released.

If enable_press_up is not checked, the set custom return value is invalid.

press_up_value:	☐
-----------------	---

- Default value: false/
- Data type: bool/hex
- Whether it is editable: yes

5.2.12 enable_click/click_value

Whether to allow the return value to be sent when the button is clicked and released. / Custom return value sent when the button is clicked and released.

If enable_click is not checked, the set custom return value is invalid.

click_value:	☒
--------------	--

- Default value: true/
- Data type: bool/hex
- Whether it is editable: yes

5.2.13 enable_long_press/long_press_value

Whether to allow the return value to be sent when long-pressed. / Custom return value sent when long press.

If enable_click is not checked, the set custom return value is invalid.

long_press_value:	☐
-------------------	---

- Default value: false/
- Data type: bool/hex
- Whether it is editable: yes

5.3 ☒ check_button

The check_button is mostly used for different options whether to check the corresponding options, such as when ordering, for the specific requirements of the food, whether to add small ingredients, whether to add sauces, etc., just tick in front of the check_button. Users can also add different background images for different states by setting the icon as none to achieve the effect of visually changing the image without using an external control panel.

5.3.1 value:

The default state of the widget, unchecked by default.

value:	☒
--------	-------------------------------------

- Default value: false
- Data type: bool
- Whether it is editable: yes

5.3.2 text

The text content displayed by the widget.

text:	CheckButton
-------	-------------

- Default value: checkbutton
- Data type: text
- Whether it is editable: yes

5.4 ☒ radio_button

The radio_button is mostly used to select one of different options. For example, when filling in the information, you need to choose one of the two genders, and click to check in front of the one that matches your own, then you need to use the radio_button.

5.4.1 value:

The default state of the widget, unchecked by default.

value:	☐
--------	--------------------------

- Default value: false
- Data type: bool
- Whether it is editable: yes

5.4.2 text

The text content to display when the box is checked.

text:	RadioButton
-------	-------------

- Default value: radiobutton
- Data type: text
- Whether it is editable: yes

5.5 switch

Most of the switch widgets are used to control the on and off status of an option, for example, in the process of connecting with other devices, to control the on and off status of the corresponding device.

5.5.1 value:

The default state of the widget, unchecked by default.

value:	☐
--------	--------------------------

- Default value: false
- Data type: bool
- Whether it is editable: yes

5.6 image

image is used to display an image.

5.6.1 image

The picture displayed by the widget.

The corresponding pictures need to be imported into the resources bar on the left first.

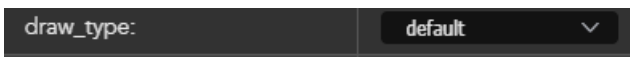
Select from the drop-down list or enter the name of the picture file in the box to search.



- Default value: num0
- Data type: text
- Whether it is editable: yes

5.6.2 draw_type:

The way the picture is displayed.



(1) default: the picture will be displayed in its original size and aspect ratio and aligned in the upper left corner of the display area of this image widget

(2) Icon: the picture will be tiled and displayed in the entire display area according to the original aspect ratio. When the aspect ratio of the actual display area is inconsistent with the aspect ratio of the original display area, the image will be enlarged or reduced according to the longer side.

(3) center: the picture will be displayed in the center of the display area with the original size and aspect ratio, and the excess part will not be scaled.

(4) scale: the picture will be displayed in the original aspect ratio. When the aspect ratio of the actual display area is inconsistent with the aspect ratio of the original display area, the picture will be enlarged or reduced according to the shorter side. It will be aligned at the top.

(5) auto: similar to the default. The picture will be displayed at its original size and aspect ratio, aligned at the top left corner of the display area of this image widget.

(6) scale_auto: the picture will be displayed in the original aspect ratio. When the aspect ratio of the actual display area is inconsistent with the aspect ratio of the original display area, the picture will be enlarged or reduced according to the shorter side.

(7) scale_down: similar to the center. The picture will be displayed in the center of the display area at its original size and aspect ratio.

(8) scale_w: width zoom display. The image is scaled by the width of the target rectangle, the width is scaled by this ratio, and the excess will not be displayed. The picture will be displayed in its original aspect ratio. When the aspect ratio of the actual display area is inconsistent with the aspect ratio of the original display area, the picture will be enlarged or reduced according to the shorter side. It will be aligned at the top.

(9) scale h: height zoom display. The image is scaled by the height of the target rectangle, and the width is scaled by this ratio. The excess part will not be displayed

(10) repeat: the picture is tiled in the space.

- Default value: default
- Data type: text
- Whether it is editable: yes

5.6.3 scale_x/y

Zoom ratio in x/y direction

scale_x:	1
scale_y:	1

- Default value: 1
- Data type: float
- Whether it is editable: yes

5.6.4 anchor_x/y:

Define the anchor point x and y coordinates. Indicates the abscissa ratio in the plane rectangular coordinate system with the upper left corner of the picture as the origin.

P.S. two formats. If set to pure numbers (0.0 - -1.0), hour/minute/second images will be scaled. If set numbers with "px", it will adjust the coordinates of the hour/minute/second image by the coordinates. (0 is the upper left corner)

anchor_x:	0.5
anchor_y:	0.5

- Default value: 0.5
- Data type: float
- Whether it is editable: yes

5.6.5 rotation

The rotation angle centered on the anchor point.

A number greater than 0 indicates the angle of clockwise rotation.

A number less than 0 represents the angle of counterclockwise rotation.

rotation:	0
-----------	---

- Default value: 0
- Data type: float
- Whether it is editable: yes

5.6.6 text

The text displayed on the image.

A dark-themed UI element consisting of a label 'text:' followed by a rectangular input field.

- Default value:
- Data type: text
- Whether it is editable: yes

5.6.7 clickable

Whether to trigger the click event when clicked.

Press_down/up_value and click_value can only be used after checking.

A dark-themed UI element consisting of a label 'clickable:' followed by a small, empty square checkbox.

- Default value: false
- Data type: bool
- Whether it is editable: yes

5.6.8 selectable

Whether to set the selected state.

A dark-themed UI element consisting of a label 'selectable:' followed by a small, empty square checkbox.

- Default value: false
- Data type: bool
- Whether it is editable: yes

5.6.9 enable_press_down/press_down_value

Whether to allow the return value to be sent when the button is pressed. / Custom return value sent when the button is pressed.

If enable_press_down is not checked, the set custom return value is invalid.

A dark-themed UI element consisting of a label 'press_down_value:', a small, empty square checkbox, and a rectangular input field.

- Default value: false/
- Data type: bool/hex
- Whether it is editable: yes

5.6.10 enable_press_up/press_up_value

Whether to allow the return value to be sent when the button is released. / Custom return value sent when the button is released.

If enable_press_up is not checked, the set custom return value is invalid.

A dark-themed UI element with the label 'press_up_value:' on the left. To its right is a small square checkbox, which is currently unchecked, followed by a dark rectangular input field.

- Default value: false/
- Data type: bool/hex
- Whether it is editable: yes

5.6.11 enable_click/click_value

Whether to allow the return value to be sent when the button is clicked and released. / Custom return value sent when the button is clicked and released.

P.S. If enable_click is not checked, the set custom return value is invalid.

A dark-themed UI element with the label 'click_value:' on the left. To its right is a small square checkbox, which is currently checked, followed by a dark rectangular input field.

- Default value: true/
- Data type: bool/hex
- Whether it is editable: yes

5.7 gif

gif is used to display a moving picture in gif format, which can be added and used according to your own needs.

Special note: Due to memory-related issues, it is recommended that the resolution be less than 30*30 and the size be less than 15kb. If it exceeds the range, it is recommended to use the image_animation widget instead.

5.7.1 image

The picture displayed by the widget.

Import the resources bar on the left first.

Select from the drop-down list or enter the name of the picture file in the box to search.

A dark-themed UI element with a label 'image:' on the left, preceded by a right-pointing chevron '>'. To the right of the label is a dark rectangular box containing the text 'none' and a small downward-pointing chevron 'v' on the right side, indicating a dropdown menu.

- Default value:
- Data type: text
- Whether it is editable: yes

5.7.2 scale_x/y:

Defines the x/y zoom ratio of the gif image file.

scale_x:	1
scale_y:	1

- Default value: 1
- Data type: float
- Whether it is editable: yes

5.7.3 anchor_x/y

Define the anchor point x and y coordinates. Indicates the abscissa ratio in the plane rectangular coordinate system with the upper left corner of the picture as the origin.

P.S. two formats. If set to pure numbers (0.0 - -1.0), hour/minute/second images will be scaled. If set numbers with "px", it will adjust the coordinates of the hour/minute/second image by the coordinates. (0 is the upper left corner)

anchor_x:	0.5
anchor_y:	0.5

- Default value: 0.5
- Type of data: float
- Whether it is editable: yes

5.7.4 rotation

The rotation angle centered on the anchor point.

A number greater than 0 represents an angle of clockwise rotation, and a number less than 0 represents an angle of counterclockwise rotation.

rotation:	0
-----------	--------------------------------

- Default value: 0
- Data type: float
- Whether it is editable: yes

5.7.5 clickable

Whether to trigger a click event when clicked.

clickable:	☐
------------	--------------------------

- Default value: false
- Data type: bool
- Whether it is editable: yes

5.7.6 selectable

Whether to set the selected state.

selectable:	☐
-------------	--------------------------

- Default value: false
- Data type: bool
- Whether it is editable: yes

5.8 svg

svg is used to display pictures in svg format, and the types of widgets and pictures support vector enlargement.

Special note: Considering the characteristics of SVG images, if some abnormal problems occur after adding SVG images and downloading them to the device, please replace the images.

5.8.1 scale_x/y

Defines the x/y zoom ratio of the gif image file.

scale_x:	1
scale_y:	1

- Default value: 1
- Data type: float
- Whether it is editable: yes

5.8.2 anchor_x/y

Define the anchor point x and y coordinates. Indicates the abscissa ratio in the plane rectangular coordinate system with the upper left corner of the picture as the origin.

P.S. two formats. If set to pure numbers (0.0 - -1.0), hour/minute/second images will be scaled. If set numbers with "px", it will adjust the coordinates of the hour/minute/second image by the coordinates. (0 is the upper left corner)

anchor_x:	0.5
anchor_y:	0.5

- Default value: 0.5
- Type of data: float
- Whether it is editable: yes

5.8.3 rotation

The rotation angle centered on the anchor point.

A number greater than 0 indicates the angle of clockwise rotation. A number less than 0 represents the angle of counterclockwise rotation.

rotation:	0
-----------	--------------------------------

- Default value: 0
- Data type: float
- Whether it is editable: yes

5.8.4 text:

Default text.

text:	
-------	----------------------

- Default value:
- Data type: text
- Whether it is editable: yes

5.8.5 clickable

Whether to trigger a click event when clicked.

clickable:	☐
------------	--------------------------

Default value: false

Data type: bool

Whether it is editable: yes

5.8.6 selectable

Whether to set the selected state.

selectable:	☐
-------------	--------------------------

Default value: false

Data type: bool

Whether it is editable: yes

5.9 image value

image_value is used to display the number with the picture form. The number supports many different types. Modify the format according to your own needs.

5.9.1 image

The picture displayed by the widget. Import the resources bar on the left first.

Select from the drop-down list or enter the name of the picture file in the box to search.

image:	num
--------	-----

- Default value: num
- Data type: text
- Whether it is editable: yes

5.9.2 min/max:

The minimum/maximum numerical limit for the widget.

min:	0
max:	100

- Default value: 0/100
- Data type: uint
- Whether it is editable: yes

5.9.3 value

Default value of the widget

value:	1.23
--------	------

- Default value: 1.23
- Data type: float
- Whether it is editable: yes

5.9.4 format

Defines the data formatting type.

%d will only display the integer part, %f will display it as a floating-point number.

format:	%.2f
---------	------

- Default value: %.2f
- Data type: text
- Whether it is editable: yes

5.10 image animation

image_animation is used to display a group of pictures and play a set of photos in a loop, set the first picture and the last picture according to your own needs, and whether to play in a loop.

P.S.The naming of the images should be based on the same prefix, the number followed should be in %d format and no other characters are allowed, such as a group of images with the names image_ani_1, image_ani_2, image_ani_3 image_ani_100, then you need to fill in all the content before the number at the image, that is, image_ani_.

5.10.1 auto_play

Whether to display playback automatically.

auto_play:	☐
------------	--------------------------

- Default value: false
- Data type: bool
- Whether it is editable: yes

5.10.2 image

The name of the picture group displayed by the widget. Import the recourse bar on the left first.

Select from the drop-down list or enter the name of the picture file in the box to search.

image:	num
--------	----------------------------------

- Default value: num
- Data type: text
- Whether it is editable: yes

5.10.3 start_index

Play the first picture of the picture group in the animation, defined by the number in the picture name.

start_index:	0
--------------	--------------------------------

- Default value: 0
- Data type: uint
- Whether it is editable: yes

5.10.4 end_index

Play the last picture of the picture group in the animation, defined by the number in the picture name.

end_index:	9
------------	---

- Default value: 9
- Data type: uint
- Whether it is editable: yes

5.10.5 interval

Playing time of each picture

The unit is ms.

interval:	16
-----------	----

- Default value: 16
- Data type: uint
- Whether it is editable: yes

5.10.6 format

Determines the name format for animation image names.

format:	%s%d
---------	------

- Default value: %s%d
- Data type: text
- Whether it is editable: no

5.10.7 delay:

Length of playback delay in autoplay mode.

The unit is ms.

delay:	0
--------	---

- Default value: 0
- Data type: unit
- Whether it can be changed: yes

5.10.8 loop

Whether the animation loops repeatedly.

If the animation is not checked, it will only play once.

loop:	☒
-------	-------------------------------------

- Default value: true
- Data type: bool
- Whether it can be changed: yes

5.10.9 auto_play

Whether the animation plays automatically. If checked, it will play automatically.

auto_play:	☒
------------	-------------------------------------

- Default value: true
- Data type: bool
- Whether it can be changed: yes

5.11 video_image



video_image is a video related widget, which is used to download and display videos on the display. It supports loop playback and stops at the last frame after playback.

Special note: Considering the overall project capacity, the video size and time length need to be selected carefully by users according to the remaining flash memory capacity of the display. In addition, the video frame rate should be less than 24.

It is recommended that the resolution of imported video clip should not exceed the resolution of the display.

5.11.1 loop

Loop playback function

ps.the video playing will stop at the last frame if loop has not been activated

loop:	☒
-------	-------------------------------------

- Default value: true
- Data Type: bool
- Whether it is editable: yes

5.11.2 video_name

Name of video

video_name:	
-------------	----------------------

- Default value:
- Data Type: text
- Whether it is editable: yes

5.11.3 auto_play

Auto play function

auto_play:	☒
------------	-------------------------------------

- Default value:true
- Data Type: bool
- Whether it is editable: yes

5.11.4 frame_interval

Interval time of frames

- Default value:
- Data Type: uint
- Whether it is editable: yes

5.11.5 audio_enable

Enable Audio playing or not

- Default value:true
- Data Type: text
- Whether it is editable: yes

5.11.6 audio_type

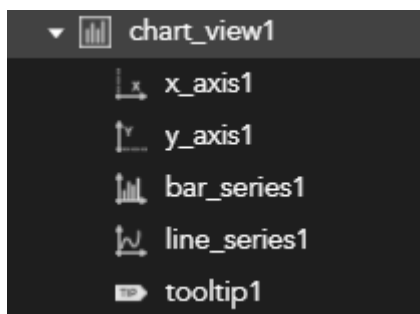
Type of audio , support *.mp3 and *.wav

- Default value:wav
- Data Type: text
- Whether it is editable: yes

5.12 chart_view

chart_view is a chart widget, which contains bar charts and line charts. Add multiple different line_series and bar_series to one widget according to your own needs. Make charts and so on according to your own needs.

The inclusion relationship of a chart_view is as follows:



5.12.1 top_series

The series index at the top level.

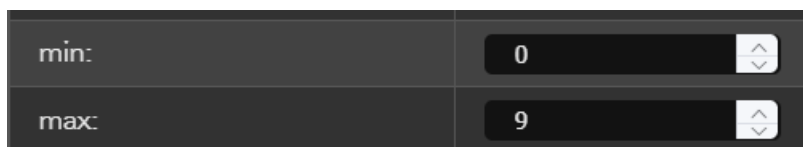


- Default value: -1
- Data type: uint
- Whether is it editable: Yes

5.12.2 x_axis

5.12.2.1 min/max

The minimum and maximum value of the widget range.



- Default value: 0/9
- Data type: uint
- Whether it is editable: Yes

5.12.2.2 axis_type

The type of axis.



- Default value: value
- Data type: text
- Whether it is editable: Yes

5.12.2.3 at

The position of the coordinate axis.

at:	auto
-----	------

- Default value: auto
- Data type: text
- Whether it is editable: Yes

5.12.2.4 data

Displayed tick value.

data:	[1,2,3,4,5,6,7,8,9,10]
-------	------------------------

- Default value: [1,2,3,4,5,6,7,8,9,10]
- Data type: text
- Whether it is editable: Yes

5.12.2.5 split_line

Parameters for dividing lines.

split_line:	{show:true}
-------------	-------------

- Default value: {show:true}
- Data type: bool
- Whether it is editable: Yes

5.12.2.6 tick

Parameters for tick marks.

tick:	{show:true}
-------	-------------

- Default value: {show:true}
- Data type: bool
- Whether it is editable: Yes

5.12.2.7 line

Parameters of the axis.

line:	{show:true}
-------	-------------

- Default value: {show:true}
- Data type: bool
- Whether it is editable: Yes

5.12.2.8 label

The parameter for the tick value.

label:	{show:true}
--------	--

- Default value: {show:true}
- Data type: bool
- Whether it is editable: Yes

5.12.2.9 inverse

Whether to display in reverse.

inverse:	☐
----------	--------------------------

- Default value: false
- Data type: bool
- Whether it is editable: Yes

5.12.2.10 offset

Offset from the starting position.

offset:	0
---------	--------------------------------

- Default value: 0
- Data type: uint
- Whether it is editable: Yes

5.12.2.11 offset_percent

Whether the offset is a percentage.

P.S. only the value filled in offset will be valid.

offset_percent:	☐
-----------------	--------------------------

- Default value: false
- Data type: bool
- Whether it is editable: Yes

5.12.3 y_axis

5.12.3.1 min/max

The minimum and maximum value of the widget range.

min:	0
max:	140

- Default value: 0/140
- Data type: uint
- Whether it is editable: Yes

5.12.3.2 axis_type

The type of axis.

axis_type:	category
------------	----------

- Default value: value
- Data type: text
- Whether it is editable: Yes

5.12.3.3 at

The position of the coordinate axis.

at:	auto
-----	------

- Default value: auto
- Data type: text
- Whether it is editable: Yes

5.12.3.4 data

Displayed tick value.

data:	[0,20,40,60,80,100,120]
-------	-------------------------

- Default value: [0,20,40,60,80,100,120,140]
- Data type: text
- Whether it is editable: Yes

5.12.3.5 split_line

Parameters for dividing lines.

split_line:	{show:true} ▼
-------------	---------------

- Default value: {show:true}
- Data type: bool
- Whether it is editable: Yes

5.12.3.6 tick

Parameters for tick marks.

tick:	{show:true} ▼
-------	---------------

- Default value: {show:true}
- Data type: bool
- Whether it is editable: Yes

5.12.3.7 line

Parameters of the axis.

line:	{show:true} ▼
-------	---------------

- Default value: {show:true}
- Data type: bool
- Whether it is editable: Yes

5.12.3.8 label

The parameter for the tick value.

label:	{show:true} ▼
--------	---------------

- Default value: {show:true}
- Data type: bool
- Whether it is editable: Yes

5.12.3.9 inverse

Whether to display in reverse.

inverse:	☐
----------	--------------------------

- Default value: false
- Data type: bool
- Whether it is editable: Yes

5.12.3.10 offset

Offset from the starting position.

offset:	0
---------	--------------------------------

- Default value: 0
- Data type: uint
- Whether it is editable: Yes

5.12.3.11 offset_percent

Whether the offset is a percentage.

P.S. only the value filled in offset will be valid.

offset_percent:	☐
-----------------	--------------------------

- Default value: false
- Data type: bool
- Whether it is editable: Yes

5.12.4 bar_series

5.12.4.1 value

value:	15,75,40,60,140,80,100,120,25,90
--------	---

- Default value: 15,75,40,60,140,80,100,120,25,90
- Data type: text
- Whether it is editable: Yes

5.12.4.2 series_axis

Axis of sequence bits, default to retrieve the first one when empty.

series_axis:	
--------------	----------------------

- Default value:
- Data type: uint
- Whether it is editable: yes

5.12.4.3 value_axis

Axis of series values, defaults to retrieve the first one when empty.

value_axis:	
-------------	----------------------

- Default value:
- Data type: uint
- Whether it is editable: yes

5.12.4.4 bar

Parameters for the bar.

bar:	
------	----------------------

- Default value:
- Data type: text
- Whether it is editable: yes

5.12.4.5 capacity

The capacity of the sequence fifo.

capacity:	10
-----------	---------------------------------

- Default value: 10
- Data type: uint
- Whether it is editable: yes

5.12.4.6 offset

The offset of the sequence fifo relative to the end.

offset:	0
---------	--------------------------------

- Default value: 0
- Data type: uint
- Whether it is editable: yes

5.12.4.7 display_mode

Display mode.

display_mode:	push
---------------	-----------------------------------

- Default value: push
- Data type: text
- Whether it is editable: yes

5.12.4.8 value_animation

The duration of the animation.

The unit is ms.

value_animation:	500
------------------	----------------------------------

- Default: 500
- Data type: uint

- Whether it is editable: yes

5.12.5 line_series

5.12.5.1 area_color

The sequence curve and coordinates are the background color of the area.

area_color:	
-------------	---

- Default value: rgba(51, 143, 255, 0.33)
- Data type: hex
- Whether it is editable: Yes

5.12.5.2 line_border_color

The color of the polyline.

line_border_color:	
--------------------	---

- Default: rgba(51, 143, 255, 1)
- Data type: hex
- Whether it is editable: Yes

5.12.5.3 symbol_bg_color

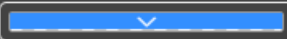
The background color of the polyline points.

symbol_bg_color:	
------------------	---

- Default value: rgba(255, 255, 255, 1)
- Data type: hex
- Whether it is editable: Yes

5.12.5.4 symbol_border_color

Border color for polyline points.

symbol_border_color:	
----------------------	---

- Default: rgba(51, 143, 255, 1)
- Data type: hex
- Whether it is editable: Yes

5.12.5.5 line_border_width

The width of the polyline.

line_border_width:	1 
--------------------	--

- Default value: 1
- Data type: uint
- Whether it is editable: yes

5.12.5.6 symbol_round_radius

The arc of the polyline point.

When it is 0, it is a square.

symbol_round_radius:	4
----------------------	---

- Default value: 4
- Data type: uint
- Whether it is editable: yes

5.12.5.7 value

value:	15,75,40,60,140,80,100,120,25,90
--------	----------------------------------

- Default value: 15,75,40,60,140,80,100,120,25,90
- Data type: text
- Whether it is editable: Yes

5.12.5.8 series_axis

Axis of sequence bits, default to retrieve the first one when empty.

series_axis:	
--------------	--

- Default value:
- Data type: uint
- Whether it is editable: yes

5.12.5.9 value_axis

Axis of series values, defaults to retrieve the first one when empty.

value_axis:	
-------------	--

- Default value:
- Data type: uint
- Whether it is editable: yes

5.12.5.10 line

The parameters of the sequence curve.

line:	{smooth:true}
-------	---------------

- Default value: {smooth:true}
- Data type: bool
- Whether it is editable: yes

5.12.5.11 area

The sequence curve and coordinates are the parameters of the area.

area:	{show:true} ▾
-------	--------------------------

- Default value: {show:true}
- Data type: bool

5.12.5.12 symbol

Sequence point parameters.

symbol:	{show:true} ▾
---------	--------------------------

- Default value:
- Data type: bool
- Whether it is editable: yes

5.12.5.13 capacity

The capacity of the sequence fifo.

capacity:	10 ▴ ▾
-----------	-------------------

- Default value: 10
- Data type: uint
- Whether it is editable: yes

5.12.5.14 offset

The offset of the sequence fifo relative to the end.

offset:	0 ▴ ▾
---------	------------------

- Default value: 0
- Data type: uint
- Whether it is editable: yes

5.12.5.15 display_mode

Display mode.

display_mode:	push ▾
---------------	-------------------

- Default value: push
- Data type: text
- Whether it is editable: yes

5.12.5.16 value_animation

The duration of the animation.

The unit is ms.

value_animation:	500
------------------	-----

- Default: 500
- Data type: uint
- Whether it is editable: yes

5.12.6 tooltip

5.12.6.1 symbol_bg_color

The background color of the polyline points.

symbol_bg_color:	
------------------	----------------------

- Default value: rgba(255, 255, 255, 1)
- Data type: hex
- Whether it is editable: Yes

5.12.6.2 symbol_border_color

Border color for polyline points.

symbol_border_color:	
----------------------	----------------------

- Default: rgba(51, 143, 255, 1)
- Data type: hex
- Whether it is editable: Yes

5.12.6.3 line_border_width

The width of the polyline.

line_border_width:	1
--------------------	---

- Default value: 1
- Data type: uint
- Whether it is editable: yes

5.12.6.4 symbol_round_radius

The arc of the polyline point.

When it is 0, it is a square.

symbol_round_radius:	4
----------------------	---

- Default value: 4
- Data type: uint
- Whether it is editable: yes

5.12.6.5 line

The parameters of the sequence curve.

line:	
-------	--

- Default value:
- Data type: bool
- Whether it is editable: yes

5.12.6.6 symbol

Sequence point parameters.

symbol:	
---------	--

- Default value:
- Data type: bool
- Whether it is editable: yes

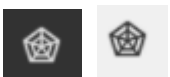
5.12.6.7 tip

The parameter for the prompt text.

tip:	
------	--

- Default value:
- Data type: bool
- Whether it is editable: yes

5.13 radar chart



radar_chart is a widget for applying radar chart visual effect based on user's customizations. It is a multidimensional data visualization tool for displaying the values of various variables by drawing multiple axes. It is suitable for comparing the performance and trends of different parameters in various aspects.

5.13.1 area_color

The background color of the area enclosed by the curves and coordinates.

area_color:	<input type="color" value="#3182ce"/>
-------------	---------------------------------------

- Default value: rgba(51, 143, 255, 0.33)
- Data Type: hex
- Whether it is editable : yes

5.13.2 line_border_color

Color of border line

line_border_color:	<input type="color" value="#3182ce"/>
--------------------	---------------------------------------

- Default value: rgba(51, 143, 255, 1)
- Data Type: hex
- Whether it is editable: yes

5.13.3 symbol_bg_color

Background color of symbol

symbol_bg_color:	<input type="color" value="white"/>
------------------	-------------------------------------

- Default value: rgba(255, 255, 255, 1)
- Data Type: hex
- Whether it is editable : yes

5.13.4 symbol_border_color

Color of symbol border

symbol_border_color:	<input type="color" value="#3182ce"/>
----------------------	---------------------------------------

- Default value: rgba(51, 143, 255, 1)
- Data Type: hex
- Whether it is editable : yes

5.13.5 line_border_width

Width of line border

line_border_width:	1
--------------------	---

- Default value: 1
- Data Type:uint
- Whether it is editable: yes

5.13.6 symbol_round_radius

Round radius of symbol

symbol_round_radius:	4
----------------------	---

- Default value: 4
- Data Type:uint
- Whether it is editable: yes

5.13.7 min/max:

Min and max range value

min:	0
max:	100

- Default value: 0/100
- Data Type:float
- Whether it is editable: yes

5.13.8 origin_offset

Widget rotation angle offset

origin_offset:	0
----------------	---

- Default value: 0
- Data Type:float
- Whether it is editable: yes

5.13.9 point_num

Total number of parameters (points)

point_num:	8
------------	---

- Default value: 8
- Data Type:uint
- Whether it is editable: yes

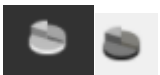
5.13.10 data

Value of each parameter

data:	[1,2,3,4,5,6,7,8,9,10]
-------	------------------------

- Default value: [1,2,3,4,5,6,7,8,9,10]
- Data Type:uint
- Whether it is editable: yes

5.14 pie_slice



The pie_slice widget present pie slice visual effect based on user's customizations, it is suitable to represent the proportion of a certain category.

5.14.1 max

Max value of widget

max:	360
------	-----

- Default value:360
- Data Type:uint
- Whether it is editable: yes

5.14.2 value

Current value of widget

value:	180
--------	-----

- Default value:180
- Data Type:uint
- Whether it is editable: yes

5.14.3 start_angle

Starting angle

start_angle:	0
inner_radius:	8

- Default value:0
- Data Type:uint
- Whether it is editable: yes

5.14.4 inner_radius

Inner radius

inner_radius:	8
---------------	---

- Default value:8
- Data Type:uint
- Whether it is editable: yes

5.14.5 uint

Unit format

unit:	
-------	--

- Default value:
- Data Type:text
- Whether it is editable: yes

5.14.6 counter_clock_wise

Clock wise or anti clock wise direction

counter_clock_wise:	☐
---------------------	--------------------------

- Default value:false
- Data Type:bool
- Whether it is editable: yes

5.14.7 show_text

Displaying text

show_text:	☐
------------	--------------------------

- Default value:false
- Data Type:bool
- Whether it is editable: yes

5.14.8 is_exploded

Shape expanding

is_exploded:	☒
--------------	-------------------------------------

- Default value:true
- Data Type:bool
- Whether it is editable: yes

5.14.9 pressed

Pressed mouse button

pressed:	☒
----------	-------------------------------------

- Default value:true
- Data Type:bool
- Whether it is editable: yes

5.14.10 is_semicircle

Generating arch shape

is_semicircle:	☒
----------------	-------------------------------------

- Default value:true
- Data Type:bool
- Whether it is editable: yes

5.14.11 x_to/y_to

Extension of X and Y coordinates

x_to:	0
y_to:	0

- Default value:0/0
- Data Type:uint
- Whether it is editable: yes

5.15 qr



QR widget is designed to generate and displaying QR code images, which user may convert text content or data to QR code for simple scanning.

5.15.1 value

The value of the QR code supports variety of text types for displaying. The input text can be converted into a QR code for direct displaying.

auto_fix:	☒
-----------	-------------------------------------

- Default value:
- Data Type:text
- Whether it is editable: yes

5.16 combo_box_ex

combo_box_ex is a drop-down box option used to select one of several different options. For example, when you need to enter personal information you need to select Mr. Ms., then the combo_box_ex can be used at this time.

5.16.1 auto_fix:

Whether the widget automatically corrects errors. Once the value changes, the system will automatically fix the content type to prevent errors.

auto_fix:	☒
-----------	-------------------------------------

- Default value: true
- Data type: bool
- Whether it is editable: yes

5.16.2 input_type

The format type of the input text.

text: input data will be displayed as ascii.

int: input data will be displayed as int type value.

float: the input data will be displayed as a float type value.

The password input data will be displayed as ascii. (Set whether the password is visible or not with password_visible.)

date: the input data will be displayed in "yyyy/mm/dd" format (data is ascii).

time: input data is displayed in "hh:mm" format (data is ascii).

input_type:	text ▼
-------------	--------

- Default value:
- Data type: text
- Whether it is editable: yes

5.16.3 min/max:

The minimum/maximum numerical limit for the widget.

min:	0
max:	100

- Default value: 0/100
- Data type: float
- Whether it is editable: yes

5.16.4 tips:

The prompt text to display when the widget text is empty.

P.S. only valid when input_type is text.

tips:	Please enter your ID
-------	----------------------

- Default value: please enter your text
- Data type: text
- Whether it is editable: yes

5.16.5 text

The default text displayed by the widget.

P.S. it needs to correspond to the input_type set earlier. .

> text:	Combo Box Ex
---------	--------------

- Default value: combo box ex
- Data type: text
- Whether it is editable: yes

5.16.6 options:

The text content of each option in the text picker.

Fixed format: "1:xxx1;2:yyy;3:zzz;...."

options:	
----------	--

- Default value:
- **Data type: text**
- Whether it is editable: yes

5.16.7 selected_index:

The initial default display option index number.

P.S. option index numbers are assigned by the system, starting from 0.

selected_index:	0
-----------------	--------------------------------

- Default value: 0
- Data type: unit
- Whether it is editable: yes

5.16.8 item_height:

The height of the dropdown option.

item_height:	30
--------------	---------------------------------

Unit: pixel

- Default value: 30
- Data type: unit
- Whether it is editable: yes

5.16.9 action_text:

Determines the action for entering the keyboard.

action_text:	done
--------------	-----------------------------------

- Default value: done
- Data type: text
- Whether it is editable: yes

5.16.10 readonly:

Whether it is read-only.

If checked, the keyboard input function is unavailable, if not checked, it is available.

readonly:	☐
-----------	--------------------------

- Default value: false
- Data type: bool
- Whether it is editable: yes

5.16.11 password_visible:

When the input type is set to "password", whether the content is displayed as "*" when it is not checked, and the input information is displayed when it is checked.

password_visible:	☐
-------------------	--------------------------

- Default value: false
- Data type: bool
- Whether it is editable: yes

5.16.12 open_kb_when_focused/close_kb_when_blured

Turn on the input method when the focus is obtained/close the input method when the focus is lost.

open_kb_when_focused:	☒
close_kb_when_blured:	☒

- Default value: true/true
- Data type: bool
- Whether it is editable: yes

5.17 spin_box

spin_box is mostly used to manually select numbers. For example, control the size of the final value by clicking the up and down buttons on the right side of the widget.

5.17.1 auto_fix:

Whether the widget automatically corrects errors. Once the value changes, the system will automatically fix the content type to prevent errors.

auto_fix:	☒
-----------	-------------------------------------

- Default value: true
- Data type: bool
- Whether it is editable: yes

5.17.2 input_type

The format type of the input text.

text: input data will be displayed as ascii.

int: input data will be displayed as int type value.

float: the input data will be displayed as a float type value.

The password input data will be displayed as ascii. (Set whether the password is visible or not with password_visible.)

date: the input data will be displayed in "yyyy/mm/dd" format (data is ascll).

time: input data is displayed in "hh:mm" format (data is ascll).

input_type:	text
-------------	------

- Default value:
- Data type: text
- Whether it is editable: yes

5.17.3 min/max:

The minimum/maximum numerical limit for the widget.

min:	0
max:	100

- Default value: 0/100
- Data type: float
- Whether it is editable: yes

5.17.4 tips:

The prompt text to display when the widget text is empty.

P.S. only valid when input_type is text.

tips:	Please enter your ID
-------	----------------------

- Default value: please enter your text
- Data type: text
- Whether it is editable: yes

5.17.5 text

The default text displayed by the widget.

P.S. it needs to correspond to the input_type set earlier. .

> text:	Combo Box Ex
---------	--------------

- Default value: combo box ex
- Data type: text
- Whether it is editable: yes

5.17.6 options:

The text content of each option in the text picker.

Fixed format: "1:xxx1;2:yyy;3:zzz;...."

options:	
----------	--

- Default value:
- **Data type: text**
- Whether it is editable: yes

5.17.7 selected_index:

The initial default display option index number.

Option index numbers are assigned by the system, starting from 0.

selected_index:	0	↑ ↓
-----------------	---	--------

- Default value: 0
- Data type: unit
- Whether it is editable: yes

5.17.8 item_height:

The height of the dropdown option.

Unit: pixel.

item_height:	30	↑ ↓
--------------	----	--------

- Default value: 30
- Data type: unit
- Whether it is editable: yes

5.17.9 action_text

Determines the action for entering the keyboard.

action_text:	done
--------------	------

- Default value: done
- Data type: text
- Whether it is editable: yes

5.17.10 readonly

Whether it is read-only.

If checked, the keyboard input function is unavailable, if not checked, it is available.

readonly:	☐
-----------	--------------------------

- Default value: false
- Data type: bool
- Whether it is editable: yes

5.17.11 password_visible

When the input type is set to "password", whether the content is displayed as "*" when it is not checked, and the input information is displayed when it is checked.

password_visible:	☐
-------------------	--------------------------

- Default value: false
- Data type: bool
- Whether it is editable: yes

5.17.12 open_kb_when_focused/close_kb_when_blured

Turn on the input method when the focus is obtained/close the input method when the focus is lost.

open_kb_when_focused:	☒
close_kb_when_blured:	☒

- Default value: true/true
- Data type: bool
- Whether it is editable: yes

5.18 text selector

text_selector widget is used for the selection of multiple different options. For example, the effect of a password lock can be achieved with text_selector.

5.18.1 selected_index

Define the option index number for the initial highlighting.

The index number is not the same as the number. The number is the number that precedes the text content.

selected_index:	6
-----------------	--------------------------------

- Default value: 0
- Data type: unit
- Whether it is editable: yes

5.18.2 visible_num

The number of options visible on the same page.

visible_num:	5
--------------	---

- Default value: 3
- Data type: unit
- Whether it is editable: yes

5.18.3 y_speed_scale:

Define the sliding speed.

The unit is $x \times \text{default speed}$.

y_speed_scale:	1
----------------	---

- Default value: 1
- Data type: unit
- Whether it is editable: yes

5.18.4 localize_options

Whether to localize translation.

localize_options:	☒
-------------------	-------------------------------------

- Default value: false
- Data type: unit
- Whether it is editable: yes

5.18.5 loop_options

Whether to cycle through all options.

loop_options:	☐
---------------	--------------------------

- Default value: false
- Data type: unit
- Whether it is editable: yes

5.19 slide_menu

slide_menu is a sliding view, which can be turned left and right to select the page to be displayed.

5.19.1 min_scale

Minimum zoom ratio.

min_scale:	0.8	
------------	-----	---

- Default value: 0.8
- Data type: float
- Whether it is editable: yes

5.19.2 value

The value of slide_menu.

value:	1	
--------	---	---

- Default value: 1
- Data type: uint
- Whether it is editable: yes

5.19.3 align_v

Vertical alignment.

align_v:	bottom	
----------	--------	---

- Default value: bottom
- Data type: text
- Whether it is editable: yes

5.19.4 x/y_offset

Offset in x/y direction.

x_offset:	0	
y_offset:	0	

- Default value: 0/0
- Data type: uint
- Whether it is editable: yes

5.20 label

The label is a plain text box. The difference from edit is that the label cannot be entered on the HMI screen manually, and the value can only be modified through instructions. Mostly used for fixed prompt text description.

* When designing widgets, be careful not to make the font size larger than the length or width of the widget.

5.20.1 length

The number of characters that can be displayed.

P.S.-1 is to display all characters.

length:	-1
---------	---------------------------------

- Default value: -1
- Data type: int
- Whether it is editable: yes

5.20.2 line_wrap

Whether to wrap lines automatically. Need to be used with word_wrap.

line_wrap:	☐
------------	--------------------------

- Default value: false
- Data type: bool
- Whether it is editable: yes

5.20.3 word_wrap

Whether to wrap lines automatically. Need to be used with line_wrap.

word_wrap:	☐
------------	--------------------------

- Default value: false
- Data type: bool
- Whether it is editable: yes

5.21 hscroll label

hscroll_label is a scrolling text box, which can be set to determine whether the widget is looped, rolled back, etc. It can also be set to play a time, which can be set according to your own needs.

5.21.1 loop:

The text content will start scrolling automatically. If unchecked, the text will only scroll once automatically.

loop:	☐
-------	--------------------------

- Default value: false
- Data type: bool
- Whether it is editable: yes

5.21.2 yoyo:

Whether the text content will be scrolled back after scrolling ends.

yoyo:	☐
-------	--------------------------

- Default value: false
- Data type: bool
- Whether it is editable: yes

5.21.3 duration

The time required to complete a scroll.

The unit is ms.

duration:	5000	↑ ↓
-----------	------	--

- Default value: 5000
- Data type: uint
- Whether it is editable: yes

5.21.4 lull

The interval between each scroll.

The unit is ms.

lull:	500	↑ ↓
-------	-----	--

- Default value: 500
- Data type: uint
- Whether it is editable: yes

5.21.5 text

Scrollbar realistic text content.

> text:	HScroll Label
---------	---------------

- Default value: hscrolltexthscrolltexthscrolltext
- Data type: text
- Whether it is editable: yes

5.22 edit

edit widget is an inputable text box, and the `input_type` can be used to determine the content that can be entered in the widget.

Note: 1. For the specific input type of `input_type`, see [6.6.5 input_type](#).

2. After the content of the input box is determined, only the content of a specific format can be input.

5.22.1 auto_fix

Determine whether to automatically correct errors. Once the value changes, the system will automatically fix the content type to prevent errors.

auto_fix:	☒
-----------	-------------------------------------

- Default value: true
- Data type: bool
- Whether it is editable: yes

5.22.2 input_type:

The format type of the widget input.

text: input data will be displayed as ascii. int: input data will be displayed as int type value. float: the input data will be displayed as a float type value.

The password input data will be displayed as ascii. (Set whether the password is visible or not with `password_visible`.) date: the input data will be displayed in "yyyy/mm/dd" format (data is ascii). time: input data is displayed in "hh:mm" format (data is ascii).

input_type:	text ▼
-------------	--------

- Default value: text
- Data type: text
- Whether it is editable: yes

5.22.3 min/max:

The minimum/maximum numerical limit of the corresponding widget.

min:	0
max:	100

- Default value: 0/100
- Data type: float
- Whether it is editable: yes

5.22.4 tips:

The prompt text to display when text is empty.

Only valid when `input_type` is text.

tips:

Please enter your text

- Default value: please enter your text
- Data type: text
- Whether it is editable: yes

5.22.5 text

Default text.

It needs to correspond to the input_type set earlier. .

> text:

- Default value:
- **Data type: text**
- Whether it is editable: yes

5.22.6 action_text:

Enter the action of the keyboard.

action_text:

done

- Default value: done
- Data type: text
- Whether it is editable: yes

5.22.7 readonly

Whether it is read-only, that is, whether the keyboard input function is available.

readonly:

☐

- Default value: false
- Data type: bool
- Whether it is editable: yes

5.22.8 password_visible

When the input type is set to "password", whether the content is displayed as "*" when it is not checked, and the input information is displayed when it is checked.

password_visible:

☐

- Default value: false
- Data type: bool
- Whether it is editable: yes

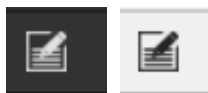
5.22.9 close_kb_when_blured

Closes the input method when the focus is lost.

close_kb_when_blured:	☒
-----------------------	-------------------------------------

- Default value: true
- Data type: bool
- Whether it is editable: yes

5.23 mledit



mledit is a multi-line text displaying widget, which means that it can display countless lines of text in the same widget. Users may enter needed content in one line according to their needs.

Special note: single line wrapping displaying effect determined by the size of the current widget, but all “lines” are literally at the same single line(Number) if no line breaks been entered. You may also refer to the guide video demonstration.

5.23.1 tips

Hint tip information

tips:	
-------	----------------------

- Default value: true
- Data Type: text
- Whether it is editable: Yes

5.23.2 keyboard

Name of customized keyboard

keyboard:	kb_default ▼
-----------	--------------

- Default value: kb_default
- Data Type: text
- Whether it is editable: Yes

5.23.3 max_lines

Maximum line numbers

max_lines:	100
------------	--------------------------

- Default value: 100
- Data Type: uint
- Whether it is editable: Yes

5.23.4 wrap_word

Auto line wrapping

wrap_word:	☒
------------	-------------------------------------

- Default value: true
- Data Type: bool
- Whether it is editable: Yes

5.23.5 scroll_line

Scroll line numbers by single time mouse action

scroll_line:	100	↑↓
--------------	-----	-----------------------------------

- Default value: 100
- Data Type: uint
- Whether it is editable: Yes

5.23.6 cancelable

Whether to support undo editing. If TRUE, all changes can be undone before losing focus point (restoring the content before gaining focus).

Special notes:

- 1) Generally used with the cancel button of the keyboard.
- 2) When TRUE, if the content changes, the editor state will be set to “changed”, so the style of “changed” state need to be configured accordingly.

cancelable:	☐
-------------	--------------------------

- Default value: false
- Data Type: bool
- Whether it is editable: Yes

5.23.7 close_im_when_blured

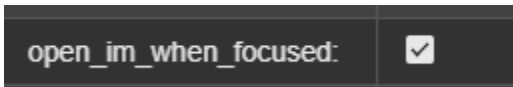
Closing input method when losing focus point

close_im_when_blured:	☒
-----------------------	-------------------------------------

- Default value: true
- Data Type: bool
- Whether it is editable: Yes

5.23.8 open_im_when_focused

Opening input method when acquiring focus point



- Default value: true
- Data Type: bool
- Whether it is editable: Yes

5.24 slider

slider widget is mostly used to adjust the volume, brightness, etc., that can be precisely controlled by the value of the slider widget.

Unit: uint.

5.24.1 bar_size

The width/height of the widget axis. Displayed vertically centered.

Unit: pixel.

- (1) There is no upper limit on the width/height, but when it exceeds the width/height of the selected area, it will not be displayed completely vertically.
- (2) When the widget size is 0, the width/height of the toolbar will automatically adjust to half the width/height of the display area.



- Default value: 0
- Data type: unit
- Whether it is editable: yes

5.24.2 dragger_size

The width/height of the widget slider.



- Default value: 14
- Data type: unit

- Whether it is editable: yes

5.24.3 slide_with_bar

The sliding function is realized by touching the corresponding slider.

If it is not checked, the slide function can be sliding by touching anywhere on the screen.

slide_with_bar:	☒
-----------------	-------------------------------------

- Default value: true
- Data type: bool
- Whether it is editable: yes

5.24.4 min/max:

The minimum/maximum numerical limit for the widget.

min:	0	
max:	100	

- Default value: 0/100
- Data type: float
- Whether it is editable: Yes

5.24.5 step

The minimum amount of change each time the widget is triggered to adjust the value.

step:	1	
-------	---	----------------------

- Default value: 1
- Data type: float
- Whether it is editable: yes

5.24.6 value

The default value of the widget.

value:	40	
--------	----	----------------------

- Default value: 40
- Data type: float
- Whether it is editable: yes

5.25 slider_circle



Slider_circle is a circular slidable widget. It can be slid on the circular track for adjusting values, volume or brightness.

5.25.1 min/max:

Minium and maximum value range of widget

min:	0
max:	100

- Default value: 0/100
- Data Type:float
- Whether it is editable : yes

5.25.2 step

Step value

step:	1
-------	---

- Default value: 1
- Data Type: float
- Whether it is editable: yes

5.25.3 start/end_angle

Starting and ending angle of widget

start_angle:	30
end_angle:	330

- Default value: 30/330
- Data Type: float
- Whether it is editable: yes

5.25.4 bg_line_width

Width of background line

bg_line_width:	8
----------------	---

- Default value: 8
- Data Type: float
- Whether it is editable: yes

5.25.5 fg_line_width

Width of foreground line

fg_line_width:	8
----------------	---

- Default value: 8
- Data Type: float

- Whether it is editable: yes

5.25.6 header_size

Radius size of the widget

header_size:	10
--------------	----

- Default value: 10
- Data Type: float
- Whether it is editable: yes

5.25.7 dragger_size

Radius size of sliding controller

dragger_size:	10
---------------	----

- Default value: 10
- Data Type: float
- Whether it is editable: yes

5.25.8 header_color

Color of sliding controller

dragger_color:	
----------------	--

- Default value: #000000
- Data Type: hex
- Whether it is editable: yes

5.25.9 unit

Unit format

unit:	
-------	--

- Default value:
- Data Type: text
- Whether it is editable: yes

5.25.10 line_cap

Shape of tip

line_cap:	round
-----------	-------

- Default value: round
- Data Type: text
- Whether it is editable: yes

5.25.11 counter_clock_wise

Clock wise or anti clock wise direction

counter_clock_wise:	☒
---------------------	-------------------------------------

- Default value: true
- Data Type: bool
- Whether it is editable: yes

5.25.12 show_text

Showing text content

show_text:	☐
------------	--------------------------

- Default value: false
- Data Type: bool
- Whether it is editable: yes

5.26 progress circle

progress_circle widget is a circular bar that can be used to display percentages, specific values, etc. For example, when loading progress, from 0 to 100% can be achieved with progress_circle.

It can also be used to set the addition and subtraction of the value by clicking the button after being bundled with the button.

5.26.1 start angle

The starting angle of the zero point. The starting point of the x-axis is 0 degrees.

P.S. the angle is calculated clockwise.

start_angle:	-90
--------------	----------------------------------

- Default value: -90
- Data type: float
- Whether it is editable: yes

5.26.2 line width

The thickness of the loop wire.

Unit: pixel.

line_width:	7
-------------	--------------------------------

- Default value: 8
- Data type: unit
- Whether it is editable: yes

5.26.3 unit

unit:	
-------	--

- Default value:
- Data type: text
- Whether it is editable: yes

5.26.4 line_cap

The shape of the line cap is rounded/square.

line_cap:	round ▼
-----------	---------

- Default value: round
- Data type: text
- Whether it is editable: yes

5.26.5 counter_clock_wise

Whether the direction of the progress bar is clockwise.

counter_clock_wise:	☐
---------------------	--------------------------

- Default value: false
- Data type: bool
- Whether it is editable: yes

5.26.6 show_text

Whether to display specific progress text.

show_text:	☒
------------	-------------------------------------

- Default value: true
- Data type: bool
- Whether it is editable: yes

5.26.7 max

The maximum value the widget can achieve.

P.S. must be greater than or equal to value.

max:	100 ▲▼
------	--------

- Default value: 100
- Data type: float
- Whether it is editable: yes

5.26.8 value

The default value of the widget.

value:	65
--------	----

- Default value: 40
- Data type: float
- Whether it is editable: yes

5.27 progress bar

progress_bar is a bar-shaped progress bar, which is mostly used to show the percentage of current progress to total progress.. For example, the progress of the installation process can be realized through the progress bar of this progress_bar.

5.27.1 vertical

Whether the progress bar is displayed vertically.

vertical:	☐
-----------	--------------------------

- Default value: false
- Data type: bool
- Whether it is editable: yes

5.27.2 show text

Displays the current value.

show_text:	☒
------------	-------------------------------------

- Default value: true (checked)
- Data type: bool
- Whether it is editable: yes

5.27.3 reverse

Whether the widgets are displayed in reverse, that is, from right to left.

reverse:	☒
----------	-------------------------------------

- Default value: false
- Data type: bool
- Whether it is editable: yes

5.27.4 max

The maximum value of the widget.

P.S. needs to be greater than or equal to value.

max:	100
------	-----

- Default value: 100
- Data type: float
- Whether it is editable: yes

5.27.5 value

The default value of the widget.

value:	40
--------	----

- Default value: 40
- Data type: float
- Whether it is editable: yes

5.28 🕒 gauge

gague is a dial, which is mostly used for dashboards in many different scenarios, such as cars, fuel tanks, etc.

5.28.1 image:

The picture displayed by the widget. Import the recourse bar on the left first.

Select from the drop-down list or enter the name of the picture file in the box to search.

image:	guage_bg
--------	----------

- Default value: gauge_bg
- Data type: text
- Whether it is editable: yes

5.28.2 draw_type:

draw_type:	center
------------	--------

- Default value: center
- Data type: text
- Whether it is editable: yes

5.29 🔧 gauge pointer

gague_pointer is usually used with gague and exists as a pointer.

5.29.1 image

The picture displayed by the widget. Import the resource bar on the left first.

Select from the drop-down list or enter the name of the picture file in the box to search.

image:	guage_pointer ▼
--------	-----------------

- **Default value:** guage_pointer
- **Data type:** text
- Whether it is editable: yes

5.29.2 anchor_x/y:

Define the anchor point x/y.

P.S. two formats. If set to pure numbers (0.0 - -1.0), hour/minute/second images will be scaled. If set numbers with "px", it will adjust the coordinates of the hour/minute/second image by the coordinates. (0 is the upper left corner)

anchor_x:	0.5
anchor_y:	0.5

- Default value: 0.5
- **Data type:** float
- Whether it is editable: yes

5.29.3 angle:

The starting angle of the widget.

P.S. Calculate the angle clockwise.

angel:	0
--------	---

- Default value: 0
- Data type: float
- Whether it is editable: yes

5.30 🕒 time clock

time_clock is used to display the dial clock, and set different dials and hours, minutes and seconds according to your needs.

5.30.1 image

The picture displayed by the widget. Import the recourse bar on the left first.

Select from the drop-down list or enter the name of the picture file in the box to search.

center_image:	none ▼
---------------	--------

- Default value:
- Data type: text
- Whether it is editable: yes

5.30.2 dial_image

Defines the widget's background image by name. If the image is not defined, the background will appear transparent. The background image should be uploaded first in the resources list on the left.

dial_image:	clock_bg ▼
-------------	------------

- Default value: clock_bg
- Type of data: text
- Whether it is editable: Yes

5.30.3 hour/minute/second_image:

Defines the icon image for the hour/minute/second hand. The image must be uploaded to the resource list before it can be used.

P.S. cannot be set to blank.

hour_image:	clock_hour ▼
minute_image:	clock_minute ▼
second_image:	clock_second ▼

- Default value: clock_hour/clock_minute/clock_second
- Data type: text
- Whether it is editable: yes

5.30.4 hour/minute/second_anchor_x|y:

Defines the anchor point for the hour/minute/second hand.

P.S. two formats. If set to pure numbers (0.0 - -1.0), hour/minute/second images will be scaled. If set numbers with "px", it will adjust the coordinates of the hour/minute/second image by the coordinates. (0 is the upper left corner)

hour_image:	clock_hour
hour_anchor_x:	0.5
hour_anchor_y:	0.5
minute_image:	clock_minute
minute_anchor_x:	0.5
minute_anchor_y:	0.5
second_image:	clock_second
second_anchor_x:	0.5
second_anchor_y:	0.5

- Default value: 0.5
- Data type: float
- Whether it is editable: yes

5.31 digit clock

digit_clock is used to display time, date, etc. For the specific format, see 4.20.1format.

5.31.1 format:

The display format of the widget.

Format:

format:	Y-MM-DD hh:mm:ss
---------	------------------

* Clock display format.

>Y: completely display for the year in digit format. Example: 2021

>M: Month in digit format. (1-12)

>D: Date in digit format. (1-31)

>h: Hours in digit format. (0-23)

>m: Minutes in digit format. (0-59)

>s: Seconds in digit format. (0-59)

>W: Day of the week in digit format. (1-6)

>W: Day of the week in English abbreviation format. (Example: Tue)

>YY: Last 2 digits of the year (21)

>MM: Month in digit format (2 byte). (01-12)

>DD: Date in digit format (2 byte). (01-31)

>hh: Hours in digit format (2 byte). (00-23)

>mm: Minutes in digit format (2 byte). (00-59)

>ss: Seconds in digit format (2 byte). (00-59)

>MMM: Month in English abbreviation format. (Example: May)

- Default value: y-mm-dd hh:mm:ss
- Data type: text
- Whether it is editable: yes

5.32 shape



The shape widget is a tool for drawing basic graphics. It can create geometric shapes such as points, lines, rectangles, and circles. It is not displayed by default and is drawn through sending relevant instructions.

5.32.1 line_color

Color of line



- Default value: 0x00000000
- Data Type: hex
- Whether it is editable: yes

5.32.2 line_width

Width of line



- Default value: 1
- Data Type: uint
- Whether it is editable: ye

5.32.3 line_solid

Pixels numbers of solid line



- Default value: 0
- Data Type: uint
- Whether it is editable: yes

5.32.4 line_dotted

Pixels numbers of dotted line

line_dotted:	0
--------------	--------------------------------

- Default value: 0
- Data Type: uint
- Whether it is editable: yes

5.32.5 fill_color

The fill color of the shape's interior

fill_color:	
-------------	-------------------------------

- Default value: 0xFFFFFFFF
- Data Type: hex
- Whether it is editable: yes

5.33 table_view



table_view is a list function widget used to display data in a table format. It supports multi-row and multi-column layouts, and it can display and modify batch information.

5.33.1 table_client

table_client are the content area in the table, which are used to display multi rows with specific data. Users need to edit this area in accordance with the actual content of the project and table header information.

5.33.1.1 row_height:

Height of each row of table

row_height:	40
-------------	---------------------------------

- Default value: 40
- Data Type: uint
- Whether it is editable: yes

5.33.1.2 Rows

Default number of rows in table

rows:	10
-------	---------------------------------

- Default value: 10
- Data Type: uint

- Whether it is editable: yes

5.33.2 scroll_bar_d

Scroll bars are generally used on the right side of the widget to indicate the position of current page. When the content exceeds the displaying area of the widget, vertical scrolling function will be activated for users to know the approximate location of the current displaying area.

5.33.2.1 max

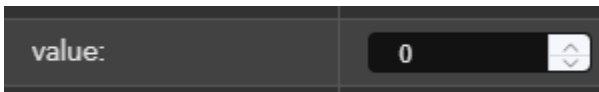
Max height of scroll bar column



- Default value:1000
- Data Type:uint
- Whether it is editable: yes

5.33.2.2 value

Current position value of scroll bar

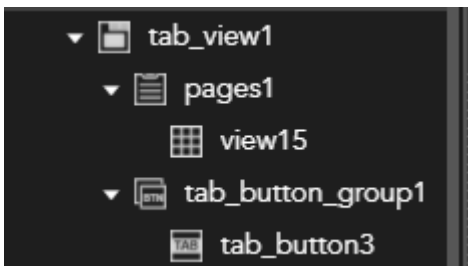


- Default value:0
- Data Type:uint
- Whether it is editable: yes

5.34 tab_view

tab_view is a view, similar to a browser. View different views by switching between different tabs.

The inclusion relationship of a tab_view is as follows:



5.34.1 pages

Pages generally exist as child widgets and are generally not used alone.

5.34.1.1 value

The value of page.

value:	0
--------	--------------------------------

- Default value: 0
- Data type: uint
- Whether it is editable: yes

5.34.1.2 x/y_offset

Offset in x/y direction.

x_offset:	0
y_offset:	0

- Default value: 0/0
- Data type: uint
- Whether it is editable: yes

5.34.2 View

view generally exists as a view of multiple child widgets, and is generally not used alone.

5.34.2.1 x/y_offset

Offset in x/y direction.

x_offset:	0
y_offset:	0

- Default value: 0/0
- Data type: uint
- Whether it is editable: yes

5.34.3 tab_button_group

tab_button_group is a tab button group, generally not used alone.

5.34.3.1 compact

Whether to choose compact typography.

compact:	☐
----------	--------------------------

- Default value: false
- Data type: bool
- Whether it is editable: yes

5.34.3.2 scrollable

Whether to support scrolling.

scrollable:	☐
-------------	--------------------------

- Default value: false
- Data type: bool
- Whether it is editable: yes

5.34.4 tab_button

tab_button is the tab button and is used in the tab_button_group group.

5.34.4.1 value

The value of tab_button.

value:	☐
--------	--------------------------

- Default value: false
- Data type: bool
- Whether it is editable: yes

5.34.4.2 load_ui

The ui to load after activation.

active_icon:	
--------------	----------------------

- Default value:
- Data type: text
- Whether it is editable: yes

5.34.4.3 active_icon

The name of the current icon item.

active_icon:	
--------------	----------------------

- Default value:
- Data type: text
- Whether it is editable: yes

5.34.4.4 icon

The name of the item that is not the current icon.

icon:	
-------	----------------------

- Default value:

- Data type: text
- Whether it is editable: yes

5.34.4..5 x/y_offset

Offset in x/y direction.

x_offset:	0	
y_offset:	0	

- Default value: 0/0
- Data type: uint
- Whether it is editable: yes

5.35 scroll_view

5.35.1 x/yoffset

The offset in x/y direction

yoffset:	0		xoffset:	0	
----------	---	--	----------	---	--

- Default value: 0/0
- Data type: uint
- Whether it is editable: yes

5.35.2 x/yslidable

Whether to allow sliding in x/y direction.

yslidable:	☒	xslidable:	☒
------------	-------------------------------------	------------	-------------------------------------

- Default value: 0/0
- Data type: uint
- Whether it is editable: yes

5.35.3 x/yspeed_scale

x/y offset velocity scale.

yspeed_scale:	0		xspeed_scale:	0	
---------------	---	---	---------------	---	---

- Default value: 0/0
- Data type: uint

- Whether it is editable: yes

5.35.4 snap_to_page

Whether to align by page when scrolling.

snap_to_page:	☐
---------------	--------------------------

- Default value: false
- Data type: bool
- Whether it is editable: yes

5.35.5 move_to_page

Whether to turn one page at a time.

move_to_page:	☐
---------------	--------------------------

- Default value: false
- Data type: bool
- Whether it is editable: yes

5.35.6 recursive

Whether to support recursive search for all child widgets.

recursive:	☐
------------	--------------------------

- Default value: false
- Data type: bool
- Whether it is editable: yes

5.35.7 focusable

Whether to support focus stay.

focusable:	☐
------------	--------------------------

- Default value: false
- Data type: bool
- Whether it is editable: yes

5.35.8 x/y_offset

Offset in x/y direction.

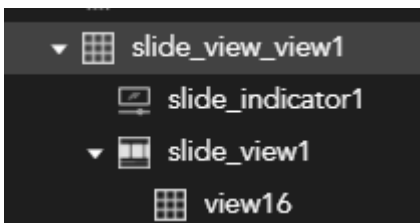
x_offset:	0
y_offset:	0

- Default value: 0/0
- Data type: uint
- Whether it is editable: yes

5.36 slide_view_view

slide_view_view is a sliding view, switch between different pages to view the design effect of each page.

The inclusion relationship of a slide_view_view is as follows:



5.36.1 slide_indicator

5.36.1.1 max

Maximum value.

max:	0
------	---

- Default value: 0
- Data type: uint
- Whether it is editable: yes

5.36.1.2 size

Indicator size.

size:	8
-------	---

- Default value: 8
- Data type: uint
- Whether it is editable: yes

5.36.1.3 anchor_x/y

Anchor point x/y coordinates.

anchor_x:	100	
anchor_y:	0	

- Default value: 100/0
- Data type: uint
- Whether it is editable: yes

5.36.1.4 auto_hide

Auto hide. When it is 0, it is forbidden.

auto_hide:	0	
------------	---	---

- Default value: 0
- Data type: uint
- Whether it is editable: yes

5.36.1.5 default_paint

Type of indicator.

default_paint:	auto	
----------------	------	--

- Default value: auto
- Data type: text
- Whether it is editable: yes

5.36.1.6 spacing

Spacing of indicators.

spacing:	15	
----------	----	---

- Default value: 15
- Data type: uint
- Whether it is editable: yes

5.36.1.7 indicated_target

The indicated target widget name.

indicated_target:	
-------------------	----------------------

- Default value:
- Data type: text
- Whether it is editable: yes

5.36.1.8 x/y_offset

Offset in x/y direction.

x_offset:	0
y_offset:	0

- Default value: 0/0
- Data type: uint
- Whether it is editable: yes

5.36.2 slide_view

As a subsidiary widget of slide_view_view, slide_view is generally not used alone.

5.36.2.1 x/y_offset

Offset in x/y direction.

x_offset:	0
y_offset:	0

- Default value: 0/0
- Data type: uint
- Whether it is editable: yes

5.36.2.2 loop

Whether to loop.

loop:	☐
-------	--------------------------

- Default value: false
- Data type: bool
- Whether it is editable: yes

5.36.2.3 vertical

Whether to slide up and down mode.

vertical:	☐
-----------	--------------------------

- Default value: false
- Data type: bool
- Whether it is editable: yes

5.36.2.4 anim_hint

Page switching effect.

anim_hint:	translayte ▼
------------	--------------

- Default value: translayte
- Data type: text
- Whether it is editable: yes

5.36.2.5 auto_play

Whether to play automatically.

auto_play:	0	↕
------------	---	---

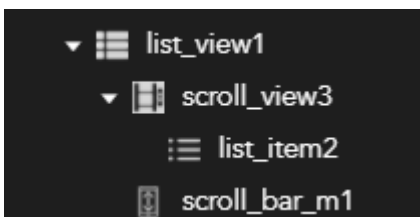
- Default value: 0
- Data type: uint
- Whether it is editable: yes

5.37 list_view

5.37.1 list_view

list_view is a list view, which can be designed in the list and viewed by selecting the corresponding page.

The inclusion relationship of a list_view is as follows:



5.37.1.1 item_height

The height of the list item.

item_height:	0	↕
--------------	---	---

- Default value: 0
- Data type: uint
- Whether it is editable: yes

5.37.1.2 default_item_height

The height of the default list item.

default_item_height:	60	↕
----------------------	----	---

- Default value: 60
- Data type: uint
- Whether it is editable: yes

5.37.1.3 auto_hide_scroll_bar

Auto-hide when scrollbars are not needed.

auto_hide_scroll_bar:	☐
-----------------------	--------------------------

- Default value: false
- Data type: bool
- Whether it is editable: yes

5.37.1.4 floating_scroll_bar

Whether the scroll bar is suspended above the scroll_view.

floating_scroll_bar:	☐
----------------------	--------------------------

- Default value: false
- Data type: bool
- Whether it is editable: yes

5.37.2 scroll_view

5.37.2.1 x/yoffset

The offset in x/y direction

yoffset:	0	xoffset:	0
----------	--------------------------------	----------	--------------------------------

- Default value: 0/0
- Data type: uint
- Whether it is editable: yes

5.37.2.2 x/yslidable

Whether to allow sliding in x/y direction.

yslidable:	☒	xslidable:	☒
------------	-------------------------------------	------------	-------------------------------------

- Default value: 0/0
- Data type: uint
- Whether it is editable: yes

5.37.2.3 x/yspeed_scale

x/y offset velocity scale.

yspeed_scale:	0	xspeed_scale:	0
---------------	---	---------------	---

- Default value: 0/0
- Data type: uint
- Whether it is editable: yes

5.37.2.4 snap_to_page

Whether to align by page when scrolling.

snap_to_page:	☐
---------------	--------------------------

- Default value: false
- Data type: bool
- Whether it is editable: yes

5.37.2.5 move_to_page

Whether to turn one page at a time.

move_to_page:	☐
---------------	--------------------------

- Default value: false
- Data type: bool
- Whether it is editable: yes

5.37.2.6 recursive

Whether to support recursive search for all child widgets.

recursive:	☐
------------	--------------------------

- Default value: false
- Data type: bool
- Whether it is editable: yes

5.37.2.7 focusable

Whether to support focus stay.

focusable:	☐
------------	--------------------------

- Default value: false
- Data type: bool
- Whether it is editable: yes

5.37.2.8 x/y_offset

Offset in x/y direction.

x_offset:	0	↕
y_offset:	0	↕

- Default value: 0/0
- Data type: uint
- Whether it is editable: yes

5.37.3 scroll_bar_m

scroll_bar_m is a scroll bar, which is generally used on the right side of a widget or page

5.37.3.1 max

Virtual width or height.

max:	1000	↕
------	------	---

- Default value: 1000
- Data type: uint
- Whether it is editable: yes

5.37.3.2 value

Current value.

value:	0	↕
--------	---	---

- Default value: 0
- Data type: uint
- Whether it is editable: yes

5.37.4 list_item

list_item is used as a sub-widget of list_view and is generally not used alone.

5.37.4.1 x/y_offset

Offset in x/y direction.

x_offset:	0	↕
y_offset:	0	↕

- Default value: 0/0

- Data type: uint
- Whether it is editable: yes

5.37.4.2 scroll_bar_m

scroll_bar_m is a scroll bar, which is generally used on the right side of a widget or page

5.37.4.3 max

Virtual width or height.

max:	1000	
------	------	---

- Default value: 1000
- Data type: uint
- Whether it is editable: yes

5.37.4.4 value

Current value.

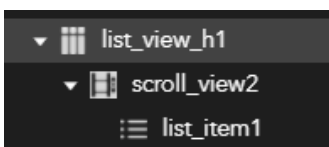
value:	0	
--------	---	---

- Default value: 0
- Data type: uint
- Whether it is editable: yes

5.38 list_view_h

list_view is a horizontal list view, which can be designed in the list and viewed by selecting the corresponding page.

The inclusion relationship of a list_view_h is as follows:



5.38.1 scroll_view

5.38.1.1 x/yoffset

The offset in x/y direction

yoffset:	0		xoffset:	0	
----------	---	---	----------	---	---

- Default value: 0/0
- Data type: uint
- Whether it is editable: yes

5.38.1.2 x/yslidable

Whether to allow sliding in x/y direction.

yslidable:	☒	xslidable:	☒
------------	-------------------------------------	------------	-------------------------------------

- Default value: 0/0
- Data type: uint
- Whether it is editable: yes

5.38.1.3 x/yspeed_scale

x/y offset velocity scale.

yspeed_scale:	0	xspeed_scale:	0
---------------	---	---------------	---

- Default value: 0/0
- Data type: uint
- Whether it is editable: yes

5.38.1.4 snap_to_page

Whether to align by page when scrolling.

snap_to_page:	☐
---------------	--------------------------

- Default value: false
- Data type: bool
- Whether it is editable: yes

5.38.1.5 move_to_page

Whether to turn one page at a time.

move_to_page:	☐
---------------	--------------------------

- Default value: false
- Data type: bool
- Whether it is editable: yes

5.38.1.6 recursive

Whether to support recursive search for all child widgets.

recursive:	☐
------------	--------------------------

- Default value: false

- Data type: bool
- Whether it is editable: yes

5.38.1.7 focusable

Whether to support focus stay.

focusable:	☐
------------	--------------------------

- Default value: false
- Data type: bool
- Whether it is editable: yes

5.38.1.8 x/y_offset

The offset in x/y direction

yoffset:	0	xoffset:	0
----------	---	----------	---

- Default value: 0/0
- Data type: uint
- Whether it is editable: yes

5.38.2 list_item

list_item is used as a sub-widget of list_view and is generally not used alone.

5.38.2.1 x/y_offset

Offset in x/y direction.

x_offset:	0
y_offset:	0

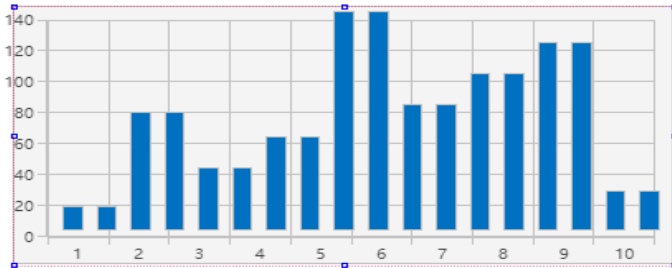
- Default value: 0/0
- Data type: uint
- Whether it is editable: yes

6. Some Composite Widgets Usage

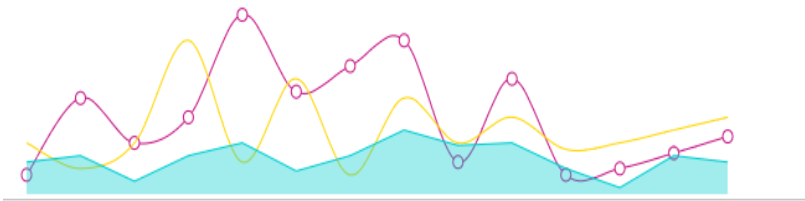
6.1 chart_view

chart_view is a chart widget that allows you to add multiple bar charts and multiple line charts to a table at the same time to achieve a effect similar to a data display.

The following is the effect of having multiple different bar_series in the same widget.



The following is the effect of having multiple different line_series in the same widget.



The x and y axis data of chart_view need to be set manually, and the data should be set one by one according to your needs.

The specific operation steps are as follows.

Expand the chart_view widget in the project area on the left and click on the x_axis widget.



(1) Find the property column on the right, and modify max and data according to your actual needs.

If $\text{data} > \text{max}$, you can achieve the effect of horizontal sliding; $\text{data} < \text{max}$, you can achieve the effect of scaling.

Properties

type:

x_axis

x:

104

y:

113

w:

400

h:

200

enable:

☒

visible:

☒

style:

default

state:

normal

> text:

> border:

> bg_image:

-- select --

> fg_image:

-- select --

> round_radius:

0

> margin:

0

min:

0

max:

9

axis_type:

value

at:

auto

data:

[1,2,3,4,5,6,7,8]

split_line:

{show:true}

tick:

{show:true}

line:

{show:true}

label:

{show:true}

inverse:

☐

offset:

0

offset_percent:

☐

> animation:

+

key_tone:

☐

(2) Find the corresponding line_series widget and bar_series widget and modify the value and capacity according to your needs.

value:

15,75,40,60,141

series_axis:

value_axis:

line:

{smooth:true}

area:

{show:true}

symbol:

{show:true}

capacity:

10

offset:

0

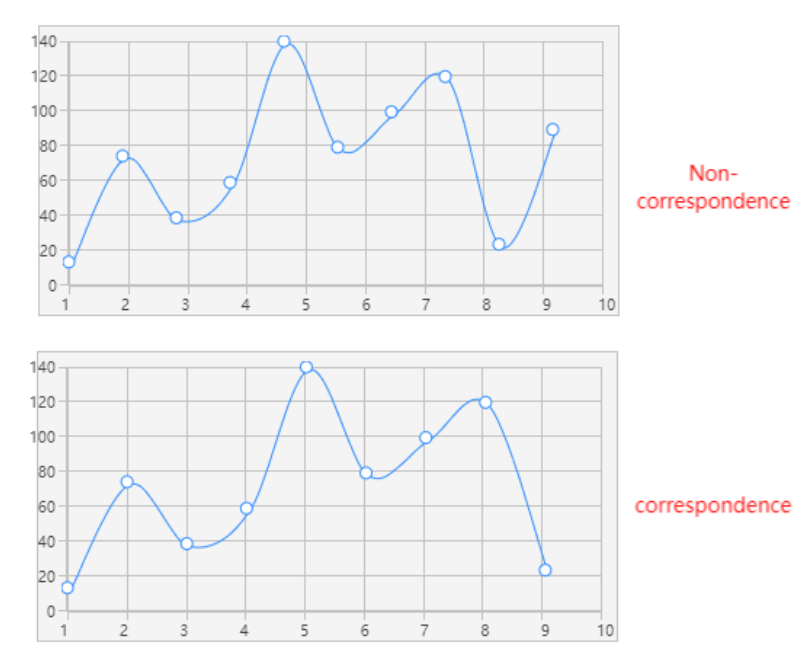
display_mode:

push

value_animation:

500

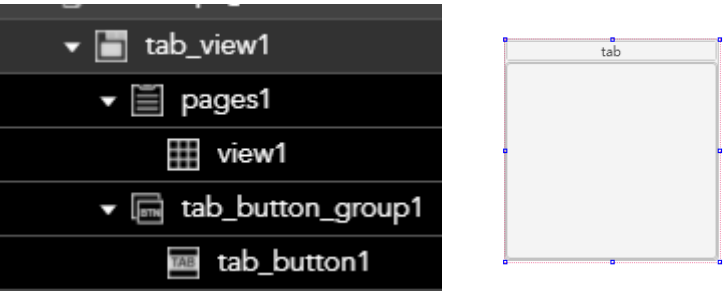
Special note: In the process of setting, if you need the data of line_series and the split_line of x_axis to achieve one-to-one correspondence, that is, all data are on the axis of x_axis, you need to set the data and max of x_axis, the value of max is the amount of data-1, such as data is [1,2,3,4,6,7,8,9,10], then max=9. Otherwise, it may not correspond with the data.



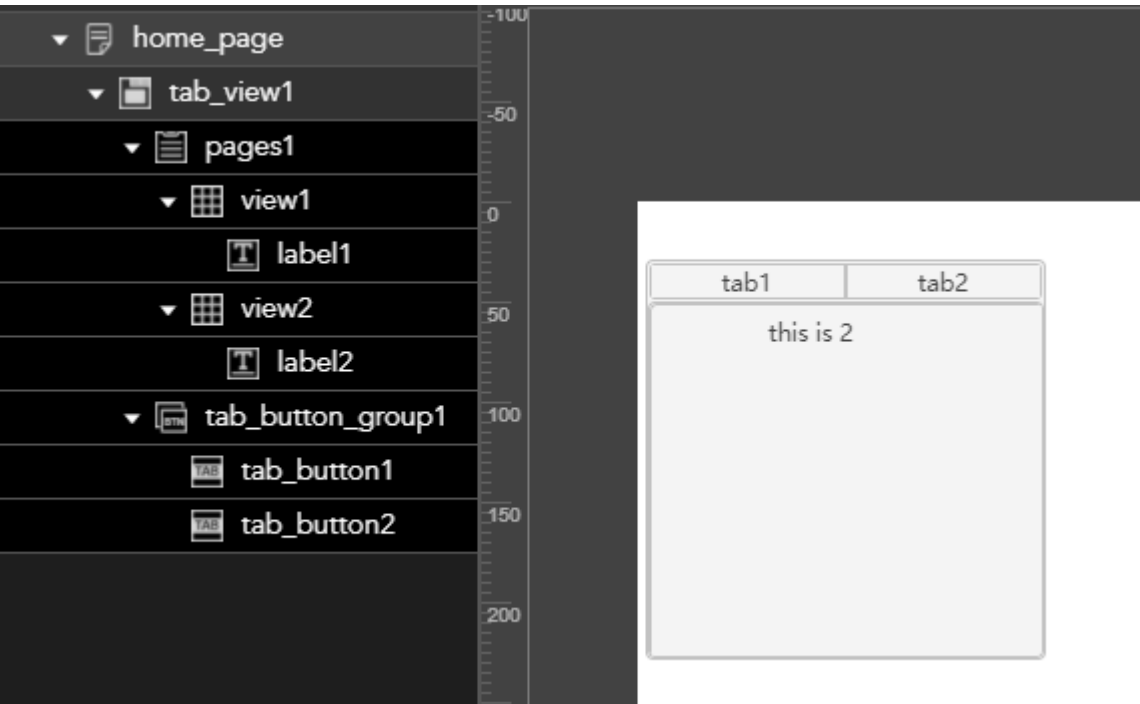
6.2 tab_view

tab_view is a tab view, you can add tab_button by your own actual needs, and add the corresponding view to achieve clicking different tab_button to switch to different pages.

1. The initial state is shown in the figure.



2. The effect after setting is completed

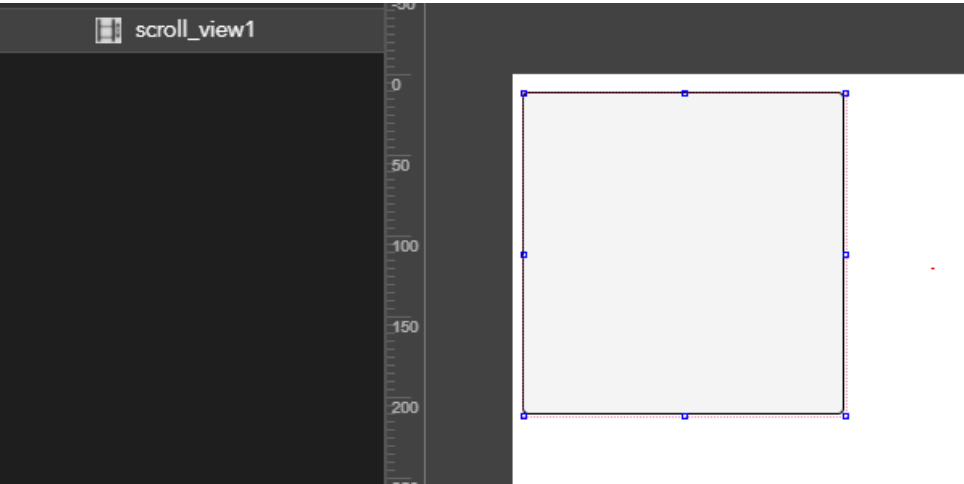


6.3 scroll_view

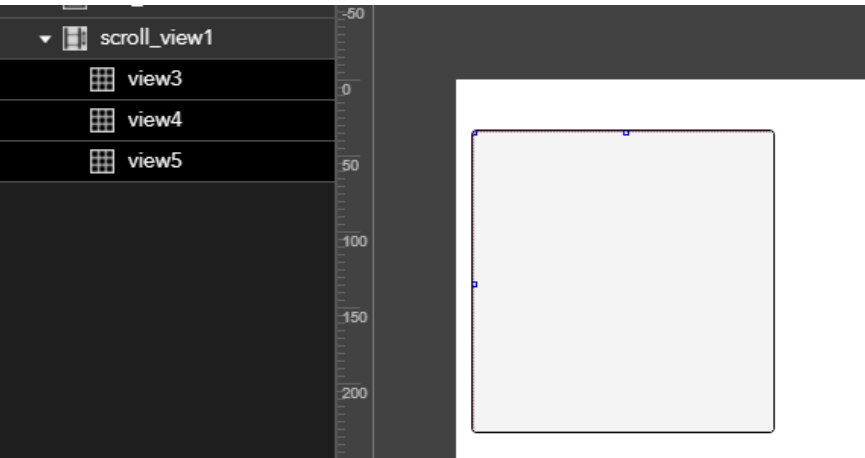
scroll_view is a scrolling view, which can be scrolled up and down or left and right.

The layout is done internally (you need to decide whether it is a list_item or a view according to your actual needs), and after the layout is done, you can add the required sub-widgets in the corresponding view (view or list_item).

1. The initial state is shown in the figure.



2. Drag the view to the sub-widget (same as list_item) :



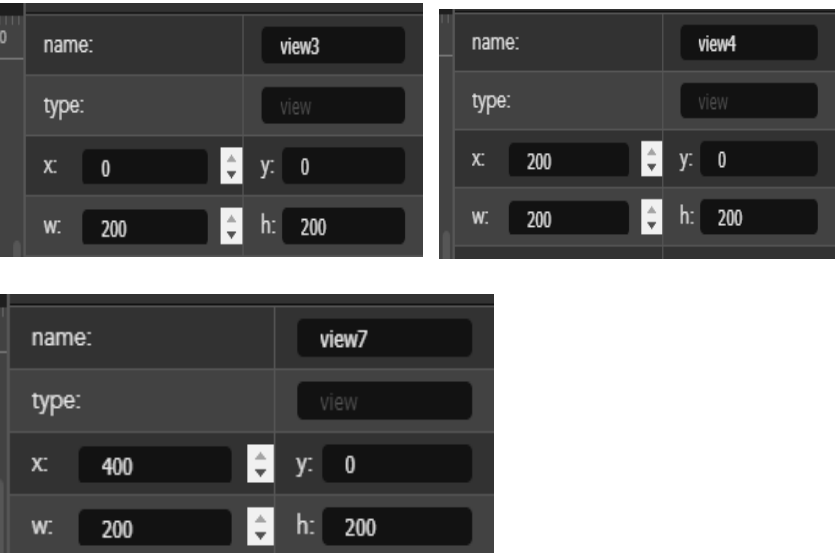
3. according to your needs for layout, such as the need for vertical sliding effect, you need to check the yslideable of scroll_view and set similar to the following settings, that is, the stack of view in the y direction.

Properties			
name:	view3		
type:	view		
x:	0	y:	0
w:	200	h:	200

Properties			
name:	view4		
type:	view		
x:	0	y:	200
w:	200	h:	200

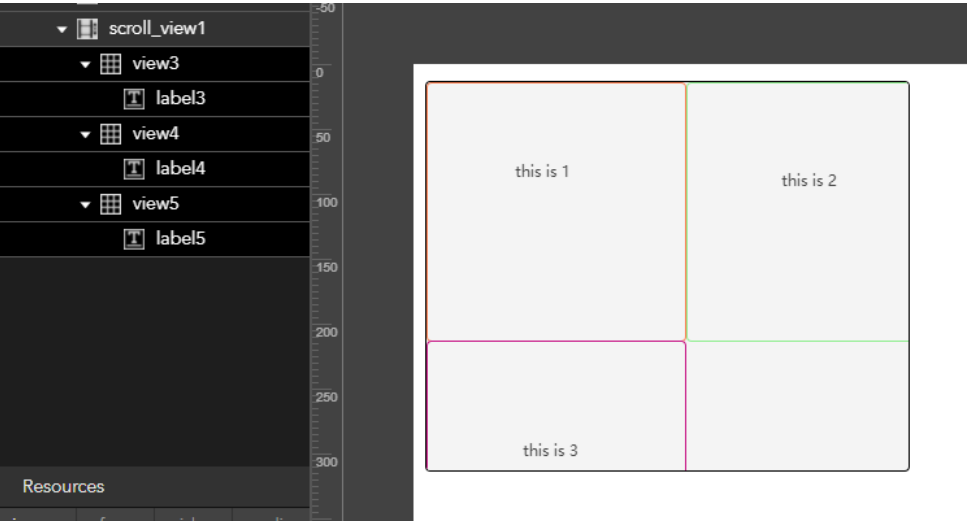
Properties			
name:	view5		
type:	view		
x:	0	y:	400
w:	200	h:	200

If you need a horizontal sliding effect, you need to check the `xslideable` of the `scroll_view` and set a setting similar to the following, that is, the stack of views in the x-direction.



4. If you want to achieve sliding in x and y directions at the same time, you can check both `xslideable` and `yslideable`, and set the position of the coordinates and the size of the view or `list_item`.

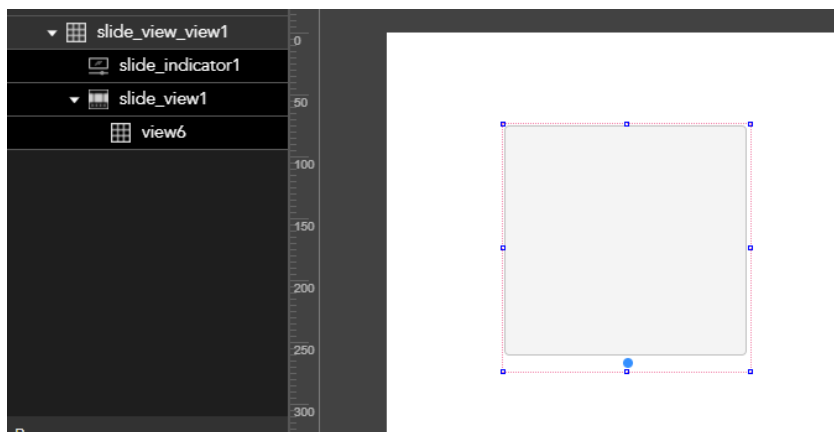
5. The effect after setting :



6.4 slide_view_view

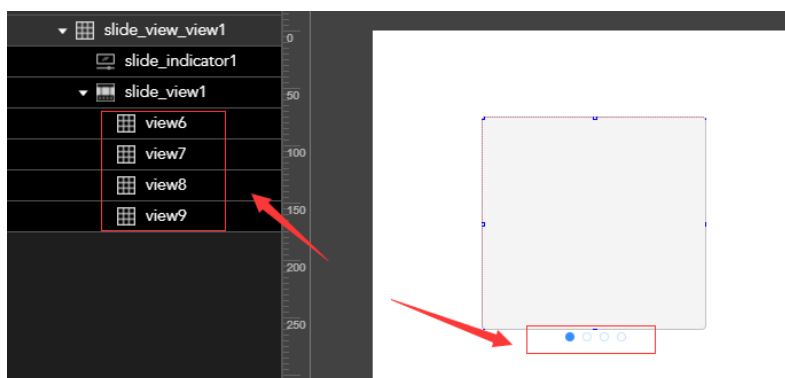
slide_view_view is a sliding view, you can add several different views according to your needs, and switch pages by sliding left and right or up and down, and you can also add an indicator below to display the position of the current page (the corresponding value of the current page can be set to the highlighted color).

1. The initial state is shown in the figure :

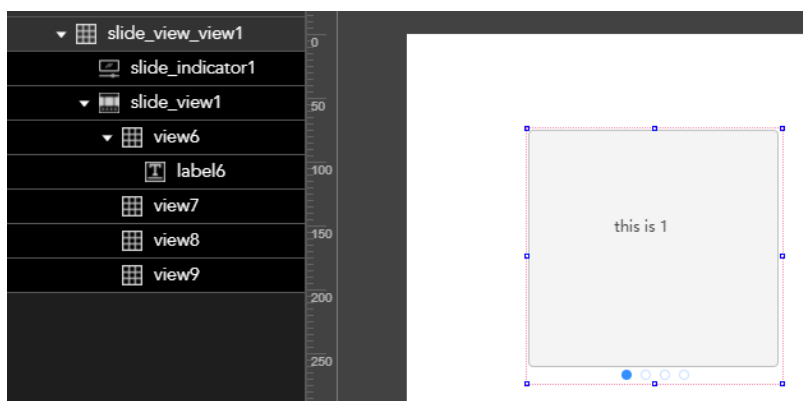


2. Add a sub-widget view, when adding the view, the slide_indicator will also synchronize the corresponding amount of views at the same time.

As shown in the figure, there are currently 4 views and the amount of slide_indicator is 4.



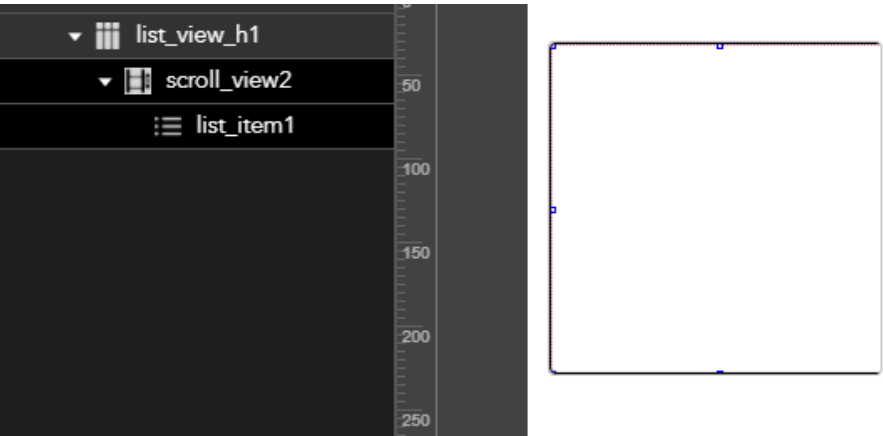
3. The effect after setting :



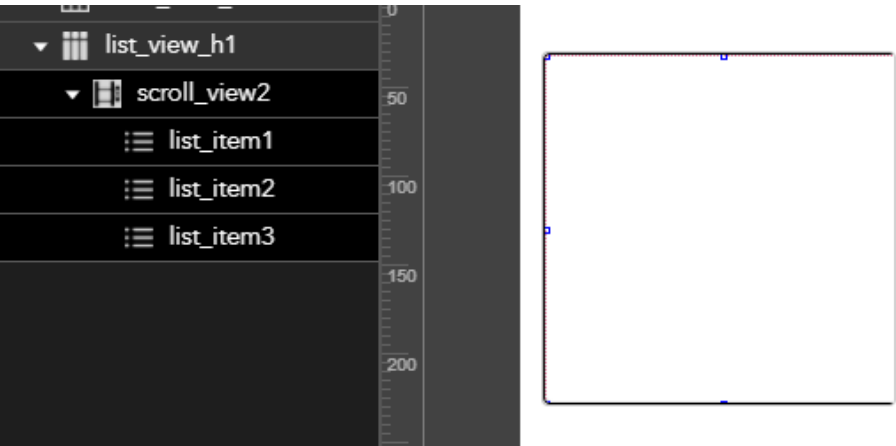
6.5 list_view_h

list_view_h widget is horizontal list view, horizontal list view can switch to different view interface by sliding left and right, unlike slide_view_view, list_view widget has list_item and view as sub-widgets, list_item has some UI interaction difference compared to view, you can choose the widgets according to your needs and add widgets and layouts inside the selected sub-widgets.

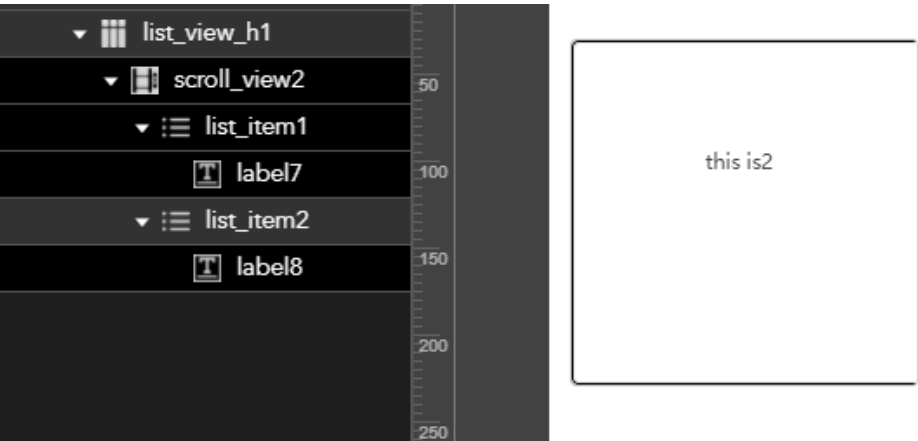
1. The initial state is shown in the figure :



2. Add sub-widgets, you can add list_item and view according to your needs, here is list_item as an example



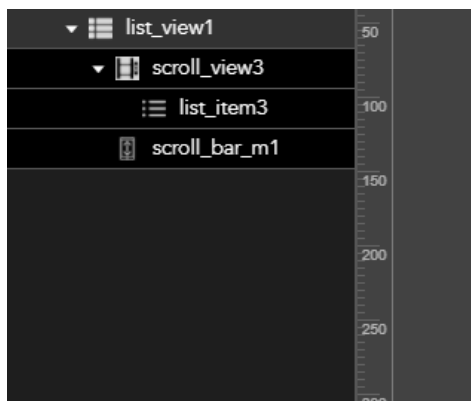
3. The effect after setting :



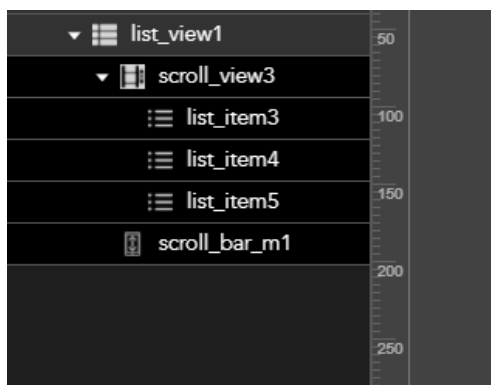
6.6 list_view

The list_view widget is a list view with a corresponding scroll bar on the right side to show the current position of the list_item compared to the entire widget, but requires corresponding settings in the software (e.g. max and value properties of scroll_bar_m). As list_view_h, you can choose whether to add a view or a list_item according to your needs, and add widgets and layouts inside the selected sub-widget.

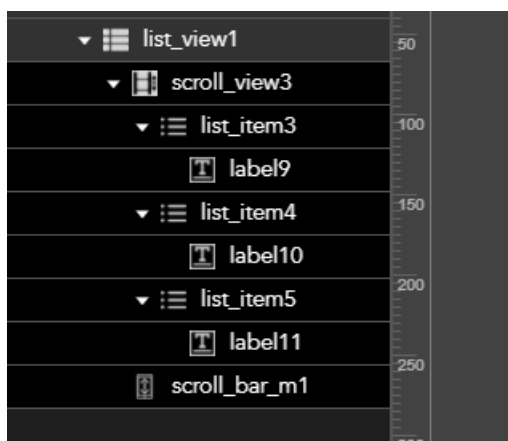
1. The initial state is shown in the figure:



2. Add sub-widgets, you can add list_item and view according to your needs, here is list_item as an example.



3. The effect after setting :



6.7 Add subsidiary widgets

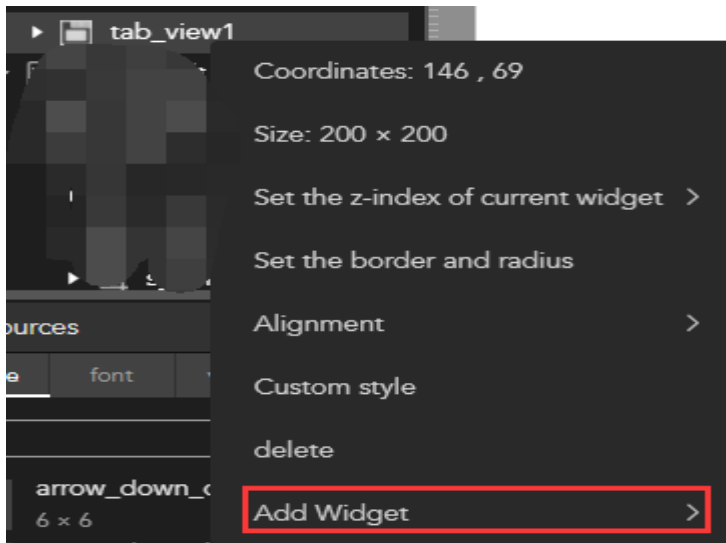
For the current project, considering a variety of different requirements, it may be necessary to add the option of subsidiary widgets to some widgets, that is, widget 1 contains widgets 2, 3, 4.... The affiliation of these widgets belongs to the parent widget, rather than windows.

Generally speaking, all widgets can have their own subsidiary widgets, but most of the following widgets need to be used with subsidiary widgets:

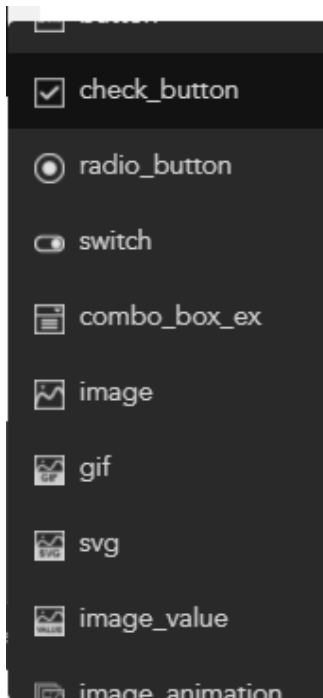
tab_view/slide_menu/slide_view_view/list_view/list_view_h/chart_view

6.7.1 right click to add

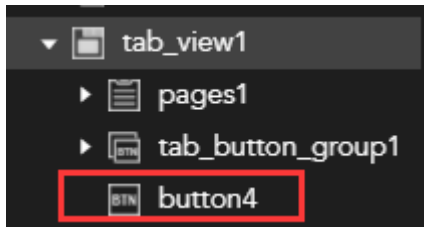
1. With the widget selected, right-click add_widget.



2. Select the widget and right-click to add.

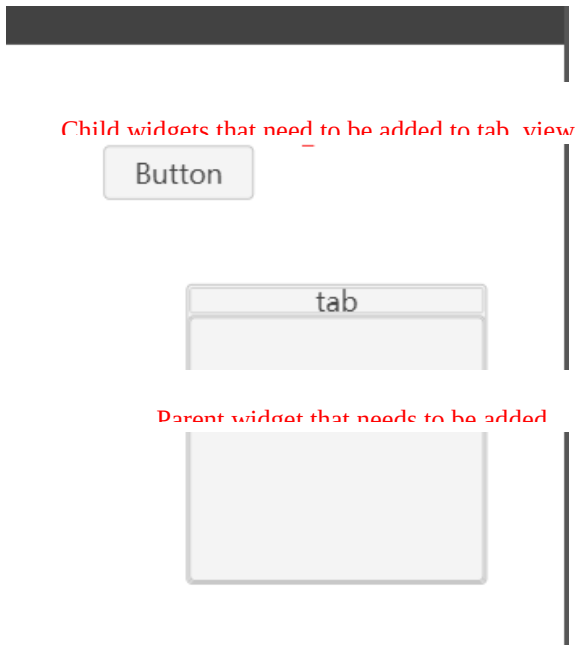


3. Add successfully. This operation is common to other widgets.

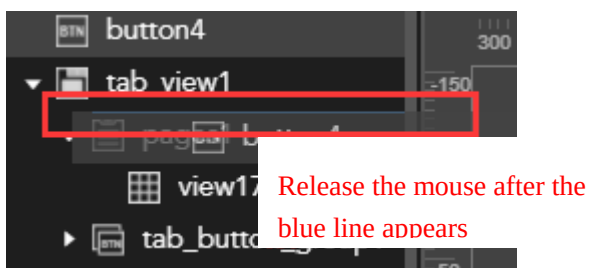
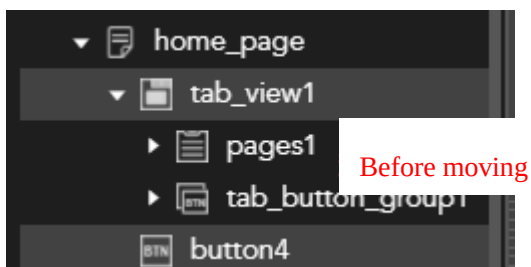


6.7.2drag to add

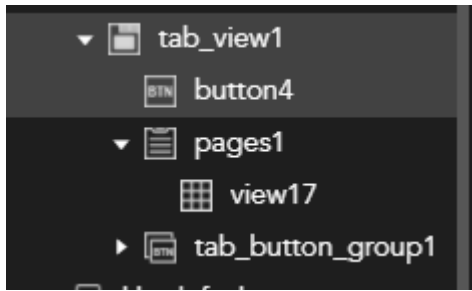
1. Drag the widget to the window first.



2. Move the widget to the desired parent widget on the left.

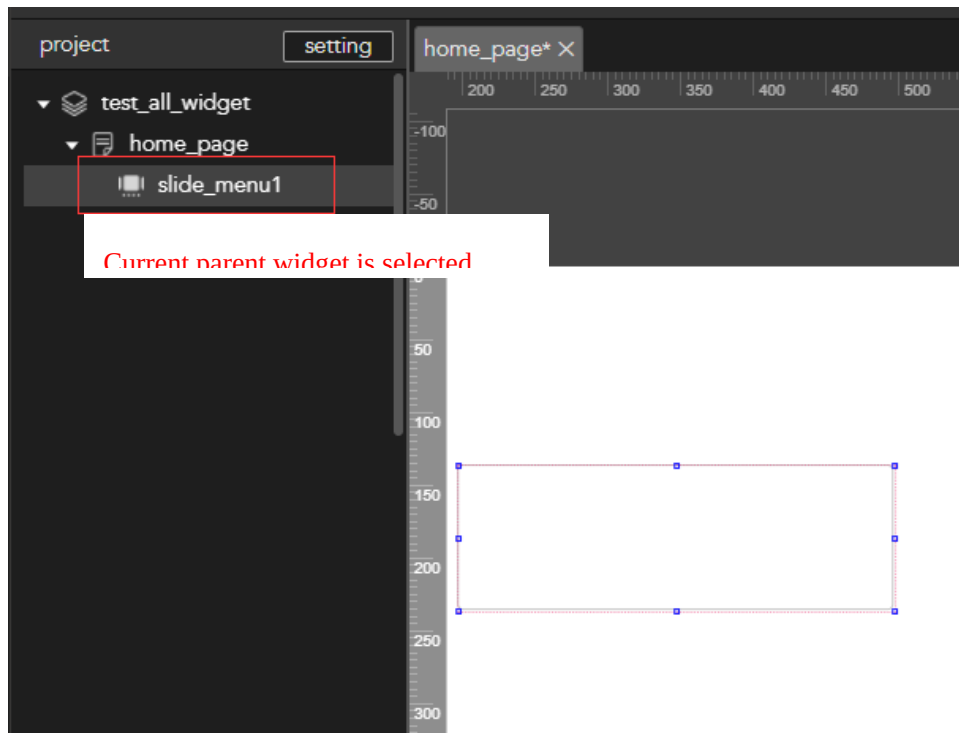


3. Added successfully.

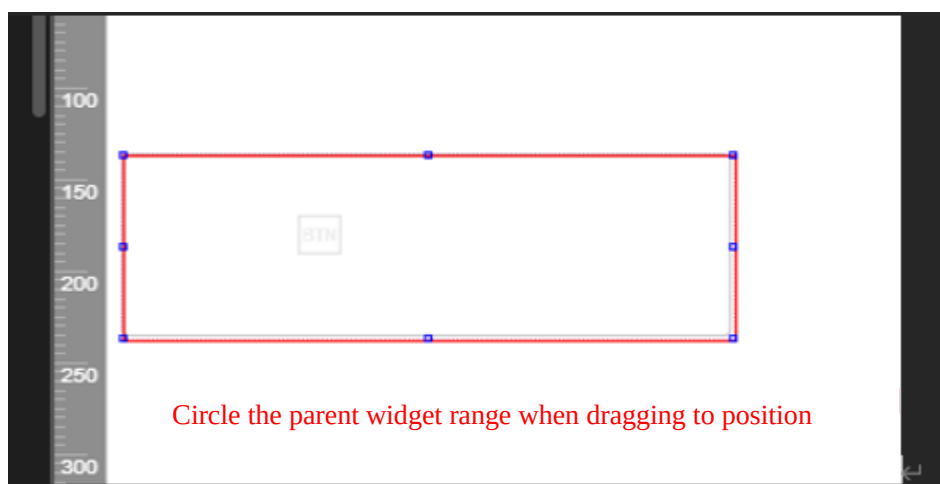


6.7.3 drag and drop to add

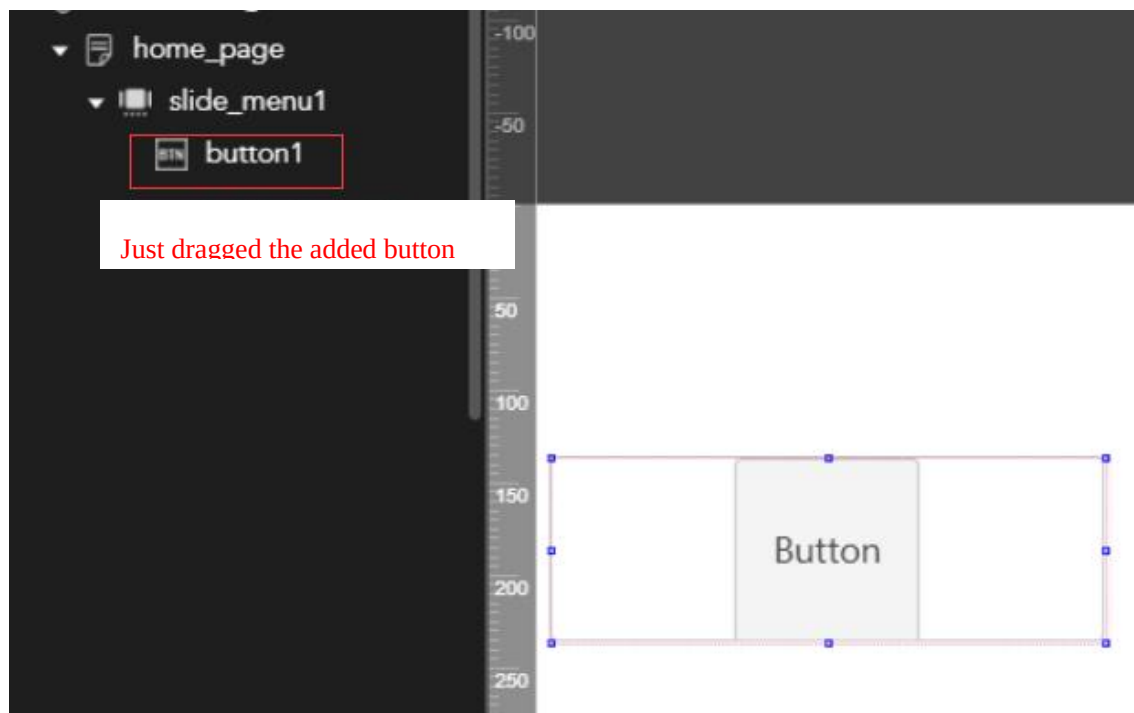
1. Select the parent widget in the left project column.



2. Drag the widget to the selected parent widget.



3. Release the mouse, the widget will appear in the selected parent widget, and the addition is successful.



7. Network port configuration instructions

7.1 Network port configuration

7.1.1 Network port configuration conditions

Net port communication enables communication between the PC and the HMI side by configuring the network and connecting the network cable at the net port instead of connecting through the serial port in the area.

Communication can be achieved as long as the Internet or local area network exists. To achieve network port communication, the following conditions need to be met first:

1. There is a host or controller in the current area that can be used as the server.
2. Update the HMI firmware to the latest version (firmware version V3.0.01.230303RC and newer)
3. app_conf.json file needs to add new network-related configuration options, refer to [7.3 json field specific meaning](#).

7.1.2 Network port configuration use range

The network port configuration is compatible with all instructions in the serial instruction set, such as system instructions, general instructions and instructions for different widgets.

After the configuration of the net port is completed, you can send instructions and receive data back from the HMI remotely through the server to achieve remote communication.

Special Note: Considering the current network environment and other issues, there may be some delays.

7.2 Network port configuration steps

After the GUI project is completed, open the app_conf.json file in the \default\raw\data path and manually add the following configuration:

```
"lan": {  
    "dhcp": true.  
    "mac_addr": "AA:BB:CC:DD:EE"  
}.  
"cmd_server": {  
    "server_addr": "www.google.com".  
    "server_port": 1668  
}
```

The final example is shown in the following figure:

```
{
  "baud_rate": 115200,
  "brightness": 100,
  "lcd_rotation": 0,
  "lan": {
    "dhcp": true,
    "mac_addr": "AA:BB:CC:DD:EE"
  },

  "cmd_server": {
    "server_addr": "www.google.com",
    "server_port": 1668
  }
}
```

7.3 json field specific meaning

lan: LAN configuration.

dhcp: Currently only supports dynamic ip acquisition, i.e. true.

mac_addr: the physical address of the network port.

cmd_server: instruction server.

server_addr: address of the server. Support ip and domain name setting, such as 192.168.0.0, i.google.com, etc.

server_port: port number of the server.

With LAN configuration enabled, below is an example of the correct format.

```
"lan": {
  "dhcp": true.
  "mac_addr": "AA:BB:CC:DD:EE"
}.
"cmd_server": {
  "server_addr": " 192.168.0.0 ".
  "server_port":1668
}
```

7.4 Communication implementation operation description

7.4.1 HMI's GUI project update

1. Write the above configuration into the app_conf.json file under \default\raw\data and save it.
2. Copy the default folder where you saved the configuration file to the HMI, and you can choose the method of connecting with computer or u-disk to update the project according to the actual demand.

Special note: If the current firmware version is old, you need to update the firmware first, otherwise the network port configuration will not take effect and communication cannot be realized.

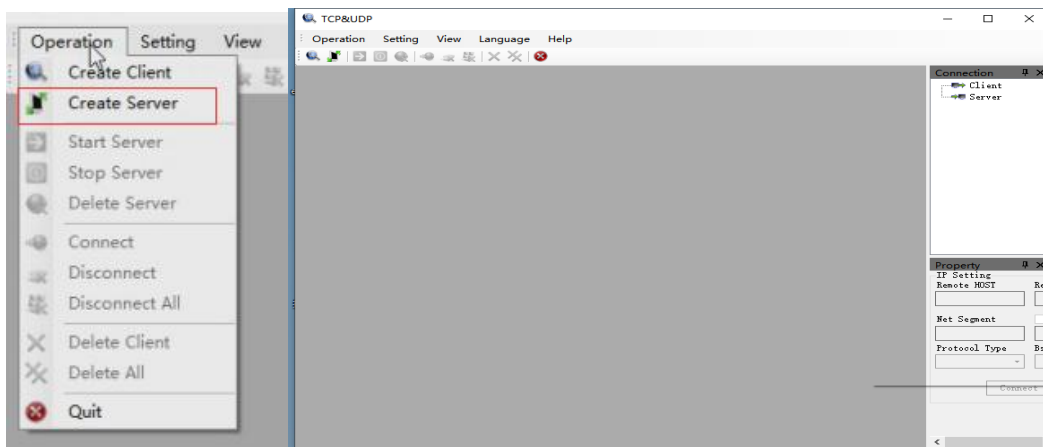
The method of updating via computer connection can be referred to the download of Project.

7.4.2 Server connection establishment and communication

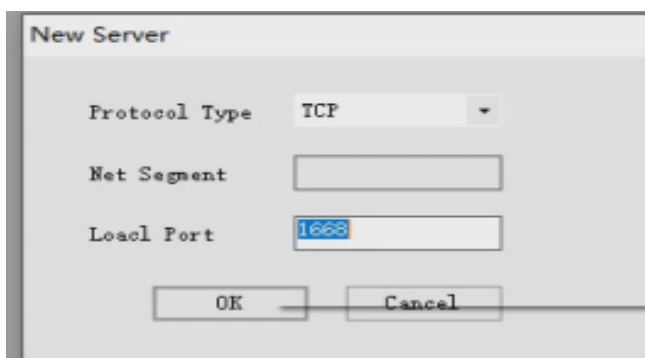
1. Download the software to choose the path according to your actual needs and install it in PC, taking Conextop TCPUDP as an example.

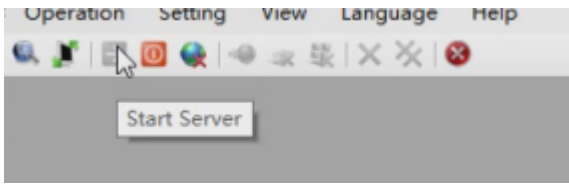
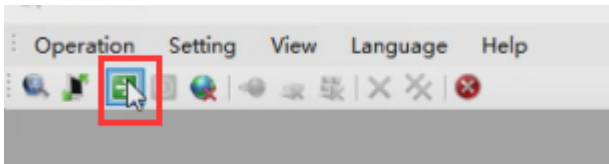


2. After installation, open the software and configure it to create a server according to your actual needs.

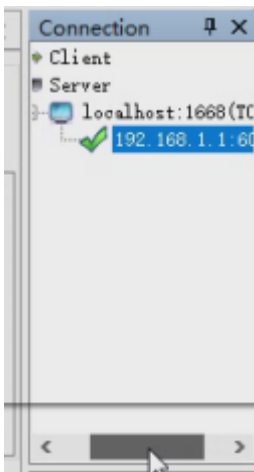


3. After clicking OK, start the server and let the server run normally.

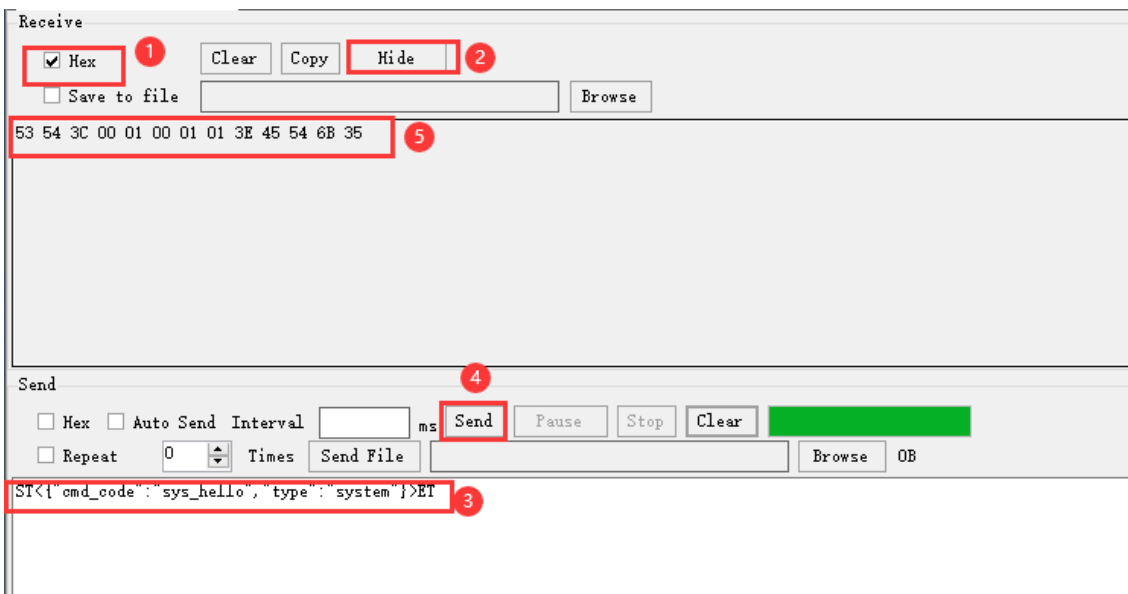




4. Wait for the device to be connected successfully. After the device is properly connected to the server via the network cable, it can send and receive instructions for communication.



5. Test sending and receiving instructions. After checking the corresponding option, you will receive the return value from the HMI, and the HMI will make corresponding changes or responses after sending instructions.



8. MQTT configuration instructions

8.1 MQTT introduction and development significance

MQTT (Message Queuing Telemetry Transport) is a lightweight, open, publish/subscribe model-based message transmission protocol. It was originally developed by IBM and is designed to connect devices with low bandwidth, high latency or unstable networks. The MQTT protocol is simple, flexible, and scalable, and is suitable for various scenarios. The basic working principle of MQTT is based on the publish/subscribe model. The combined device can be a **Publisher**, publishing messages to a specific **Topic**, while other devices can be **Subscriber**, subscribing to topics of interest. Receive related news. This model makes communication between devices more loosely coupled, improving flexibility and scalability.

The publish/subscribe model of MQTT makes the communication between the host computer and the external device very flexible, and messages can be exchanged and transmitted through network cables. The MQTT protocol has certain message delivery guarantees to ensure the reliability and sequence of messages. When used on MQTT devices, network bandwidth consumption is very low, which is critical for the device, and the MQTT protocol supports real-time message passing, which means that for the device, real-time monitoring, control, and communication can be quickly carried out. For example, after sending the JSON code in the project, you can receive the HEX return value in real time, which facilitates data monitoring and abnormal situation monitoring.

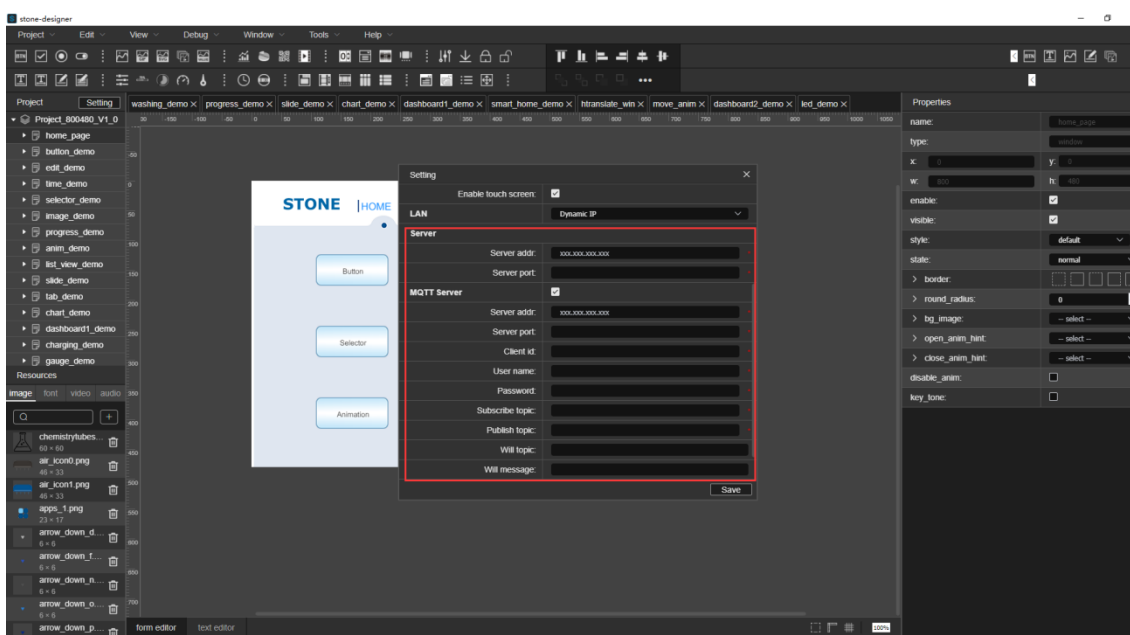
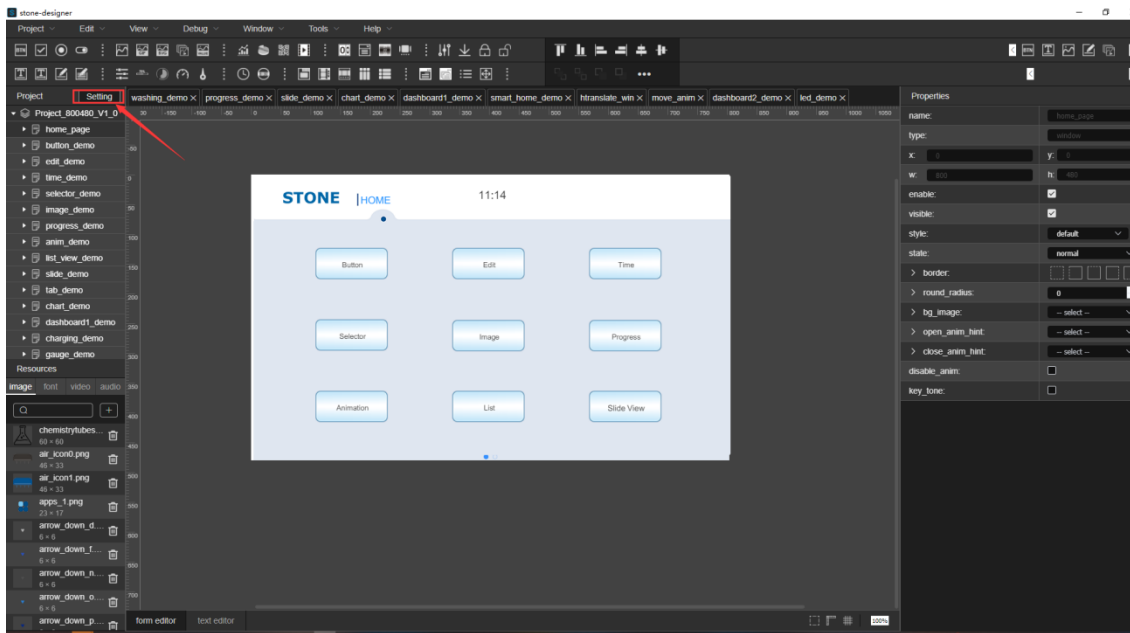
8.2 MQTT configuration on Stone Designer software

MQTT exists to provide a lightweight and efficient communication protocol for connecting and exchanging data IoT

devices, facilitating reliable communication and interoperability between devices.

First, you need to perform relevant configurations in the setting interface of the GUI software. The specific configuration needs to be manually configured based on the current project and some relevant information about the computer. After manual configuration, download the project to the device, ensure the normal connection of the network cable, and ensure that the MQTTX software starts normally.

MQTT setting:



After filling in, save the project. You can refer to the following filling examples.

Setting ×

LAN

Dynamic IP ▼

Server

Server addr:

192.168.1.0 *

Server port:

1666 *

MQTT Server ☒

Server addr:

192.168.1.1 *

Server port:

1666 *

Client id:

SN2304251883 *

User name:

test_mqtt1 *

Password:

test_mqtt1 *

Subscribe topic:

stone/hmi/SN202304081668 *

Publish topic:

stone/mcu/SN202304081668 *

Will topic:

*

Will message:

*

Screen Saver

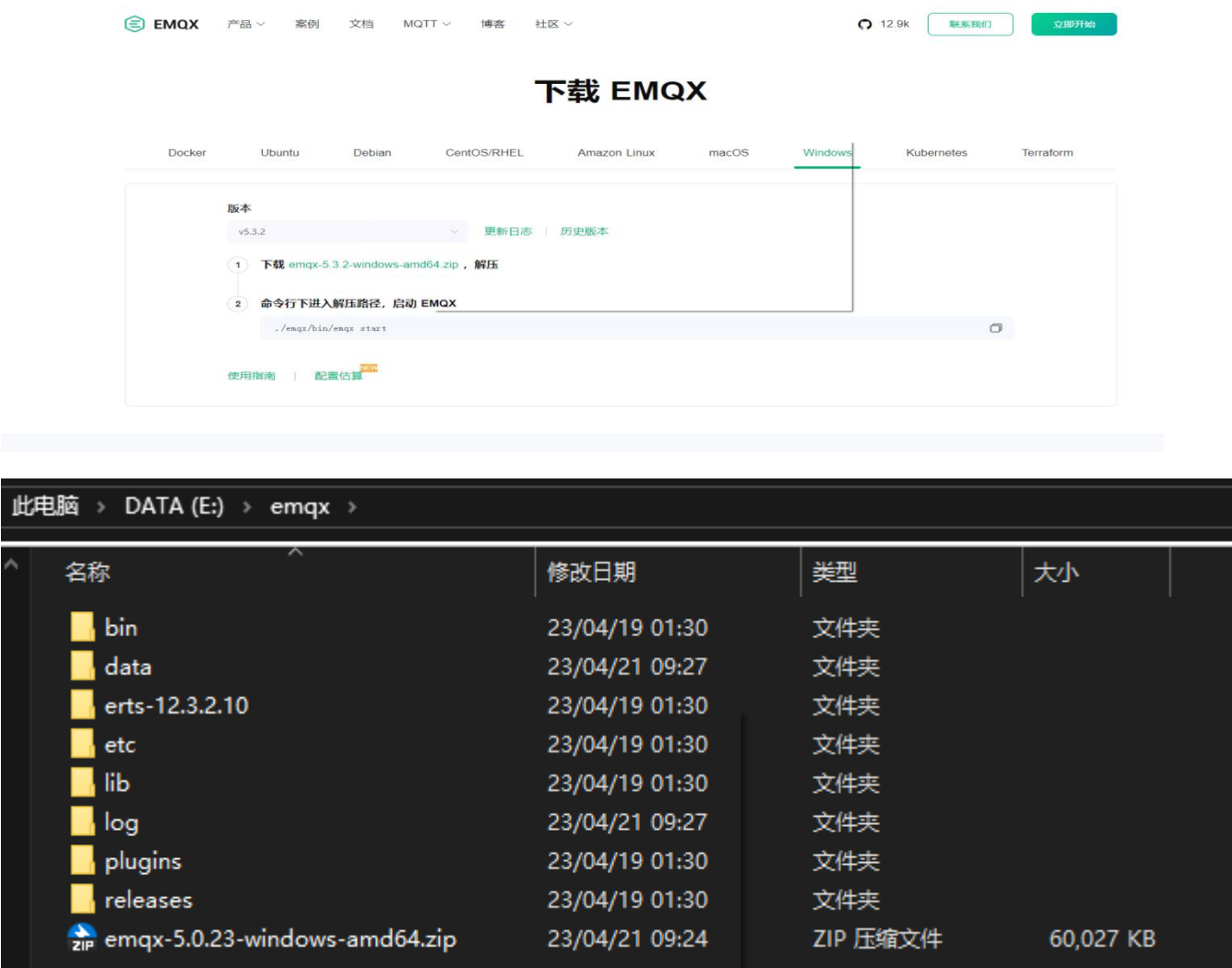
Save

During the process of downloading projects and sending instructions and receiving data through the MQTT protocol, try to ensure that the network of the HMI is smooth and try to avoid re-powering on. If a restart is required, it is recommended to send instructions to restart and make regular backups of your instructions.

8.3 Download and start of EMQX

<https://www.emqx.io/zh/downloads?os=Windows>

Enter the EMQX download page, take windows as an example, and extract the downloaded file to the required path.



Enter the installation directory of EMQ X Broker from cmd, as shown in the figure, use the command emqx start to start EMQ X Broker and wait for the startup to be successful.

```
E:\>cd emqx/bin

E:\emqx\bin>emqx start

E:\>cd emqx/bin

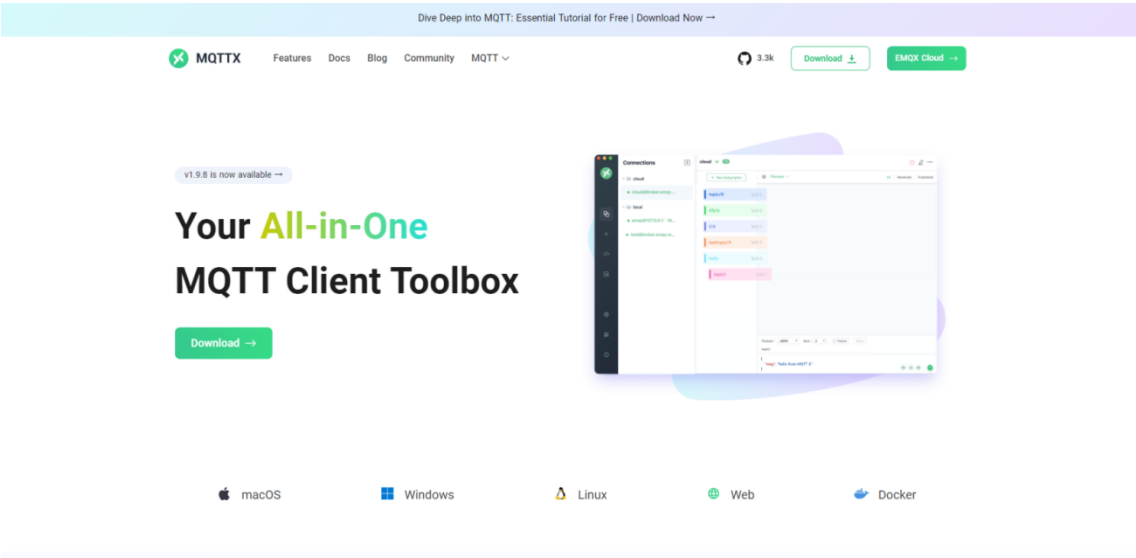
E:\emqx\bin>emqx start
EMQX_NODE__DB_ROLE [node.role]: core
EMQX_NODE__DB_BACKEND [node.db_backend]: mnesia

E:\emqx>
```

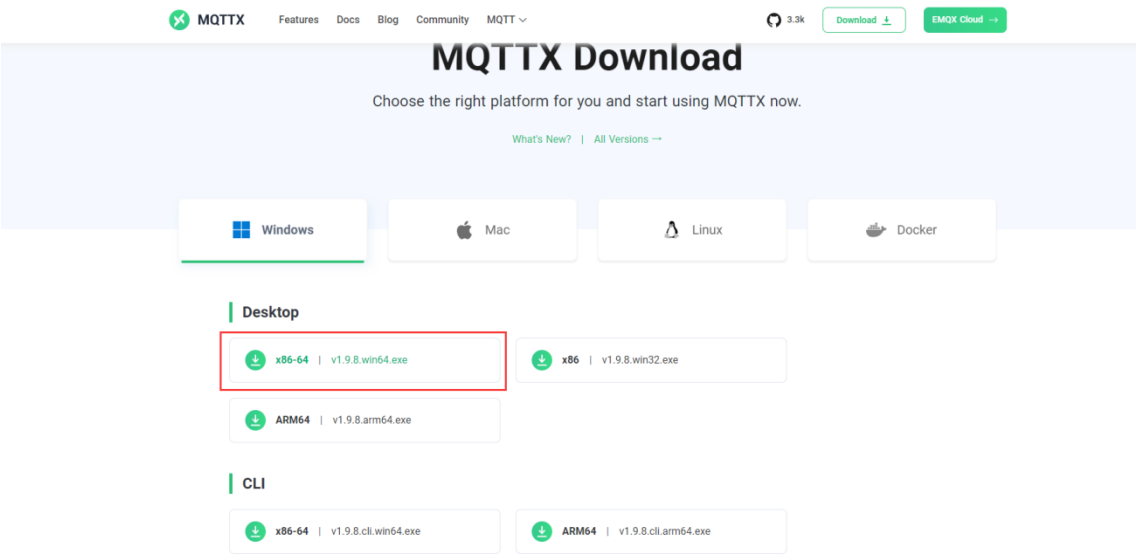
8.4 MQTTX software installation and use

8.4.1 Software download

Open the browser and enter the official website address of MQTTX <https://mqttx.app/>.



Select the applicable system version on the download page and download the software package. Windows is used as an example below.



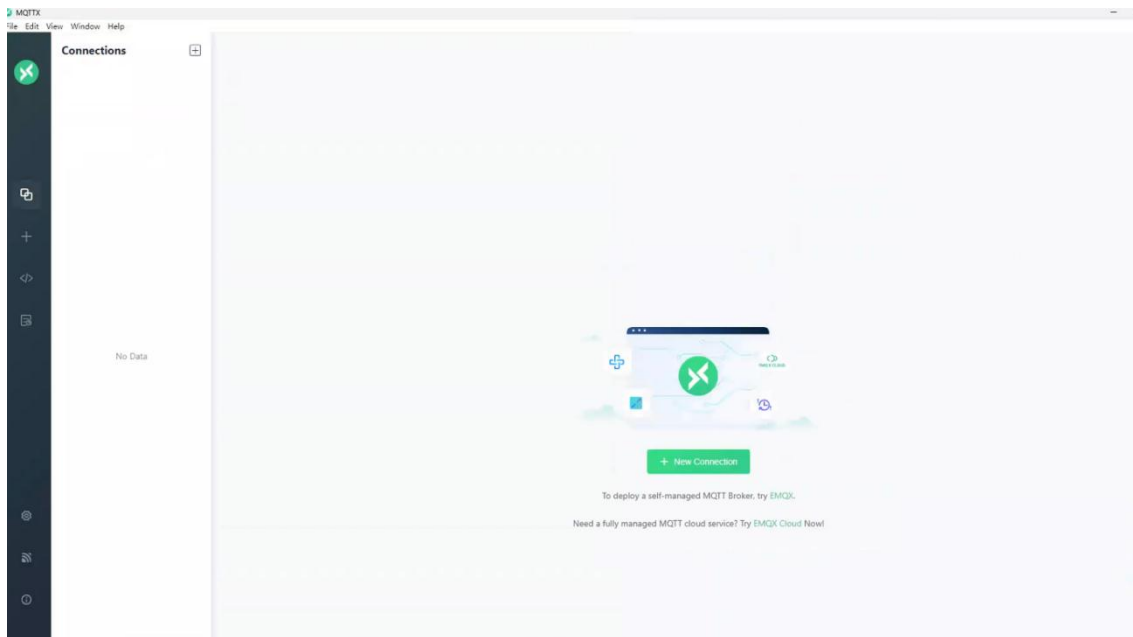
8.4.2 Software installation

MQTTX may require some runtime dependencies, which may involve installing Node.js or other runtime environments based on the computer's operating environment. Install the environment according to your own needs.

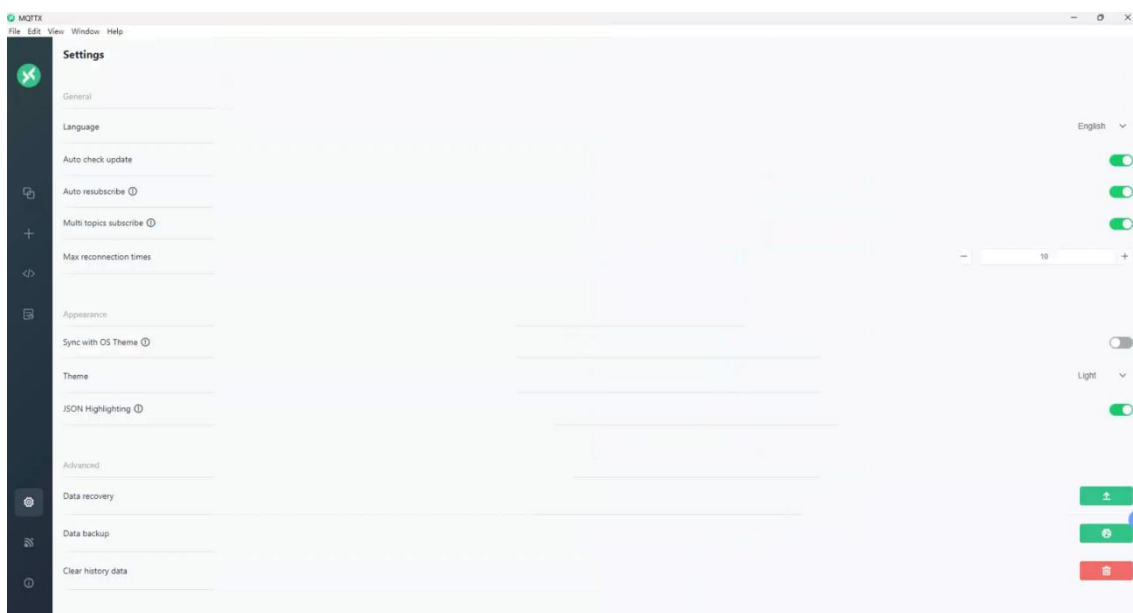
For some detailed instructions and tutorials, please refer to: <https://www.emqx.io/docs/en/latest/>

8.4.3 Software use and device communication

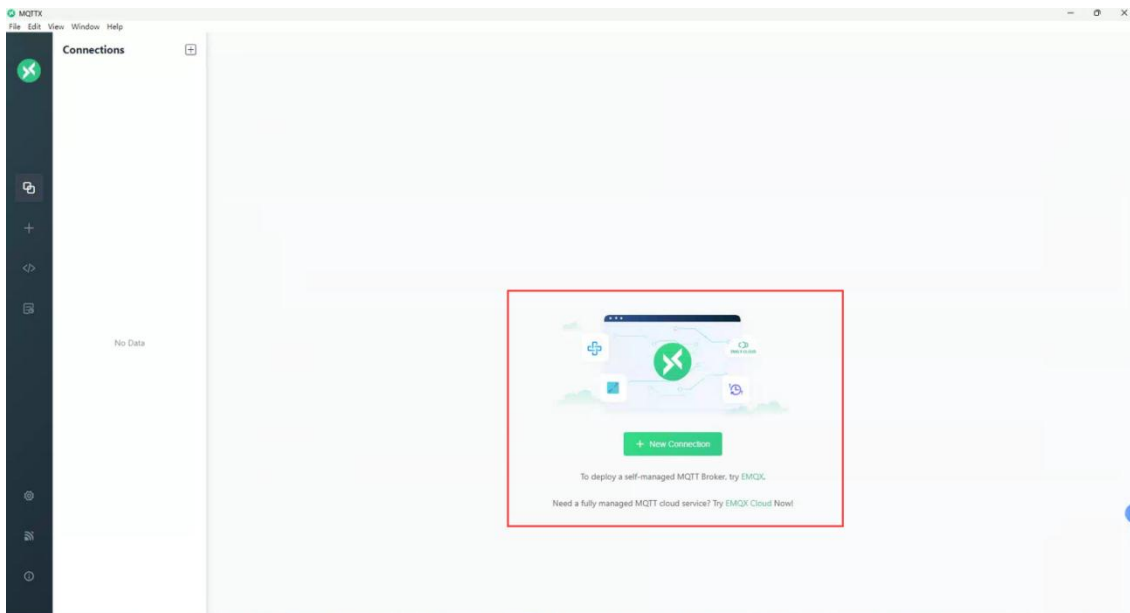
Open the installed graphical MQTTX interface.



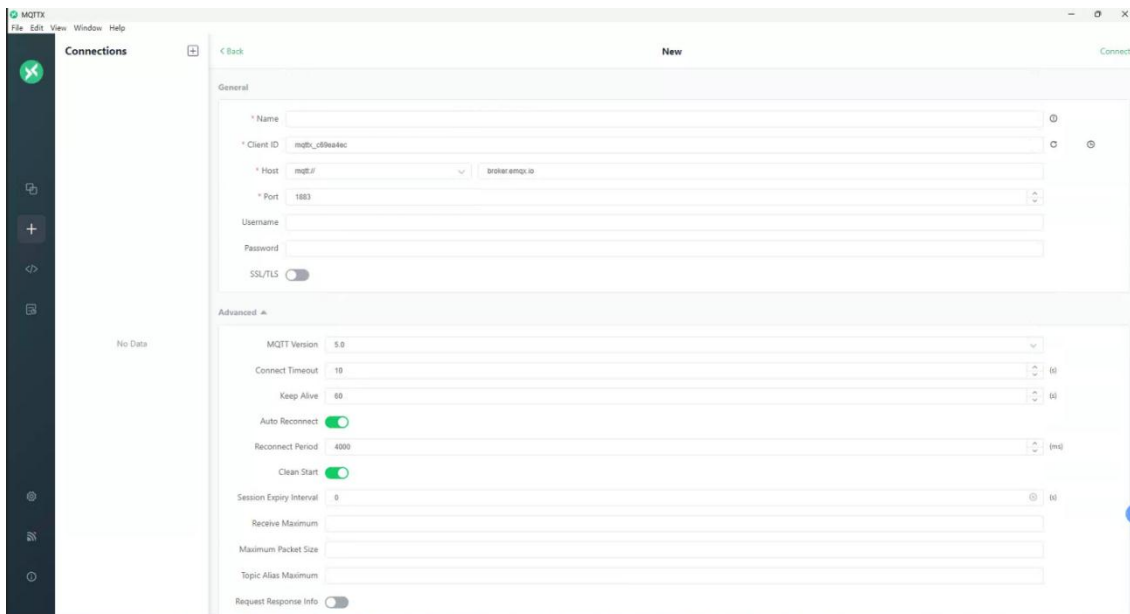
Running MQTTX for the first time requires some basic configuration, such as selecting language, interface theme, etc., and configure it according to personal preferences.



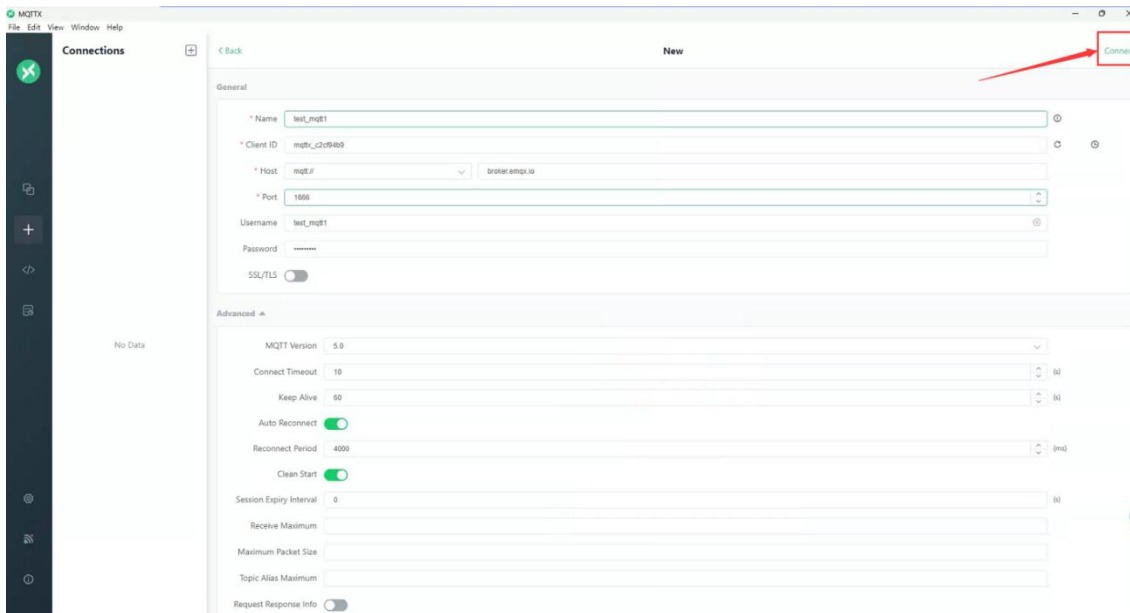
In the main interface of MQTTX, click the "Connect" button or select the "Connect" option in the menu.



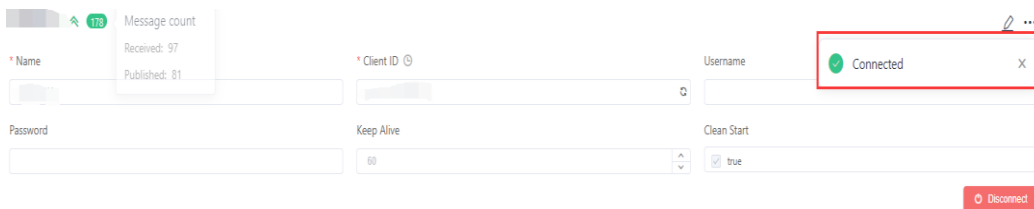
In the dialog box, fill in the MQTT broker's address, port number and other necessary information.



After filling in the connection parameters, click the "Connect" button in the connection dialog box.

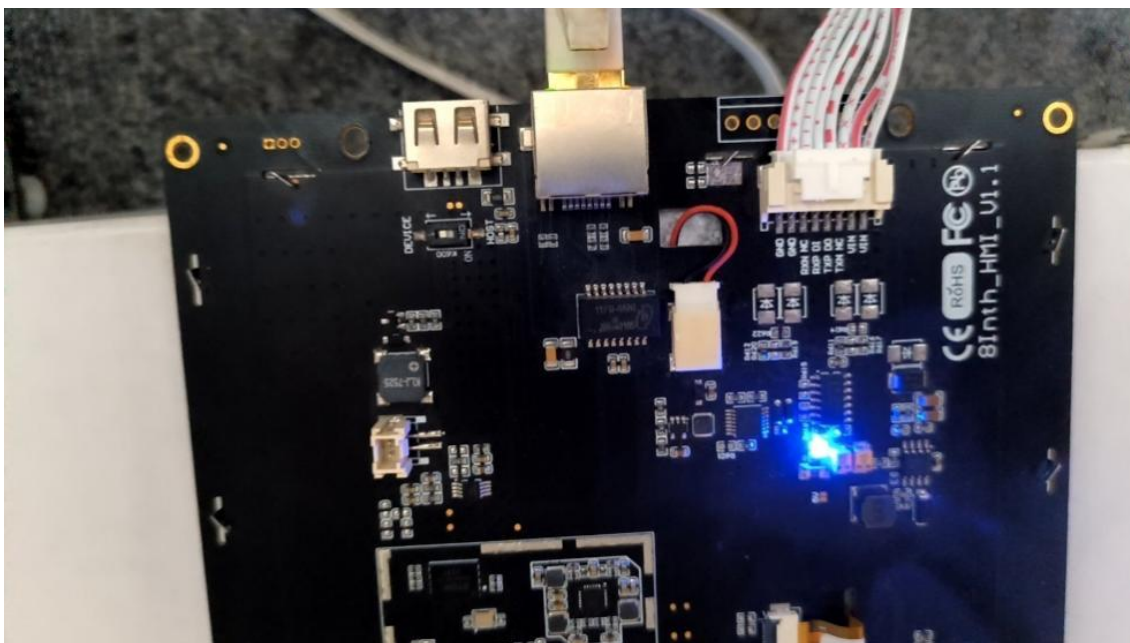


If everything is set up correctly, MQTTX will connect to the specified MQTT broker and prompt that the connection is successful.



8.4.4 Use MQTTX software to communicate with HMI

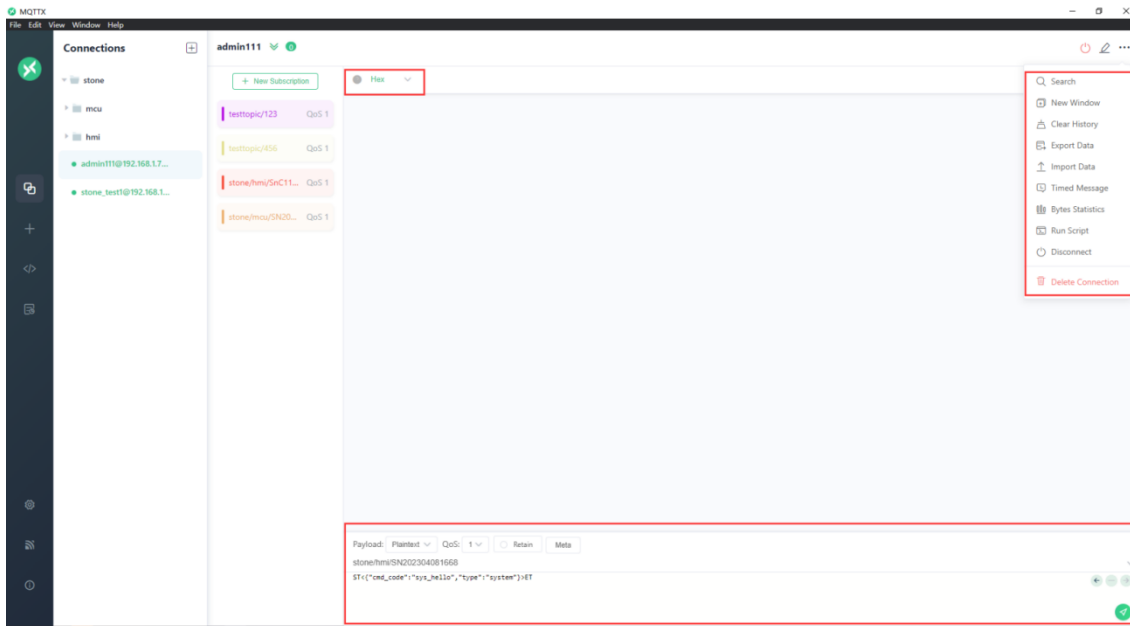
Make sure the HMI updates the newly configured items, and plug the HMI into the network cable. Please refer to the following.



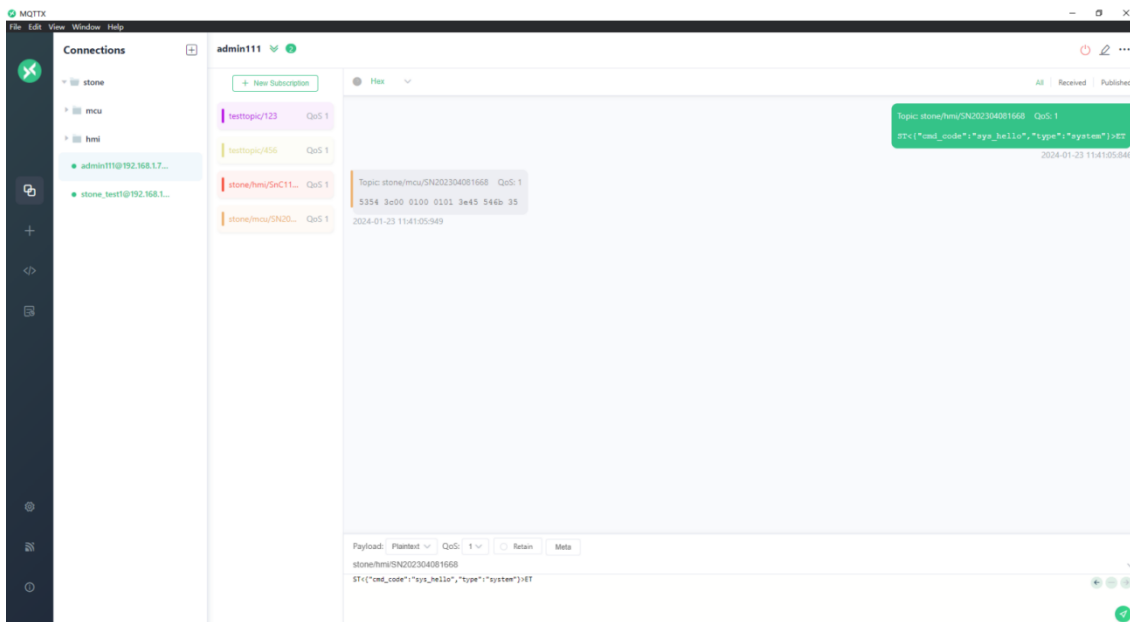
Connect the parameters just configured in the MQTTX software and select the item in the upper right corner according to your actual needs.

Considering the data return type of the instruction set, the received format needs to be selected as hex.

After setting up, enter the command to be sent in the sending area. Note that you need to set the Qos above according to actual needs. It is recommended to set it to 1 and set the sending object.



Take the **sys_hello** instruction as an example. After sending, you can receive the return value from the HMI, it means the communication is normal. After that, you can receive data and send instructions through software.

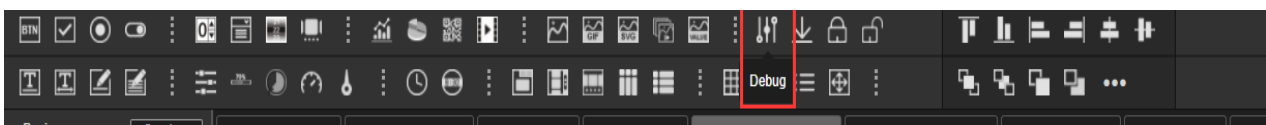


9. Debug function introduction

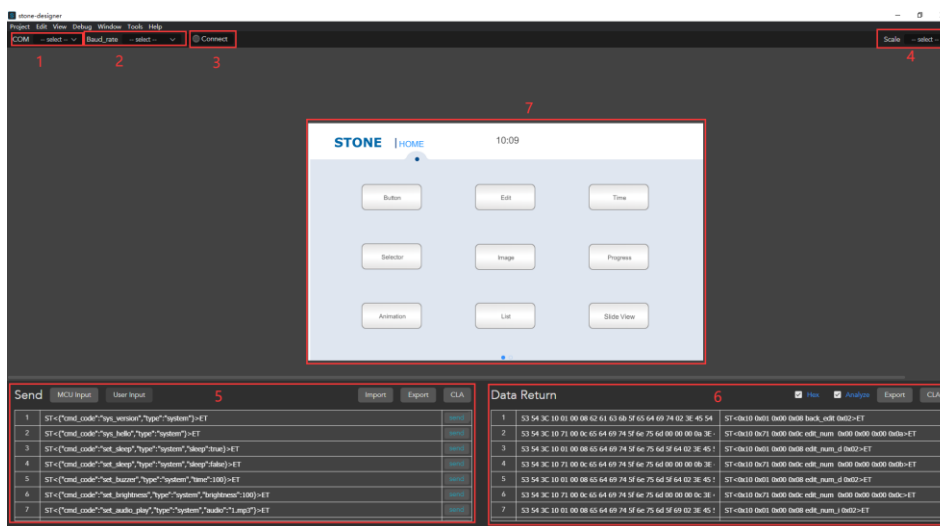
9.1 Debug page

Debug can simulate and run the current user's saved project, also can be sent and received commands, the main idea is to facilitate the user to preview their projects on the PC. Convenient for users to build projects for real-time debugging and modification. Click on the debug button will generate the debug page.

1. Users can debug alone without connecting PC with HMI device.
2. Debug page supports the user to modify the size of the debug window which can also be set to full screen. If the current page size is 1024 * 768, but the monitor resolution of 1080P, the software will be adjusted for the actual layout of the situation.
3. The Debug function supports sending and receiving JSON CODE commands while simulating data return.



Screenshot of the debug page:



A description of each section:

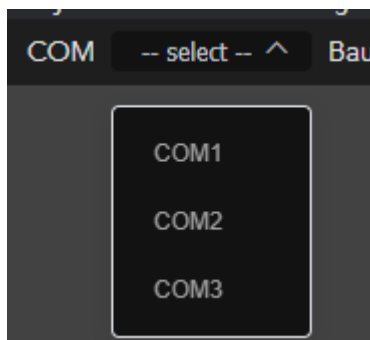
- 1: COM port Setting.
- 2: Baud Rate Setting.
- 3: Connection Status.
- 4: Display Ratio.
- 5: Command Sending Area.
- 6: Data Return Area.
- 7: Debug Window.

The overall page is divided into a number of different sections, each of which is described in more detail below.

9.2 Detailed description of each section:

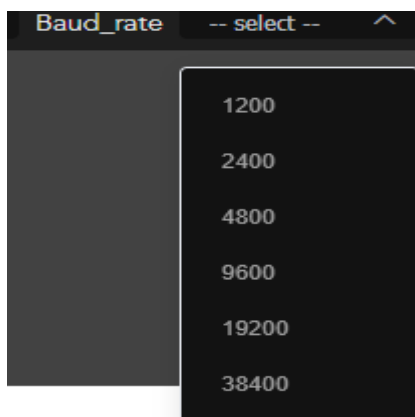
9.2.1 COM port Setting

Automatically scan the COM ports of the connected user's MCU in the PC, and you can select the required ones for connection test.



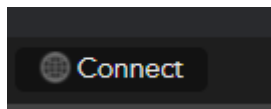
9.2.2 Baud Rate Setting

By default, the initial value of the baud rate setting is 115200. Users can select the baud rate setting that meets their project communication needs from these options. Currently the device supports 1200, 2400, 4800, 9600, 19200, 115200, 230400, 460800, 500000, 921600 and 1500000.



9.2.3 Connection Status

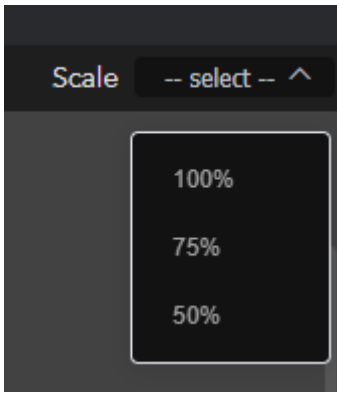
Used to show the connection status of PC and external MCU, the icon on the left side will change to bright color after connection.



9.2.4 Display Ratio

Use to show the current debug interface's zoom ratio, default is 100%.

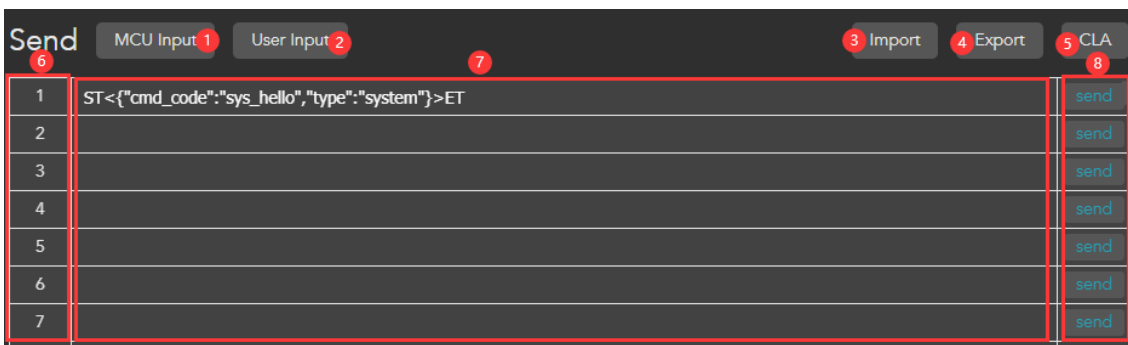
Zoom in is not supported, you can choose the value of 50%, 75%, 100%.



9.2.5 Command Input Area

Command input only retains the string input format, and all of the commands can be found in **#Instruction Sets.pdf**.

The debug can parse the commands and execute the corresponding actions in the GUI page if the commands are correct.



1 : MCU Input : Commands are sent from the MCU for the widgets in the GUI project to respond.

2 : User Input : Manually enter the JSON CODE in 7 and click 8 to make the GUI to respond.

3 : Import : Import instructions. Will automatically recognize instructions into each num.

4 : Export : Export Instructions. Consolidates and exports all existing instructions in the current num to a file.

5 : CLA : Clears all currently entered commands.

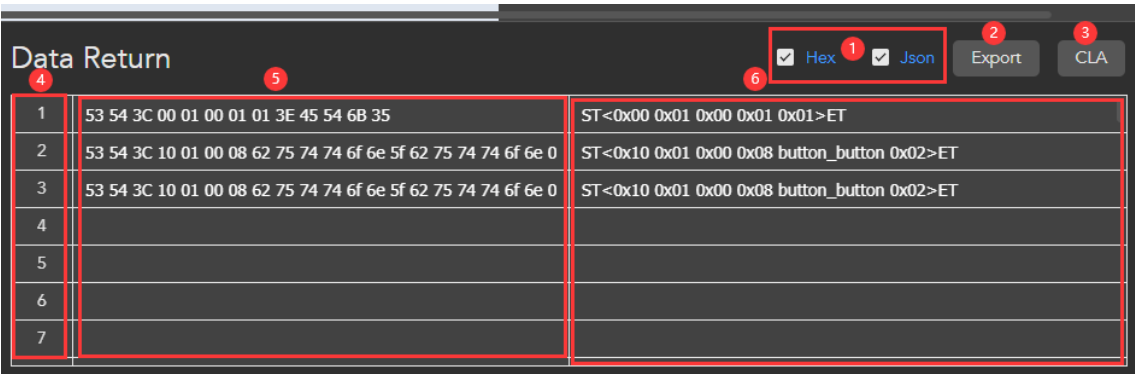
6 : Num : Serial number, not modifiable.

7 : data : The command to send, user-defined to send. Only valid in User Input state.

8 : send : Click to send the command in this line of data.

9.2.6 Data Return Area

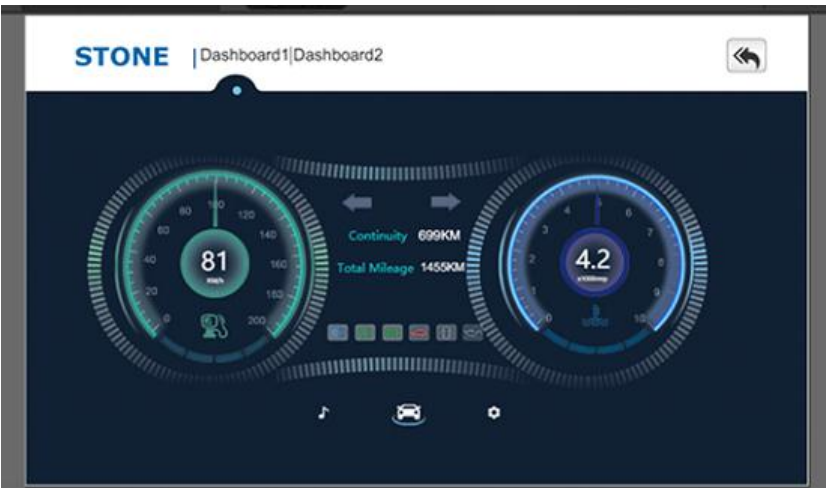
The format of the data returned is fixed in hexadecimal (hex) format and is not editable. Users can export data files according to their actual needs.



- 1: Export Type: HEX and parsed JSON CODE can choose to export one or both.
- 2 : Export : Export command. Consolidates and exports the data already in all current nums to a file.
- 3 : CLA : Clears all currently entered commands..
- 4 : Num : Serial number, not modifiable.
- 5 : data : Data return value of type hex.
- 6 : Data Parsing: Parses the hex on the left into a partial JSON CODE for easy visualization.

9.2.7 Debug Window

The Debug window is a simulation of the current GUI project.



The effects that can be supported include, but are not limited to, the following:

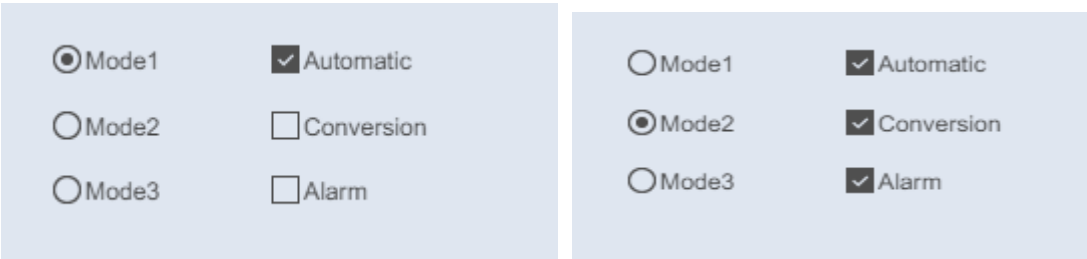
- 1. Key tone.
- 2. Button controls page jump and data increase or decrease.
- 3. Virtual keyboard (with edit and other widgets) .
- 4. Window animation and widgets animation.
- 5. Audio and video playback.

9.3 Debug Function Demo Example

9.3.1 Device-side effect simulation

Debug can realize most of the interaction effects in the HMI device, the following are some examples.

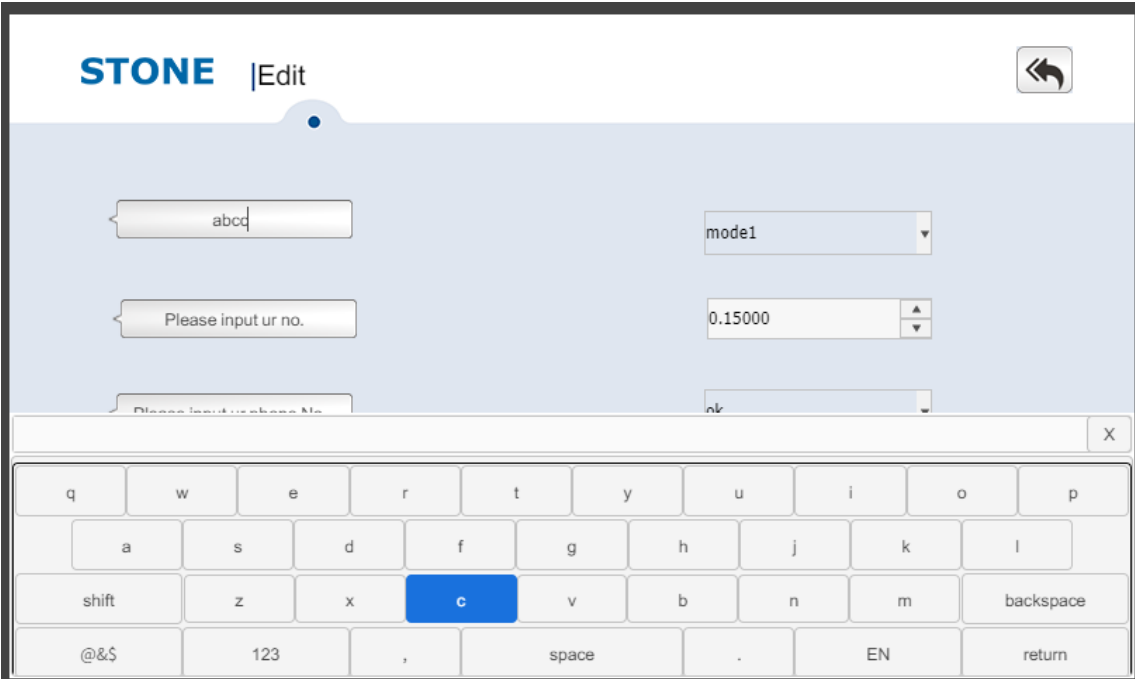
1) Simulation of checked state for check_button and radio_button widgets.

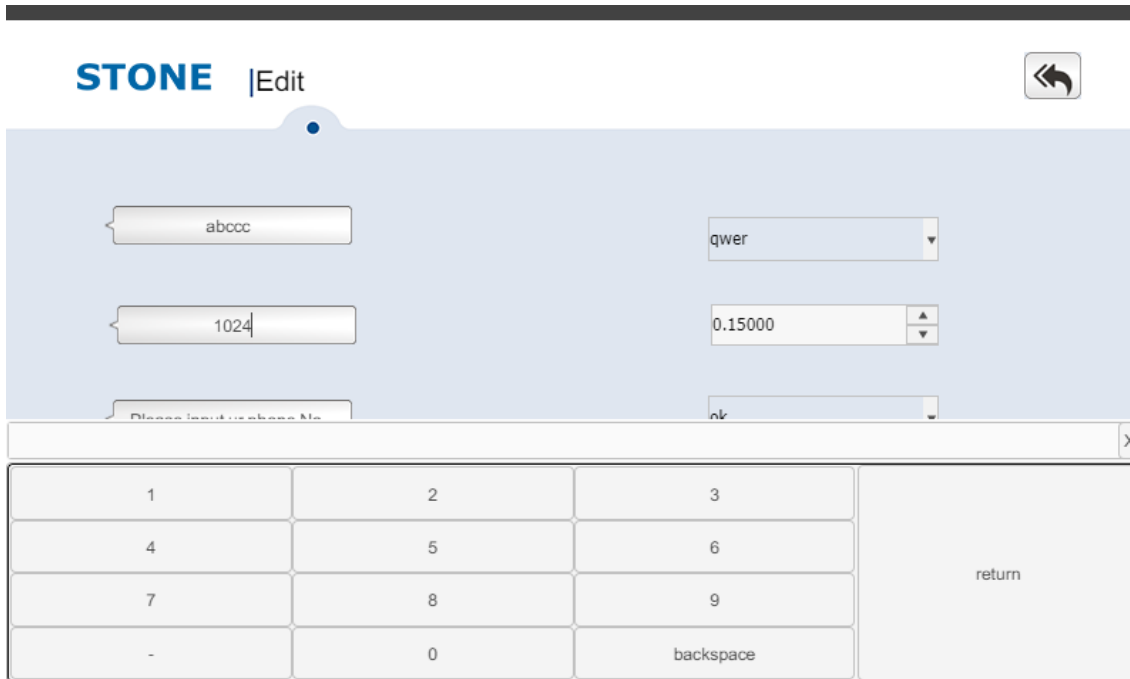


2) Simulation of digital clocks and dial clocks.

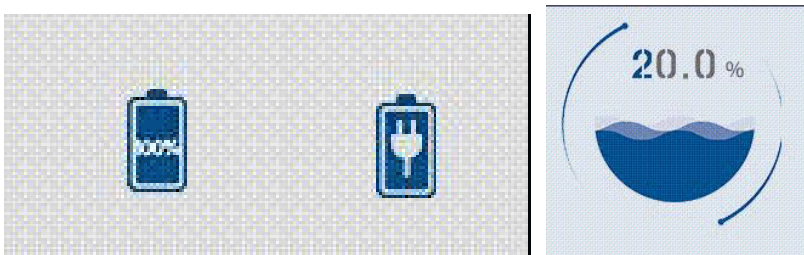


3) Simulation of keyboard. The simulated keyboard can be clicked after popping up, only widgets with keyboard property are supported.





4) Simulation of image_animation.

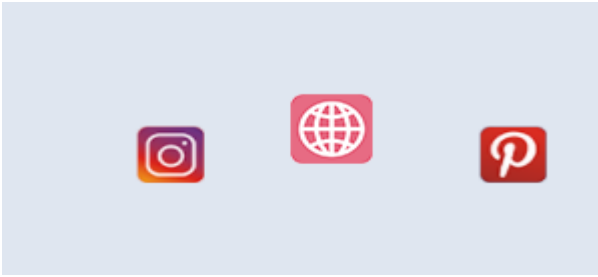


5) Simulation effects for widgets. This includes, but is not limited to, drag up and down, drag left and right, and widgets page switching.

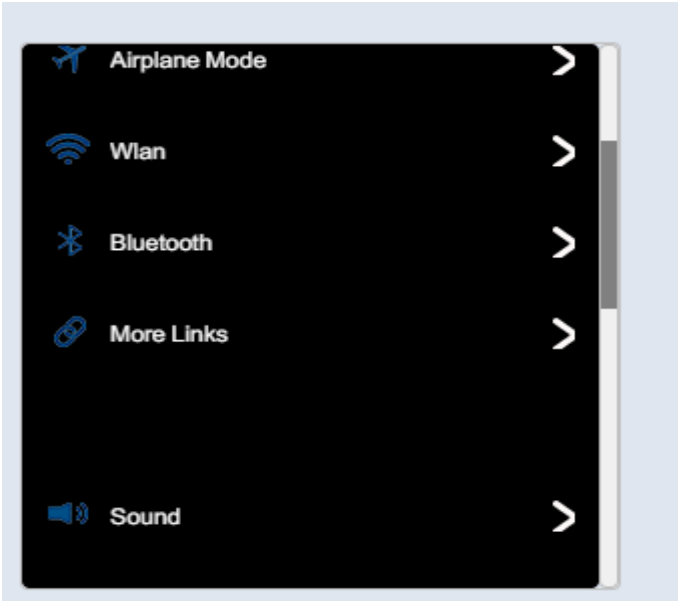
Drag up and down effects for text_selector.



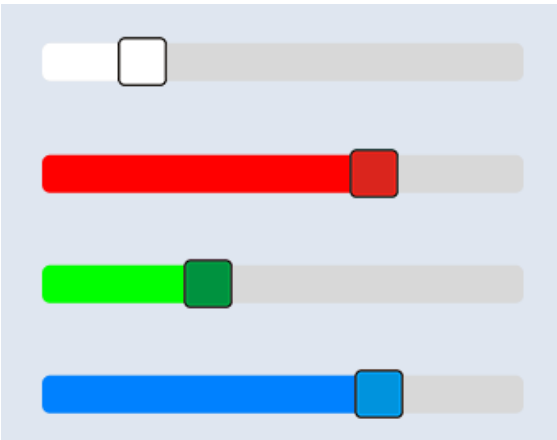
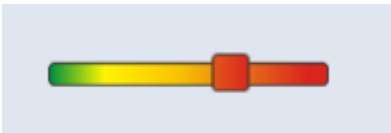
slide_menu's left/right swipe toggle effect.



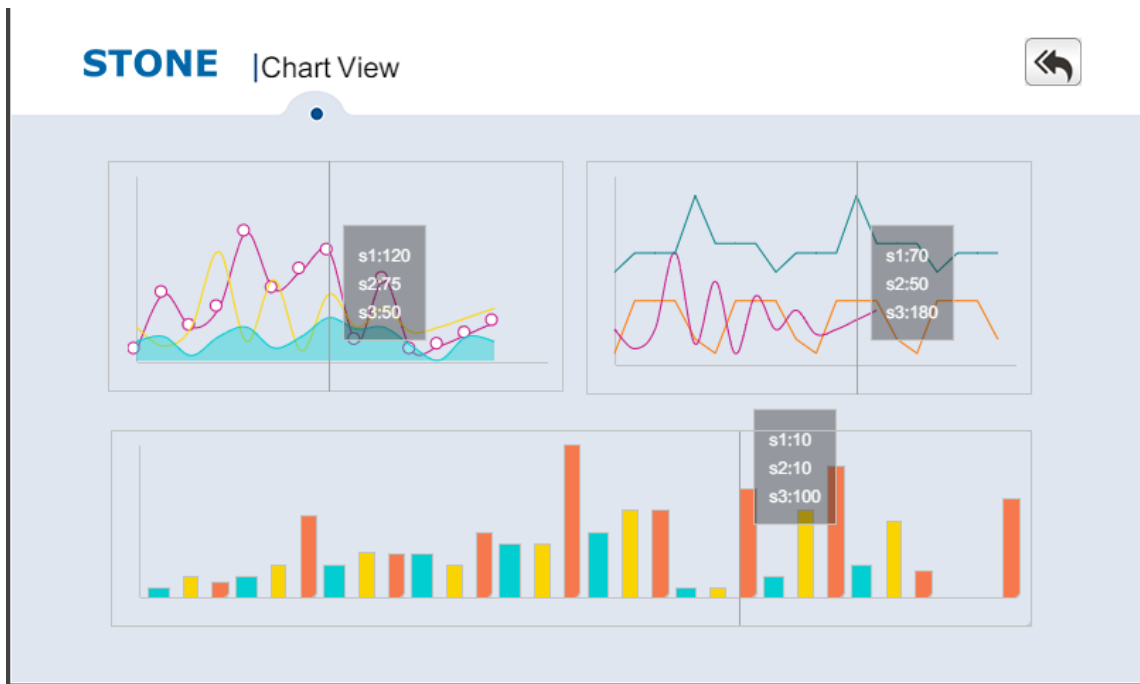
list_view's up and down slide effect.



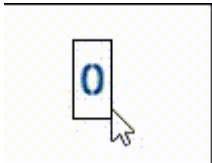
Slider's sliding effect achieved by clicking on the slider of the widget.



6) Visualization of the chart_view widget and data display of the tooltip.



7) The selectable effect of the image widget. (Clicking on it will work with switching images)



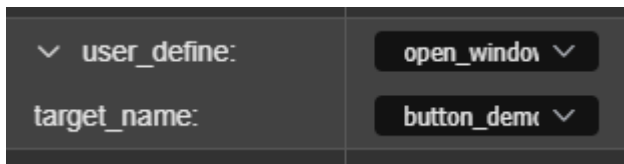
9.3.2 user_define function

6.2.1 user_define chapter mentioned that the User can set button widget's user_define function to realize the page jump, or by associating some widgets to increase or decrease their values. Here Debug function can also simulate these functions. Take the default demo as an example.

First of all, let's set up a button widget with a open_window function:

1) open_window

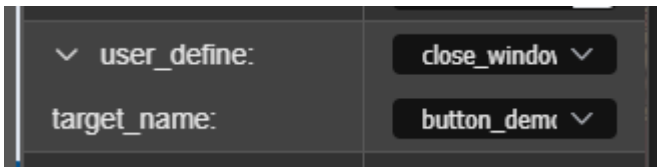
When this button is clicked, the GUI will open the page which set at target_name.



2) close_window

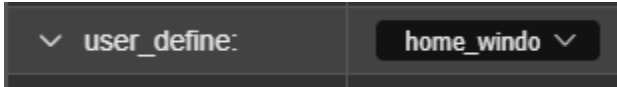
Clicking the button will close the window which set at target_name. do not close the home_page.

Normally this option corresponds to the window itself.



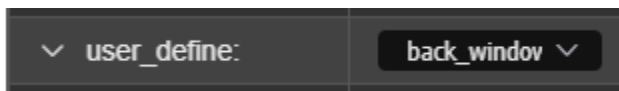
3) home_window

Clicking the button will jump to the home_page window. No need to set target_name.



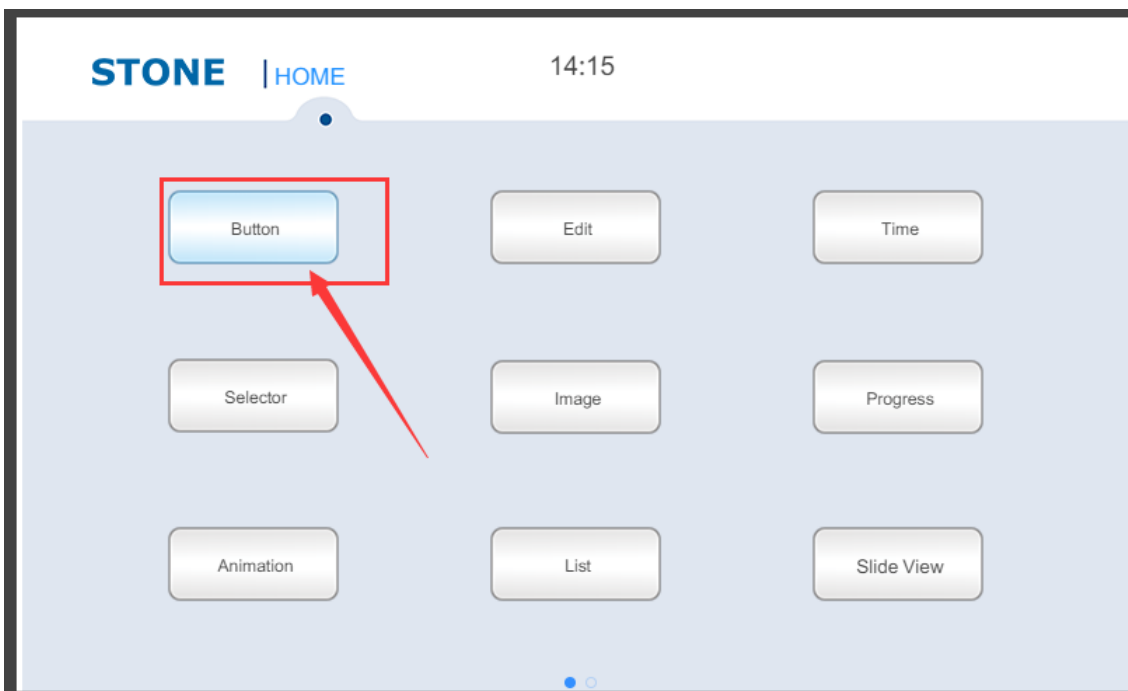
4) back_window

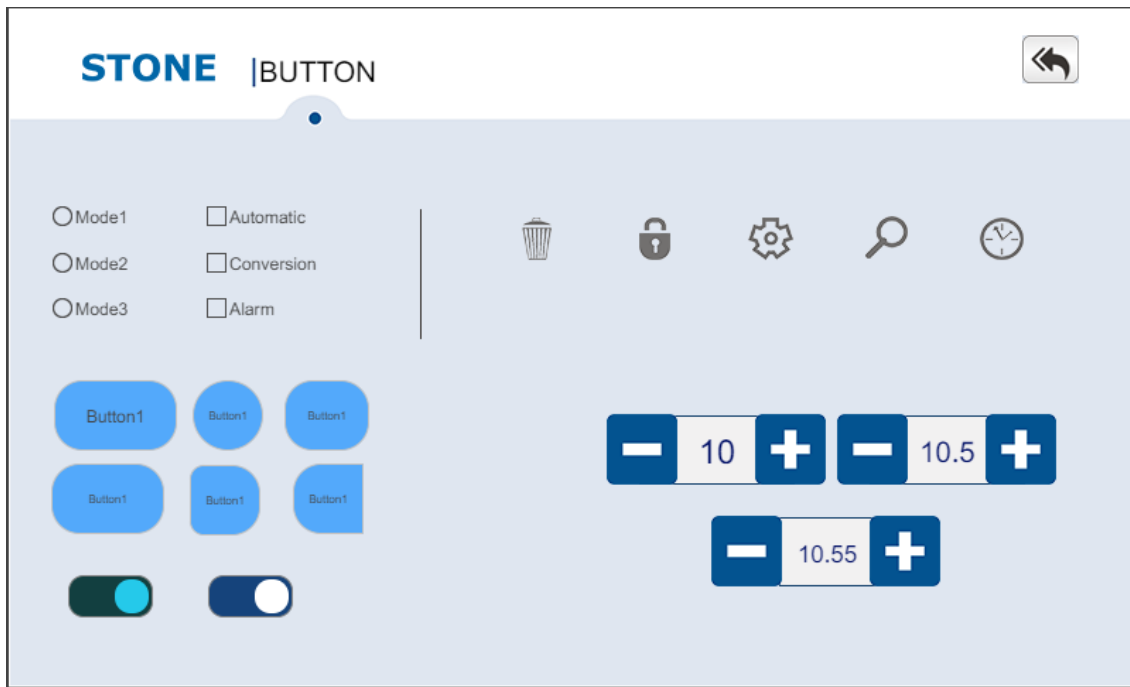
Clicking the button will return to the upper window (i.e. to the page of the previous window). No need to set target_name.



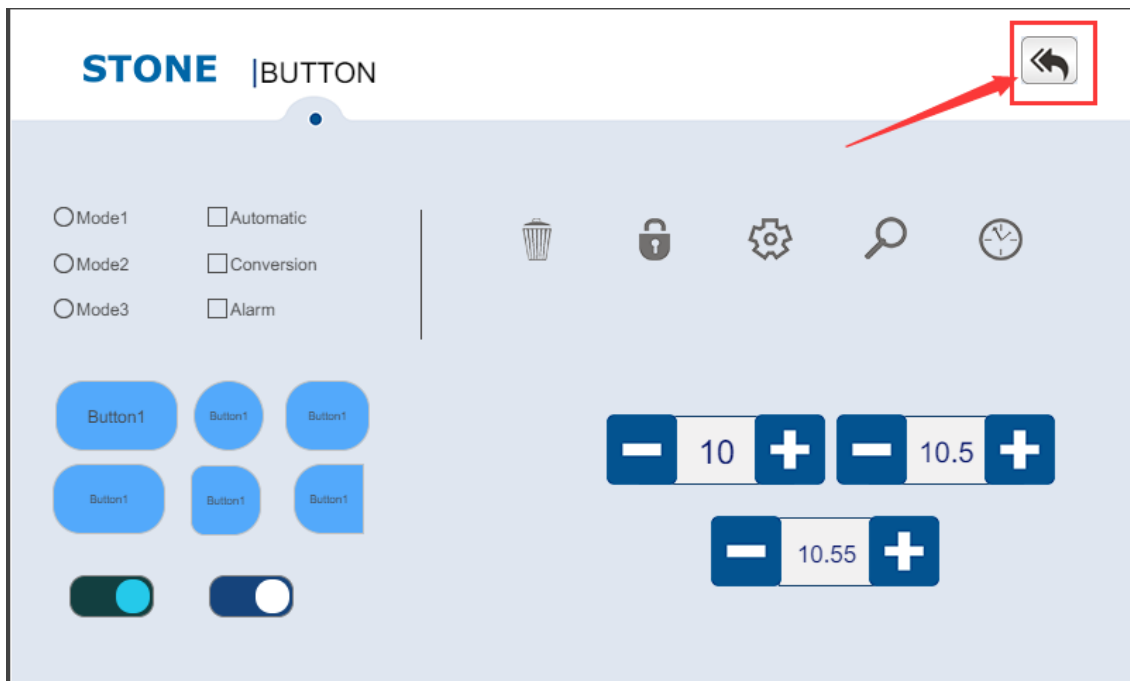
The following is the demo effect of window jump:

After clicking the button, it will first show the pressed state effect, and after releasing it, it will jump to the corresponding set page.

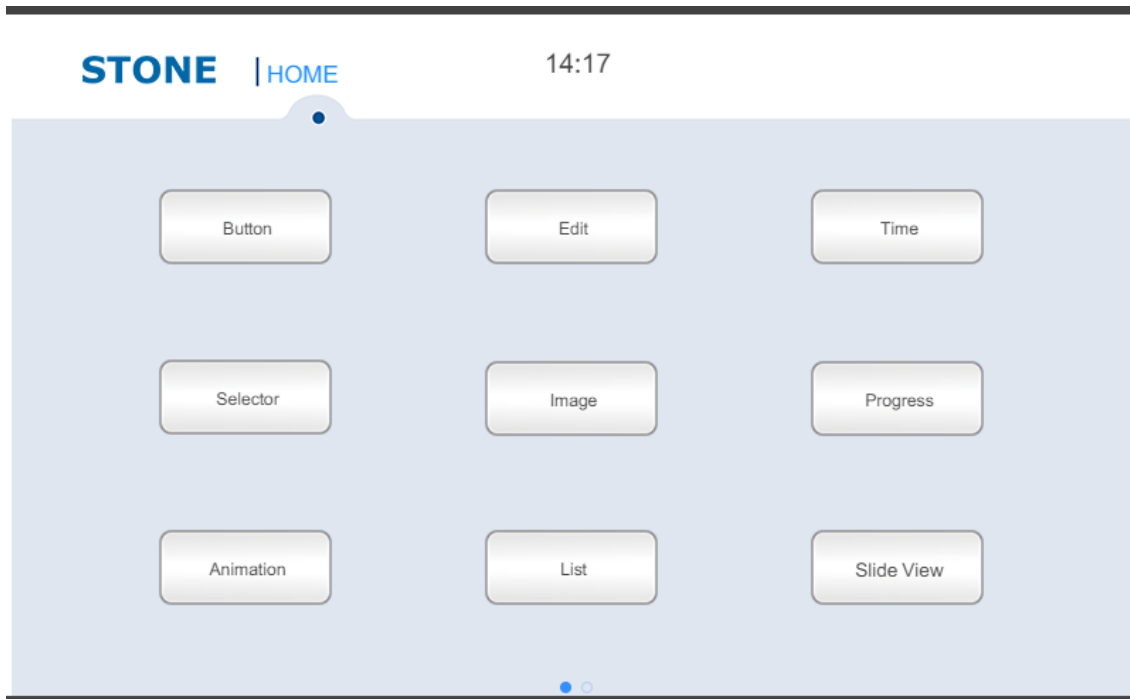




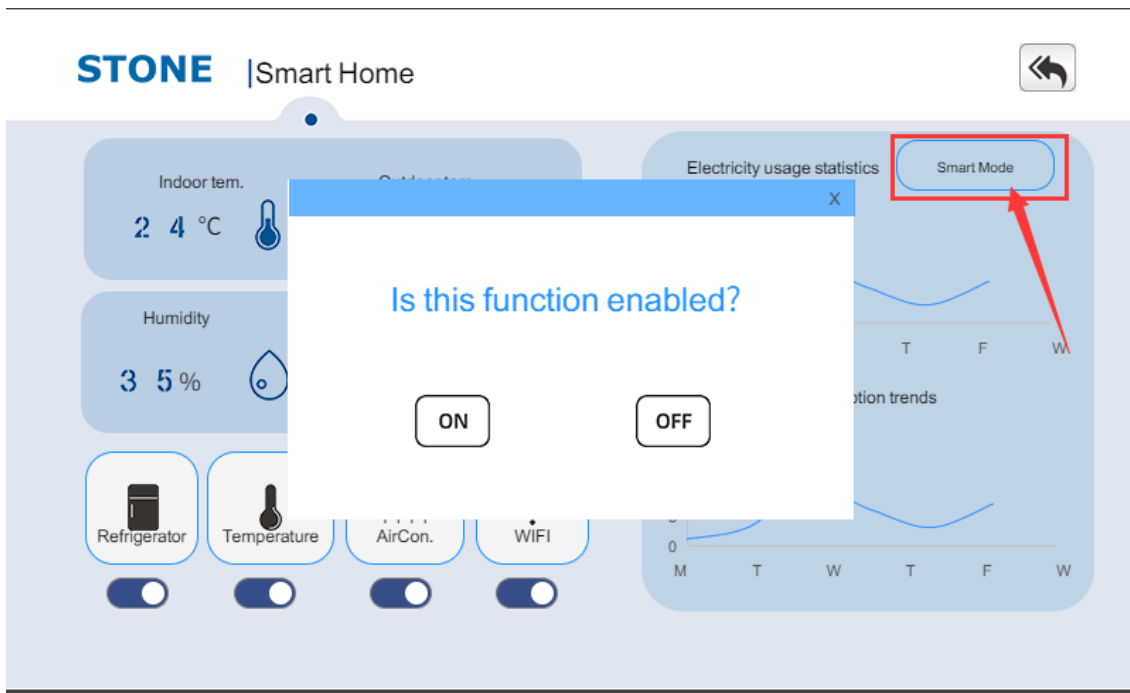
Similarly, clicking on other buttons will also realize the corresponding set up close_window、back_window、home_window.



Clicking on it will execute the corresponding jump action.



Special note, the dialog can also be set with the `user_define` function of the button widget, and then click the button to pop up the corresponding dialog.



To control the increase or decrease of the value, you need to combine the button's settings in the GUI project, and click the button to realize the increase or decrease. The following figure is an example of the properties set up in the GUI project.

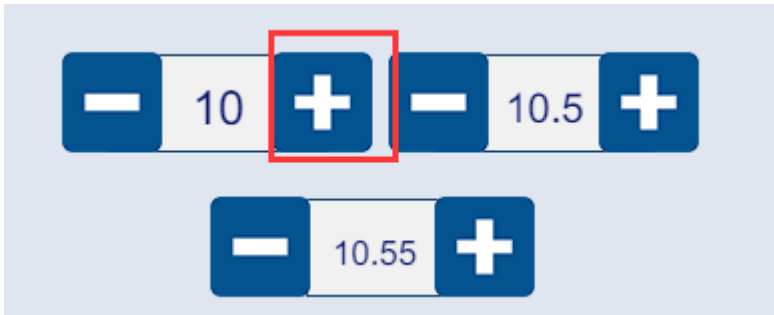
margin:	0
user_define:	set_value
target_name:	int_num
min:	0
max:	100
step:	1
format:	%d
increase_decrease:	++
enable_long_press:	☒

Similarly, the button can also control the value of progress_bar and progress_circle to increase or decrease, the following are the settings of the two buttons.

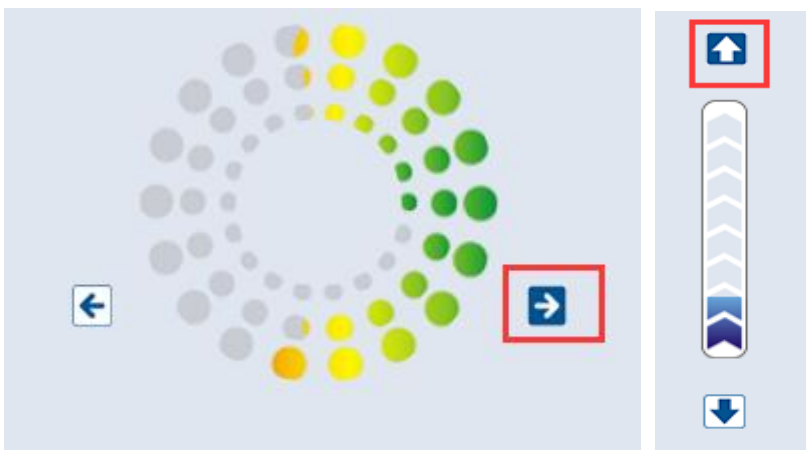
user_define:	set_value
target_name:	circle_right4
min:	0
max:	100
step:	1
format:	%d
increase_decrease:	++

user_define:	set_value
target_name:	bar2_vertical
min:	0
max:	100
step:	1
format:	%d
increase_decrease:	++

Clicking on it will increase or decrease the format according to the set format.



The above settings together with the setup of the GUI, clicking on the two buttons can be realized by increasing the values of progress_circle and progress_bar.



9.3.3 Simulation of instruction sending and receiving

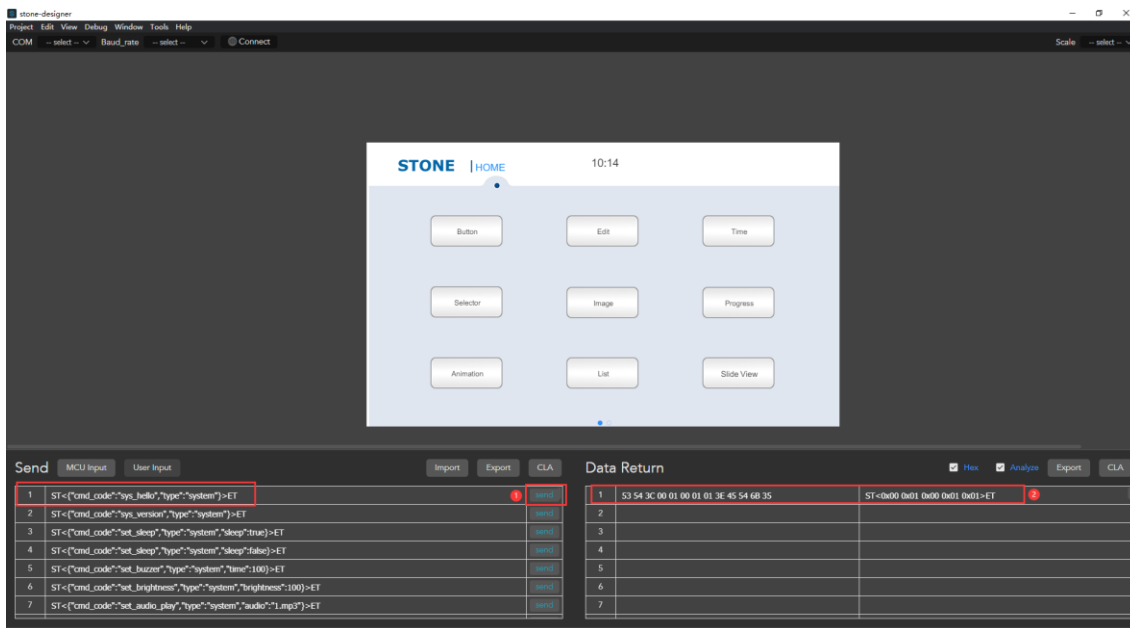
In conjunction with [9.2.5 Instruction Input Area](#) and [9.2.6 Data Return Area](#) in the above sections, a simple demonstration of the actual application is given.

A complete instruction initialization and data return requires the following steps:

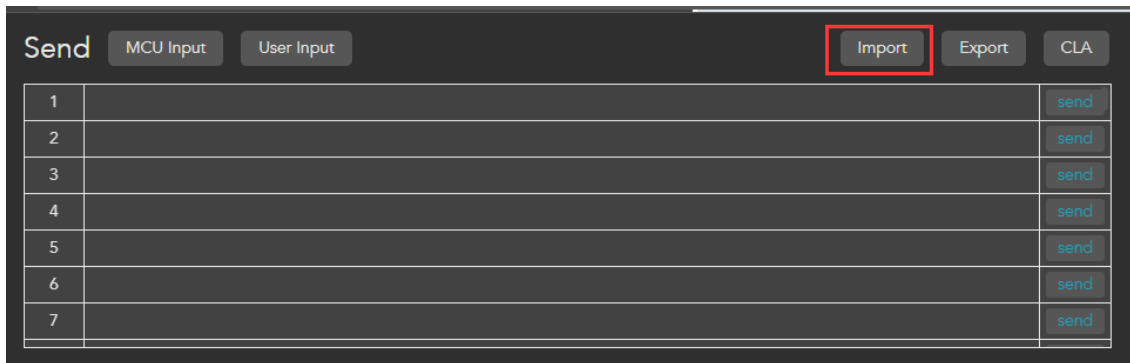
- 1) Write the instruction in the Send column on the left.
- 2) Click on the send button.
- 3) The corresponding return is given at Data Return Area.

Special Note: Please refer to the commands and responses in the **#Instruction Sets.pdf** if there is a data return.

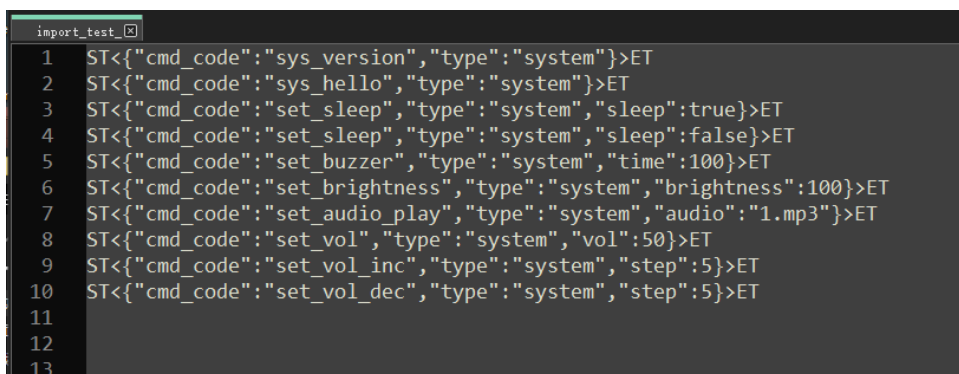
The following example simulates sending the sys_hello command and the system gives a response.



1) Command Import: Click Import to import the corresponding file.



When preparing the command file, you need to pay attention to the format of the commands inside the file, one line for each command, as shown in the following figure.



After importing, you can refer to the picture shown below.

Send

MCU Input

User Input

Import

Export

CLA

1	ST<{"cmd_code":"sys_version","type":"system"}>ET	send
2	ST<{"cmd_code":"sys_hello","type":"system"}>ET	send
3	ST<{"cmd_code":"set_sleep","type":"system","sleep":true}>ET	send
4	ST<{"cmd_code":"set_sleep","type":"system","sleep":false}>ET	send
5	ST<{"cmd_code":"set_buzzer","type":"system","time":100}>ET	send
6	ST<{"cmd_code":"set_brightness","type":"system","brightness":100}>ET	send
7	ST<{"cmd_code":"set_audio_play","type":"system","audio":"1.mp3"}>ET	send

2) Command Export: Click Export to export all the commands of the current Send column.

Send

MCU Input

User Input

Import

Export

CLA

1	ST<{"cmd_code":"sys_version","type":"system"}>ET	send
2	ST<{"cmd_code":"sys_hello","type":"system"}>ET	send
3	ST<{"cmd_code":"set_sleep","type":"system","sleep":true}>ET	send
4	ST<{"cmd_code":"set_sleep","type":"system","sleep":false}>ET	send
5	ST<{"cmd_code":"set_buzzer","type":"system","time":100}>ET	send
6	ST<{"cmd_code":"set_brightness","type":"system","brightness":100}>ET	send
7	ST<{"cmd_code":"set_audio_play","type":"system","audio":"1.mp3"}>ET	send

The exported file can be referred to the following figure, where each row is the command for the corresponding Send line.

export_test_	
1	ST<{"cmd_code":"sys_version","type":"system"}>ET
2	ST<{"cmd_code":"sys_hello","type":"system"}>ET
3	ST<{"cmd_code":"set_sleep","type":"system","sleep":true}>ET
4	ST<{"cmd_code":"set_sleep","type":"system","sleep":false}>ET
5	ST<{"cmd_code":"set_buzzer","type":"system","time":100}>ET
6	ST<{"cmd_code":"set_brightness","type":"system","brightness":100}>ET
7	ST<{"cmd_code":"set_audio_play","type":"system","audio":"1.mp3"}>ET
8	ST<{"cmd_code":"set_vol","type":"system","vol":50}>ET
9	ST<{"cmd_code":"set_vol_inc","type":"system","step":5}>ET
10	ST<{"cmd_code":"set_vol_dec","type":"system","step":5}>ET
11	
12	

3) Data Export: Combined with the options on the left, you can choose to export hex or parsed hex text. You can choose to export individually or together.

1

2

☒ Hex
 ☒ Analyze

Export

CLA

1	53 54 3C 10 71 00 0e 70 68 6f 6e 65 5f 65 64 69 74 00 00 00 (	ST<0x10 0x71 0x00 0x0e phone_edit 0x00 0x00 0x00 0x03>ET
2	53 54 3C 10 71 00 0e 70 68 6f 6e 65 5f 65 64 69 74 00 00 00 (	ST<0x10 0x71 0x00 0x0e phone_edit 0x00 0x00 0x00 0x03>ET
3	53 54 3C 10 71 00 0e 70 68 6f 6e 65 5f 65 64 69 74 00 00 00 (	ST<0x10 0x71 0x00 0x0e phone_edit 0x00 0x00 0x00 0x03>ET
4	53 54 3C 10 71 00 0e 70 68 6f 6e 65 5f 65 64 69 74 00 00 00 (	ST<0x10 0x71 0x00 0x0e phone_edit 0x00 0x00 0x00 0x03>ET
5	53 54 3C 10 71 00 0e 70 68 6f 6e 65 5f 65 64 69 74 00 00 00 (	ST<0x10 0x71 0x00 0x0e phone_edit 0x00 0x00 0x00 0x03>ET
6	53 54 3C 10 70 00 17 22 6e 61 6d 65 5f 65 64 69 74 22 3a 48	ST<0x10 0x70 0x00 0x17 "name_edit":Hello Stone>ET
7	53 54 3C 10 01 00 08 65 64 69 74 5f 62 75 74 74 6f 6e 02 3E	ST<0x10 0x01 0x00 0x08 edit_button 0x02>ET

The following are cases where both types are exported at the same time:

```
export_hex  export_analyze  export_hex_and_analyze
1  53 54 3C 10 62 00 0e 66 6c 6f 61 74 32 5f 6e 75 6d 41 20 cc cc 3E 45 54 22 91
2  ST<0x10 0x62 0x00 0x0e float2_num 0x41 0x20 0xcc 0xcc >ET
3
4  53 54 3C 10 01 00 08 66 6c 6f 61 74 32 5f 6e 75 6d 5f 64 02 3E 45 54 2A 9B
5  ST<0x10 0x01 0x00 0x08 float2_num_d 0x02>ET
6
7  53 54 3C 10 62 00 0e 66 6c 6f 61 74 32 5f 6e 75 6d 41 24 cc cc 3E 45 54 A6 90
8  ST<0x10 0x62 0x00 0x0e float2_num 0x41 0x24 0xcc 0xcc >ET
9
10 53 54 3C 10 01 00 08 66 6c 6f 61 74 32 5f 6e 75 6d 5f 64 02 3E 45 54 2A 9B
11 ST<0x10 0x01 0x00 0x08 float2_num_d 0x02>ET
12
13 53 54 3C 10 62 00 0b 69 6e 74 5f 6e 75 6d 41 40 00 00 3E 45 54 A1 5E
14 ST<0x10 0x62 0x00 0x0b int_num 0x41 0x40 0x00 0x00 >ET
15
16 53 54 3C 10 01 00 08 69 6e 74 5f 6e 75 6d 5f 69 02 3E 45 54 35 60
17 ST<0x10 0x01 0x00 0x08 int_num_i 0x02>ET
18
19 53 54 3C 10 62 00 0b 69 6e 74 5f 6e 75 6d 41 30 00 00 3E 45 54 91 55
20 ST<0x10 0x62 0x00 0x0b int_num 0x41 0x30 0x00 0x00 >ET
21
22 53 54 3C 10 01 00 08 69 6e 74 5f 6e 75 6d 5f 69 02 3E 45 54 35 60
23 ST<0x10 0x01 0x00 0x08 int_num_i 0x02>ET
24
25 53 54 3C 10 01 00 08 62 75 74 74 6f 6e 5f 62 75 74 74 6f 6e 02 3E 45 54 2A 23
26 ST<0x10 0x01 0x00 0x08 button_button 0x02>ET
27
28 53 54 3C 00 01 00 01 01 3E 45 54 6B 35
29 ST<0x00 0x01 0x00 0x01 0x01>ET
30
```

Below is just the export hex:

```
C:\Users\DELL\Desktop\export_hex - Notepad++ [Administrator]
文件(F) 编辑(E) 搜索(S) 视图(V) 编码(N) 语言(L) 设置(T) 工具(O) 宏(M) 运行(R) 插件(P) 窗口(W) ?
53 54 3C 10 62 00 0e 66 6c 6f 61 74 32 5f 6e 75 6d 41 20 cc cc 3E 45 54 22 91
53 54 3C 10 01 00 08 66 6c 6f 61 74 32 5f 6e 75 6d 5f 64 02 3E 45 54 2A 9B
53 54 3C 10 62 00 0e 66 6c 6f 61 74 32 5f 6e 75 6d 41 24 cc cc 3E 45 54 A6 90
53 54 3C 10 01 00 08 66 6c 6f 61 74 32 5f 6e 75 6d 5f 64 02 3E 45 54 2A 9B
53 54 3C 10 62 00 0b 69 6e 74 5f 6e 75 6d 41 40 00 00 3E 45 54 A1 5E
53 54 3C 10 01 00 08 69 6e 74 5f 6e 75 6d 5f 69 02 3E 45 54 35 60
53 54 3C 10 62 00 0b 69 6e 74 5f 6e 75 6d 41 30 00 00 3E 45 54 91 55
53 54 3C 10 01 00 08 69 6e 74 5f 6e 75 6d 5f 69 02 3E 45 54 35 60
53 54 3C 10 01 00 08 62 75 74 74 6f 6e 5f 62 75 74 74 6f 6e 02 3E 45 54 2A 23
53 54 3C 00 01 00 01 01 3E 45 54 6B 35
53 54 3C 00 02 00 10 56 32 2E 32 2E 31 33 2E 32 32 30 33 32 39 52 43 3E 45 54 1C 58
53 54 3C 00 01 00 01 01 3E 45 54 6B 35
53 54 3C 00 01 00 01 01 3E 45 54 6B 35
53 54 3C 00 02 00 10 56 32 2E 32 2E 31 33 2E 32 32 30 33 32 39 52 43 3E 45 54 1C 58
```

Below is just the export analyze :

```

export_hex [x]  exprot_analyze [x]
1  ST<0x10 0x62 0x00 0x0e float2_num 0x41 0x20 0xcc 0xcc >ET
2
3  ST<0x10 0x01 0x00 0x08 float2_num_d 0x02>ET
4
5  ST<0x10 0x62 0x00 0x0e float2_num 0x41 0x24 0xcc 0xcc >ET
6
7  ST<0x10 0x01 0x00 0x08 float2_num_d 0x02>ET
8
9  ST<0x10 0x62 0x00 0x0b int_num 0x41 0x40 0x00 0x00 >ET
10
11 ST<0x10 0x01 0x00 0x08 int_num_i 0x02>ET
12
13 ST<0x10 0x62 0x00 0x0b int_num 0x41 0x30 0x00 0x00 >ET
14
15 ST<0x10 0x01 0x00 0x08 int_num_i 0x02>ET
16
17 ST<0x10 0x01 0x00 0x08 button_button 0x02>ET
18
19 ST<0x00 0x01 0x00 0x01 0x01>ET
20
21 ST<0x00 0x02 0x00 0x10 V2.2.13.220329RC>ET
22
23 ST<0x00 0x01 0x00 0x01 0x01>ET
24
25 ST<0x00 0x01 0x00 0x01 0x01>ET
26
27 ST<0x00 0x02 0x00 0x10 V2.2.13.220329RC>ET

```

Note: Data return does not support import, only export.